

*MLPDA*

2023秋

作业参考答案

双语对照版

# 1 Machine Learning for Predictive Data Analytics (Exercise Solutions)

4. The following table lists a dataset from the credit scoring domain that we discussed in the chapter. Underneath the table we list two prediction models consistent with this dataset, **Model 1** and **Model 2**.

| ID | OCCUPATION   | AGE | LOAN-SALARY | OUTCOME |
|----|--------------|-----|-------------|---------|
|    |              |     | RATIO       |         |
| 1  | industrial   | 39  | 3.40        | default |
| 2  | industrial   | 22  | 4.02        | default |
| 3  | professional | 30  | 2.7 0       | repay   |
| 4  | professional | 27  | 3.32        | default |
| 5  | professional | 40  | 2.04        | repay   |
| 6  | professional | 50  | 6.95        | default |
| 7  | industrial   | 27  | 3.00        | repay   |
| 8  | industrial   | 33  | 2.60        | repay   |
| 9  | industrial   | 30  | 4.5 0       | default |
| 10 | professional | 45  | 2.78        | repay   |

## Model 1

```
if LOAN-SALARY RATIO > 3.00 then
    OUTCOME = default
else
    OUTCOME = repay
end if
```

1Machine Learning for Predictive Data Analytics (Exercise Solutions )

4. 下表列出了我们在本章中讨论的信用评分领域的数据集。在表格下方，我们列出了两个与此数据集一致的预测模型，即模型 1 和模型 2。

| L-S |       |    |       |  |    |
|-----|-------|----|-------|--|----|
| 身份证 | OA    | R  | O     |  |    |
| 1   | 工业    | 39 | 3.40  |  | 违约 |
| 2   | 工业    | 22 | 4.02  |  | 违约 |
| 3   | 专业 30 |    | 2.7 0 |  | 偿还 |
| 4   | 专业 27 |    | 3.32  |  | 违约 |
| 5   | 专业 40 |    | 2.04  |  | 偿还 |
| 6   | 专业 50 |    | 6.95  |  | 违约 |
| 7   | 工业    | 27 | 3.00  |  | 偿还 |
| 8   | 工业    | 33 | 2.60  |  | 偿还 |
| 9   | 工业    | 30 | 4.5 0 |  | 违约 |
| 10  | 专业 45 |    | 2.78  |  | 偿还 |

模型 1  
如果  $L-S \geq 3.00$  那么  
     $O =$  默认值  
else  
     $O =$  还款  
结束如果

## Model 2

```
if AGE= 50 then
    OUTCOME = default
else if AGE= 39 then
    OUTCOME = default
else if AGE= 30 and OCCUPATION = industrial then
    OUTCOME = default
else if AGE= 27 and OCCUPATION = professional then
    OUTCOME = default
else
    OUTCOME = repay
end if
```

- (a) Which of these two models do you think will generalize better to instances not contained in the dataset?

**Model 1** is more likely to generalize beyond the training dataset because it is simpler and appears to be capturing a real pattern in the data.

- (b) Propose an inductive bias that would enable a machine learning algorithm to make the same preference choice that you made in Part (a).

归纳的偏差  
相同偏好的选择

If you are choosing between a number of models that perform equally well then prefer the simpler model over the more complex models.

- (c) Do you think that the model that you rejected in Part (a) of this question is overfitting or underfitting the data?

**Model 2** is overfitting the data. All of the decision rules in this model that predict OUTCOME = *default* are specific to single instances in the dataset. Basing predictions on single instances is indicative of a model that is overfitting.

## 第 1 章：用于预测数据分析的机器学习（练习解决方案）

### 模型 2

```
如果 A " 50 那么
  O = 默认值
否则, 如果 A " 39 那么
  O = 默认值
否则, 如果 A " 30 和 O = 工业, 则
  O = 默认值
否则, 如果 A " 27 和 O = 专业, 那么
  O = 默认值
else
  O = 还款
结束如果
```

(a) 您认为这两种模型中哪一种可以更好地推广到数据集中未包含的实例？

模型 1 更有可能泛化到训练数据集之外，因为它更简单，并且似乎在数据中捕获了真实的模式。

(b) 提出一种归纳偏差，使机器学习算法能够做出与您在第 (a) 部分中所做的相同偏好选择。

如果要在许多性能相同的模型之间进行选择，则首选更简单的模型而不是更复杂的模型。

(c) 你认为你在这个问题的 (a) 部分所拒绝的模型是数据拟合过拟合还是欠拟合？

模型 2 对数据进行了过度拟合。此模型中预测 O = 默认值的所有决策规则都特定于数据集中的单个实例。基于单个实例进行预测表示模型过度拟合。

- \* 9. The following table lists a dataset of five individuals described via a set of stroke risk factors and their probability of suffering a stroke in the next five years. This dataset has been prepared by an analytics team who are developing a model as a decision support tool for doctors.<sup>1</sup> The goal of the model is to classify individuals into groups on the basis of their risk of suffering a stroke **STROKE RISK**. In this dataset there are three categories of risk: *low*, *medium*, and *high*. All the descriptive features are Boolean, taking two levels: *true* or *false*.

| ID | HIGH BLOOD<br>PRESSURE | SMOKER | DIABETES | HEART<br>DISEASE | STROKE<br>RISK |
|----|------------------------|--------|----------|------------------|----------------|
| 1  | true                   | false  | true     | true             | high           |
| 2  | true                   | true   | true     | true             | high           |
| 3  | true                   | false  | false    | true             | medium         |
| 4  | false                  | false  | false    | false            | low            |
| 5  | true                   | true   | true     | false            | high           |

- (a) How many possible models exist for the scenario described by the features in this dataset?

We begin by calculating the number of combinations of descriptive features. Their four boolean features and so there are  $2^4 = 16$  possible combinations of levels. The target feature also takes three levels which gives us a total number of possible mappings from descriptive feature combinations to target feature level of:  $3^{16} = 43046721$

- (b) How many of these potential models would be consistent with this sample of data?

There are  $16 - 5 = 11$  combinations of descriptive feature values that are not covered by the dataset. And therefore there are  $3^{11} = 177147$  potential models that are consistent with this dataset.

第 1 章：用于预测数据分析的机器学习（练习解决方案）

9. 下表列出了通过一组卒中风险描述的五个个体的数据集因素及其在未来五年内患中风的可能性。该数据集由一个分析团队准备，该团队正在开发一个模型作为医生的决策支持工具。该模型的目标是根据个人患中风 SR 的风险将个体分类为几组。在此数据集中，风险分为三类：低、中和高。所有描述性特征都是布尔值，分为两个级别：true 或 false。

|     | HB   |    |    |   | HS |   |  |  |
|-----|------|----|----|---|----|---|--|--|
| 身份证 | PS   | DD | RR |   |    |   |  |  |
| 1   | true | 假  | 真  | 真 | 真  | 高 |  |  |
| 2   | true | 真  | 真  | 真 | 真  | 高 |  |  |
| 3   | true | 假  | 假  |   | 真  | 高 |  |  |
| 4   | 假    | 假  | 假  | 假 | 真  | 高 |  |  |
| 5   | true | 真  | 真  | 假 | 高  |   |  |  |

(a) 对于该数据集中的特征所描述的场景，存在多少个可能的模型？

我们首先计算描述性特征的组合数量。它们的四个布尔特征，因此有  $2^4 = 16$  种可能的级别组合。目标特征也采用三个级别，这为我们提供了从描述性特征组合到目标特征级别的可能映射总数：  $3^{16} = 43046721$

(b) 这些潜在模型中有多少与这些数据样本一致？

有  $16^5 = 1048576$  种描述性特征值的组合未被数据集覆盖。因此，有  $3^{16} - 1048576 = 41998145$  的潜在模型与该数据集一致。

## 2 Data to Insights to Decisions (Exercise Solutions)

3. The table below shows a sample of a larger dataset containing details of policyholders at an insurance company. The descriptive features included in the table describe each policy holders' ID, occupation, gender, age, the value of their car, the type of insurance policy they hold, and their preferred contact channel.

| ID | OCCUPATION     | GENDER | AGE | MOTOR<br>VALUE | POLICY<br>TYPE | PREF<br>CHANNEL |
|----|----------------|--------|-----|----------------|----------------|-----------------|
| 1  | lab tech       | female | 43  | 42,632         | planC          | sms             |
| 2  | farmhand       | female | 57  | 22,096         | planA          | phone           |
| 3  | biophysicist   | male   | 21  | 27,221         | planA          | phone           |
| 4  | sheriff        | female | 47  | 21,460         | planB          | phone           |
| 5  | painter        | male   | 55  | 13,976         | planC          | phone           |
| 6  | manager        | male   | 19  | 4,866          | planA          | email           |
| 7  | geologist      | male   | 51  | 12,759         | planC          | phone           |
| 8  | messenger      | male   | 49  | 15,672         | planB          | phone           |
| 9  | nurse          | female | 18  | 16,399         | planC          | sms             |
| 10 | fire inspector | male   | 47  | 14,767         | planC          | email           |

- (a) State whether each descriptive feature contains numeric, interval, ordinal, categorical, binary, or textual data.

2Data to Insights to Decisions (Exercise Solutions )

3. 下表显示了一个较大数据集的示例，其中包含保险公司投保人的详细信息。表中包含的描述性特征描述了每个投保人的 ID、职业、性别、年龄、汽车价值、他们持有的保险单类型以及他们首选的联系渠道。

| MPP |        |    |        |        |       |      |  |  |  |
|-----|--------|----|--------|--------|-------|------|--|--|--|
| ID  | OGAVTC |    |        |        |       |      |  |  |  |
| 1   | 实验室技术  | 女性 | 43     | 42,632 | planC | 短信   |  |  |  |
| 2   | 农工女    | 57 | 22,096 | planA  | 电话    |      |  |  |  |
| 3   | 生物物理学家 | 男  | 21     | 27,221 | planA | 电话   |  |  |  |
| 4   | 郡治安官   | 女性 | 47     | 21,460 | planB | 电话   |  |  |  |
| 5   | 画家     | 男  | 55     | 13,976 | planC | 电话   |  |  |  |
| 6   | 经理     | 男  | 19     | 4,866  | planA | 电子邮件 |  |  |  |
| 7   | 地质学家   | 男  | 51     | 12,759 | planC | 电话   |  |  |  |
| 8   | 信使     | 男  | 49     | 15,672 | planB | 电话   |  |  |  |
| 9   | 护士     | 女性 | 18     | 16,399 | planC | 短信   |  |  |  |
| 10  | 消防督察   | 男  | 47     | 14,767 | planC | 电子邮件 |  |  |  |

(a) 说明每个描述性特征是否包含数字、区间、序数、分类、二进制或文本数据。

|             |             |            |         |
|-------------|-------------|------------|---------|
| ID          | Ordinal     | MOTORVALUE | Numeric |
| OCCUPATION  | Textual     | POLICYTYPE | Ordinal |
| GENDER      | Categorical | AGE        | Numeric |
| PREFCHANNEL | Categorical |            |         |

- (b) How many levels does each categorical, binary, or ordinal feature have?

|             |                                                                        |
|-------------|------------------------------------------------------------------------|
| ID          | 10 are present in the sample, but there is likely to be 1 per customer |
| GENDER      | 2 ( <i>male, female</i> )                                              |
| POLICYTYPE  | 3 ( <i>planA, planB, planC</i> )                                       |
| PREFCHANNEL | 3 ( <i>sms, phone, email</i> )                                         |

- \* 8. The following table shows a sample of a larger dataset that has been collected to build a model to predict whether newly released movies will be a hit or not.<sup>2</sup> The dataset contains details of movies that have already been released, including their title, running time, rating, genre, year of release, and whether or not the actor Kevin Costner had a starring role. An indicator of whether they were a success—a *hit*—or not—a *miss*—based on box office returns compared to budget is also included.

2. This dataset has been artificially created for this book, but machine learning has been used for this task, for example, by ?.

第 2 章：从数据到决策的见解（练习解决方案）

|       |           |
|-------|-----------|
| ID 序号 | MVNumeric |
| O     | Textual   |
| P     | TOrdinal  |
| GA    | 通用        |
| PC    | 封装        |

(b) 每个分类、二元或有序特征有多少个层次？

|     |                                                |
|-----|------------------------------------------------|
| ID  | 样本中存在 10 个，但每个客户可能有 1 个                        |
| G2  | (男、女)                                          |
| PT3 | ( <i>planA</i> 、 <i>planB</i> 、 <i>planC</i> ) |
| PC3 | (短信、电话、电子邮件)                                   |

- ° 8. 下表显示了已收集用于生成的较大数据集的示例
- 一个预测新上映的电影是否会成为热门模型。该数据集包含已上映电影的详细信息，包括片名、上映时间、评级、类型、上映年份以及演员凯文·科斯特纳是否担任主演。还包括一个指标，即它们是否成功——成功——或不成功——基于票房回报与预算相比。
2. 这个数据集是为本书人为创建的，但机器学习已被用于此任务，例如，由？。

Chapter 2 Data to Insights to Decisions (Exercise Solutions)

| ID | TITLE           | LENGTH | RATING | GENRE      | COSTNER | YEAR | HIT  |
|----|-----------------|--------|--------|------------|---------|------|------|
| 1  | Jaws            | 124    | PG     | action     | false   | 1975 | hit  |
| 2  | Waterworld      | 135    | PG-13  | sci-fi     | true    | 1995 | miss |
| 3  | Hudson Hawk     | 100    | R      | adventure  | false   | 1991 | miss |
| 4  | Downfall        | 156    | R      | drama      | false   | 2004 | hit  |
| 5  | The Postman     | 177    | R      | action     | true    | 1997 | miss |
| 6  | Toy Story       | 81     | G      | children's | false   | 1995 | hit  |
| 7  | Field of Dreams | 107    | G      | drama      | true    | 1989 | hit  |
| 8  | Amelie          | 122    | R      | comedy     | false   | 2001 | hit  |

- (a) State whether each descriptive feature contains numeric, interval, ordinal, categorical, binary, or textual data.

|        |         |         |             |
|--------|---------|---------|-------------|
| ID     | Ordinal | GENRE   | Categorical |
| TITLE  | Textual | COSTNER | Binary      |
| LENGTH | Numeric | YEAR    | Interval    |
| RATING | Ordinal | HIT     | Binary      |

- (b) How many levels does each categorical, binary, or ordinal feature have?

|         |                                                                                 |
|---------|---------------------------------------------------------------------------------|
| ID      | 8 are present in the sample, but there is likely to be 1 per movie              |
| RATING  | 4 at least ( <i>G, PG, PG-13, R, ...</i> )                                      |
| GENRE   | 5 at least ( <i>action, adventure, children's, comedy, drama, sci-fi, ...</i> ) |
| COSTNER | 2 ( <i>true, false</i> )                                                        |
| HIT     | 2 ( <i>hit, miss</i> )                                                          |

## 第 2 章：从数据到决策的见解（练习解决方案）

ID T LRG CYH

|   |                 |     |       |     |     |      |     |    |
|---|-----------------|-----|-------|-----|-----|------|-----|----|
| 1 | 大白鲨             | 124 | PG    | 动作  | 虚假  | 1975 | 年   | 命中 |
| 2 | 水世界             | 135 | PG-13 | 科幻  | 真实  | 1995 | 小姐  |    |
| 3 | Hudson Hawk     | 100 | R     | 冒险  | 假   | 1991 | 小姐  |    |
| 4 | 垮台              | 156 | R     | 剧   | 假   | 2004 | 年   | 命中 |
| 5 | 邮差              | 177 | R     | 动作  | 真正的 | 1997 | 小姐  |    |
| 6 | 玩具总动员           | 81  | G     | 儿童  |     | 1995 | 年的假 | 热门 |
| 7 | Field of Dreams | 107 | G     | 电视剧 | 真正的 | 1989 | 年   | 热门 |
| 8 | 艾米莉             | 122 | R     | 喜剧  | 假   | 2001 | 年   | 热门 |

(a) 说明每个描述性特征是否包含数字、区间、序数、分类、二进制或文本数据。

ID 序号 G Categorical  
T 外部 C Binary  
L Numeric Y Interval  
R R 数据库

(b) 每个分类、二元或有序特征有多少个层次？

样本中存在 ID 8，但有  
每部电影 R4 可能至少为 1 个（G、PG、PG-13、R、...）至少 G5（动作、冒险、儿童、  
喜剧、剧情、科幻.....  
C2（真、假） H2（命中、未命中）

### 3 Data Exploration (Exercise Solutions)

1. The table below shows the age of each employee at a cardboard box factory.

| ID  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-----|----|----|----|----|----|----|----|----|----|----|
| AGE | 51 | 39 | 34 | 27 | 23 | 43 | 41 | 55 | 24 | 25 |

| ID  | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|-----|----|----|----|----|----|----|----|----|----|----|
| AGE | 38 | 17 | 21 | 37 | 35 | 38 | 31 | 24 | 35 | 33 |

Based on this data, calculate the following **summary statistics** for the AGE feature:

- (a) Minimum, maximum, and range

By simply reading through the values we can tell that the minimum value for the AGE feature is: 17.

By simply reading through the values we can tell that the maximum value for the AGE feature is: 55.

The range is simply the difference between the highest and lowest value:

$$\begin{aligned} \text{range}(\text{AGE}) &= (55 - 17) \\ &= 38 \end{aligned}$$

- (b) Mean and median

### 3数据探索 (练习解决方案)

1. 下表显示了纸板箱工厂每位员工的年龄。编号 1 2 3 4 5 6 7 8 9 10

答 51 39 34 27 23 43 41 55 24 25

编号 11 12 13 14 15 16 17 18 19 20 A38  
17 21 37 35 38 31 24 35 33

根据此数据，计算 A 特征的以下汇总统计数据：

(a) 最小值、最大值和范围

通过简单地读取这些值，我们可以看出 Afeature 的最小值是：17。

通过简单地读取这些值，我们可以看出 Afeature 的最大值为：55。

范围只是最高值和最低值之间的差值：

$$\text{范围(A)} = (55 - 17)$$

“ 38

(b) 平均值和中位数

We can calculate the mean of the AGE feature as follows:

$$\begin{aligned}\overline{\text{AGE}} &= \frac{1}{20} \times (51 + 39 + 34 + 27 + 23 + 43 + 41 + 55 + 24 + 25 \\ &\quad + 38 + 17 + 21 + 37 + 35 + 38 + 31 + 24 + 35 + 33) \\ &= \frac{671}{20} = 33.55\end{aligned}$$

To calculate the median of the AGE feature we first have to arrange the AGE values in ascending order:

17, 21, 23, 24, 24, 25, 27, 31, 33, 34, 35, 35, 37, 38, 38, 39, 41, 43, 51, 55

Because there are an even number of instances in this small dataset we take the mid-point of the middle two values as the median. These are 34 and 35 and so the median is calculated as:

$$\begin{aligned}\text{median}(\text{AGE}) &= (34 + 35)/2 \\ &= 34.5\end{aligned}$$

(c) Variance and standard deviation

To calculate the variance we first sum the squared differences between each value for AGE and the mean of AGE. This table illustrates this:

| ID         | AGE | $(\text{AGE} - \overline{\text{AGE}})$ | $(\text{AGE} - \overline{\text{AGE}})^2$ |
|------------|-----|----------------------------------------|------------------------------------------|
| 1          | 51  | 17.45                                  | 304.50                                   |
| 2          | 39  | 5.45                                   | 29.70                                    |
| 3          | 34  | 0.45                                   | 0.20                                     |
| 4          | 27  | -6.55                                  | 42.90                                    |
| 5          | 23  | -10.55                                 | 111.30                                   |
| 6          | 43  | 9.45                                   | 89.30                                    |
| 7          | 41  | 7.45                                   | 55.50                                    |
| 8          | 55  | 21.45                                  | 460.10                                   |
| 9          | 24  | -9.55                                  | 91.20                                    |
| 10         | 25  | -8.55                                  | 73.10                                    |
| 11         | 38  | 4.45                                   | 19.80                                    |
| 12         | 17  | -16.55                                 | 273.90                                   |
| 13         | 21  | -12.55                                 | 157.50                                   |
| 14         | 37  | 3.45                                   | 11.90                                    |
| 15         | 35  | 1.45                                   | 2.10                                     |
| 16         | 38  | 4.45                                   | 19.80                                    |
| 17         | 31  | -2.55                                  | 6.50                                     |
| 18         | 24  | -9.55                                  | 91.20                                    |
| 19         | 35  | 1.45                                   | 2.10                                     |
| 20         | 33  | -0.55                                  | 0.30                                     |
| <b>Sum</b> |     |                                        | <b>1,842.95</b>                          |

### 第 3 章 数据探索 (练习解决方案)

我们可以按如下方式计算 Afeature 的平均值:

$$\begin{aligned} \bar{A} &= \frac{1}{20} \sum_{i=1}^{20} A_i = \frac{1}{20} (51 + 39 + 34 + 27 + 23 + 43 + 41 + 55 + 24 + 25 \\ &\quad + 38 + 17 + 21 + 37 + 35 + 38 + 31 + 24 + 35 + 33) \\ &= \frac{671}{20} = 33.55 \end{aligned}$$

要计算 Afeature 的中位数, 我们首先必须按升序排列 A 值:

17, 21, 23, 24, 24, 25, 27, 31, 33, 34, 35, 35, 37, 38, 38, 39, 41, 43, 51, 55 因为这个小数据集有偶数个实例, 所以我们取中间两个值的中点作为中位数。它们是 34 和 35, 因此中位数计算如下:

$$\text{median}(A) = (34 + 35) / 2 = 34.5$$

#### (c) 方差和标准差

为了计算方差, 我们首先将 A 的每个值与 A 的平均值之间的平方差相加。下表说明了这一点:

| ID | A  | $(A - \bar{A})$ | $(A - \bar{A})^2$ |
|----|----|-----------------|-------------------|
| 1  | 51 | 17.45           | 304.50            |
| 2  | 39 | 5.45            | 29.70             |
| 3  | 34 | 0.45            | 0.20              |
| 4  | 27 | -6.55           | 42.90             |
| 5  | 23 | -10.55          | 111.30            |
| 6  | 43 | 9.45            | 89.30             |
| 7  | 41 | 7.45            | 55.50             |
| 8  | 55 | 21.45           | 460.10            |
| 9  | 24 | -9.55           | 91.20             |
| 10 | 25 | -8.55           | 73.10             |
| 11 | 38 | 4.45            | 19.80             |
| 12 | 17 | -16.55          | 273.90            |
| 13 | 21 | -12.55          | 157.50            |
| 14 | 37 | 3.45            | 11.90             |
| 15 | 35 | 1.45            | 2.10              |
| 16 | 38 | 4.45            | 19.80             |
| 17 | 31 | -2.55           | 6.50              |
| 18 | 24 | -9.55           | 91.20             |
| 19 | 35 | 1.45            | 2.10              |
| 20 | 33 | -0.55           | 0.30              |
|    |    | 总和              | 1,842.95          |

Based on the sum of squared differences value of 1,842.95 we can calculate the variance as:

$$\begin{aligned}\text{var}(\text{AGE}) &= \frac{1,842.95}{20 - 1} \\ &= 96.9974\end{aligned}$$

The standard deviation is calculated as the square root of the variance, so:

$$\begin{aligned}\text{sd}(\text{AGE}) &= \sqrt{\text{var}(\text{AGE})} \\ &= 9.8487\end{aligned}$$

- (d) 1<sup>st</sup> quartile (25<sup>th</sup> percentile) and 3<sup>rd</sup> quartile (75<sup>th</sup> percentile)

To calculate any percentile of the AGE feature we first have to arrange the AGE values in ascending order:

17, 21, 23, 24, 24, 25, 27, 31, 33, 34, 35, 35, 37, 38, 38, 39, 41, 43, 51, 55

We then calculate the index for the percentile value as:

$$\text{index} = n \times \frac{i}{100}$$

where  $n$  is the number of instances in the dataset and  $i$  is the percentile we would like to calculate. For the 25<sup>th</sup> percentile:

$$\begin{aligned}\text{index} &= 20 \times \frac{25}{100} \\ &= 5\end{aligned}$$

Because this is a whole number we can use this directly and so the 25<sup>th</sup> percentile is at index 5 in the ordered dataset and is 24.

For the 75<sup>th</sup> percentile:

$$\begin{aligned}\text{index} &= 20 \times \frac{75}{100} \\ &= 15\end{aligned}$$

Because this is a whole number we can use this directly and so the 75<sup>th</sup> percentile is at index 15 in the ordered dataset and is 38.

- (e) Inter-quartile range

### 第 3 章 数据探索 (练习解决方案)

根据差值 1,842.95 的平方和, 我们可以计算方差为:

$$\begin{aligned} \text{varpAq} &= \frac{1,842.95}{20 - 1} \\ &= 96.9974 \end{aligned}$$

标准差计算为方差的平方根, 因此:

$$\begin{aligned} \text{sdpAq} &= \sqrt{\text{varpAq}} \\ &= 9.8487 \end{aligned}$$

(d) 1 四分位数 (25 百分位数) 和 3 四分位数 (75 百分位数)

要计算 A 特征的任何百分位数, 我们首先必须按升序排列 A 值:  
17、21、23、24、24、25、27、31、33、34、35、35、37、  
38、38、39、41、43、51、55 然后, 我们将百分位值的指数计算  
为:

$$\text{索引} = \frac{n \times i}{100}$$

其中 n 是数据集中的实例数, i 是我们要计算的百分位数。对于  
25percentile :

$$\begin{aligned} \text{索引} &= \frac{20 \times 25}{100} \\ &= 5 \end{aligned}$$

因为这是一个整数, 我们可以直接使用它, 所以 25 百分位数在有序  
数据集中的索引 5 处, 是 24。

对于 75percentile :

$$\begin{aligned} \text{索引} &= \frac{20 \times 75}{100} \\ &= 15 \end{aligned}$$

因为这是一个整数, 我们可以直接使用它, 所以 75 百分位数在有序  
数据集中的索引 15 处, 并且是 38。

(e) 四分位数间距

To calculate the inter-quartile range we subtract the lower quartile value from the upper quartile value:

$$\begin{aligned} IQR(\text{AGE}) &= (38 - 24) \\ &= 14 \end{aligned}$$

(f) 12<sup>th</sup> percentile

We can use the ordered list of values above once more. For the 12<sup>th</sup> percentile:

$$\begin{aligned} index &= 20 \times \frac{12}{100} \\ &= 2.4 \end{aligned}$$

Because index is not a whole number we have to calculate the percentile as follows:

$$i^{\text{th}}\text{percentile} = (1 - index\_f) \times a_{index\_w} + index\_f \times a_{index\_w+1}$$

Because  $index = 2.4$ ,  $index\_w = 2$  and  $index\_f = 0.4$ . Using  $index\_w = 2$  we can look up  $\text{AGE}_2$  to be 21  $\text{AGE}_{2+1}$  to be 23. Using this we can calculate the 12<sup>th</sup> percentile as:

$$\begin{aligned} 12^{\text{th}}\text{percentile of AGE} &= (1 - 0.4) \times \text{AGE}_2 + 0.4 \times \text{AGE}_{2+1} \\ &= 0.6 \times 21 + 0.4 \times 23 \\ &= 21.8 \end{aligned}$$

2. The table below shows the policy type held by customers at a life insurance company.

| ID | POLICY   | ID | POLICY   | ID | POLICY   |
|----|----------|----|----------|----|----------|
| 1  | Silver   | 8  | Silver   | 15 | Platinum |
| 2  | Platinum | 9  | Platinum | 16 | Silver   |
| 3  | Gold     | 10 | Platinum | 17 | Platinum |
| 4  | Gold     | 11 | Silver   | 18 | Platinum |
| 5  | Silver   | 12 | Gold     | 19 | Gold     |
| 6  | Silver   | 13 | Platinum | 20 | Silver   |
| 7  | Bronze   | 14 | Silver   |    |          |

(a) Based on this data, calculate the following **summary statistics** for the POLICY feature:

第 3 章 数据探索 (练习解决方案)

为了计算四分位数间范围，我们从上四分位数值中减去下四分位数值：

$$IQR(A) = (38 - 24) / 14$$

(f) 12 百分位数

我们可以再次使用上面的值的有序列表。对于 12 个百分点：

$$\text{索引} = \frac{12}{100} \times 24 = 2.4$$

因为索引不是整数，所以我们必须按如下方式计算百分位数：

$$\text{percentile} = p1 + (\text{索引} - \text{索引} \downarrow) \times (\text{索引} \uparrow - \text{索引} \downarrow)$$

因为索引 “ 2.4 ”，索引 w “ 2 ” 和索引 f “ 0.4 ”。使用索引 w “ 2 ”，我们可以查找 Ato be 21 Ato be 23。使用它，我们可以计算 12 % 为：

$$\text{12 百分位数} = p1 + 0.4 \times (A[0.4] - A[0.2]) = 0.6 \times 21 + 0.4 \times 23 = 21.8$$

2. 下表显示了客户在人寿保险公司持有的保单类型。

| ID | P    | ID | P    | ID | P    |
|----|------|----|------|----|------|
| 1  | 银    | 8  | 银    | 15 | 铂    |
| 2  | 铂    | 9  | 铂    | 16 | 银    |
| 3  | Gold | 10 | 铂    | 17 | 铂    |
| 4  | Gold | 11 | 银    | 18 | 铂    |
| 5  | 银    | 12 | Gold | 19 | Gold |
| 6  | 银    | 13 | 铂    | 20 | 银    |
| 7  | 青铜   | 14 | 银    |    |      |

(a) 根据此数据，计算 P 特征的以下汇总统计数据：

i. Mode and 2<sup>nd</sup> mode

To calculate summary statistics for a categorical feature like this we start by counting the frequencies of each level for the feature. These are shown in this table:

| Level    | Frequency | Proportion |
|----------|-----------|------------|
| Bronze   | 1         | 0.05       |
| Silver   | 8         | 0.40       |
| Gold     | 4         | 0.20       |
| Platinum | 7         | 0.35       |

The proportions are calculated as the frequency of each level divided by the sum of all frequencies.

The mode is the most frequently occurring level and so in this case is *Silver*.

The 2<sup>nd</sup> mode is the second most frequently occurring level and so in this case is *Platinum*.

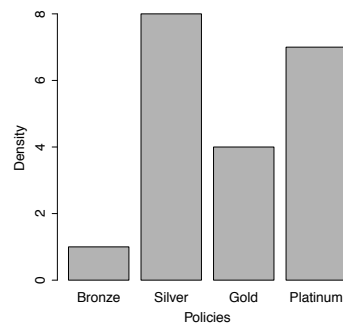
ii. Mode % and 2<sup>nd</sup> mode %

The mode % is the proportion of occurrence of the mode and in this case the proportion of occurrence of *Silver* is 40%.

The mode % of the 2<sup>nd</sup> mode, *Platinum*, is 35%.

(b) Draw a **bar plot** for the POLICY feature.

A bar plot can be drawn from the frequency table given above:



We can use proportions rather than frequencies in the plot:

### 第 3 章 数据探索 (练习解决方案)

#### i. 模式和 2 模式

为了计算此类分类特征的汇总统计数据，我们首先要计算该特征每个级别的频率。下表显示了这些内容：

| 电平 | 频率 | 比例   |
|----|----|------|
| 青铜 | 1  | 0.05 |
| 白银 | 8  | 0.40 |
| 黄金 | 4  | 0.20 |
| 铂金 | 7  | 0.35 |

比例的计算方法是每个电平的频率除以所有频率的总和。

该模式是出现频率最高的级别，因此在本例中为白银。

2mode 是第二常出现的级别，因此在这种情况下是白金。

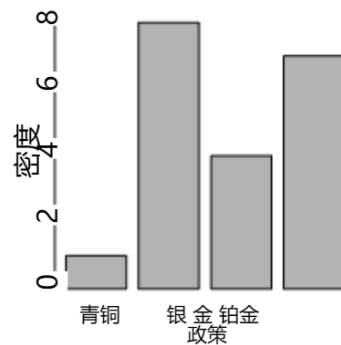
#### ii. 模式 % 和 2 模式 %

众数 % 是众数的出现比例，在这种情况下，银的出现比例为 40 %。

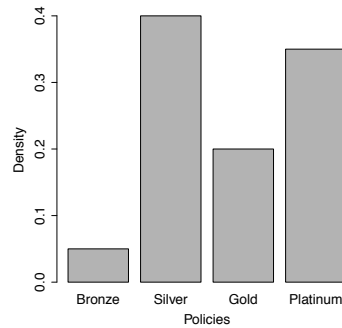
2 模式的模式百分比，铂金，是 35 %。

计算 P 特征的以下汇总统计数据： (b) 绘制 P 特征的条形图。

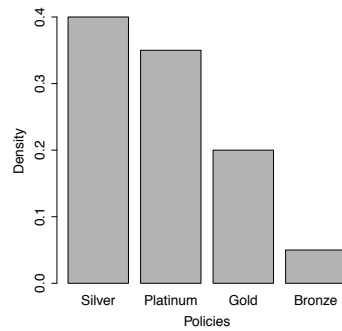
从上面给出的频率表中可以得出条形图：



我们可以在图中使用比例而不是频率：



In order to highlight the mode and 2<sup>nd</sup> mode we could order the bars in the plot by height:



This, however, is not an especially good idea in this case as the data, although categorical, has a natural ordering and changing this in a visualisation could cause confusion.

5. The table below shows the scores achieved by a group of students on an exam.

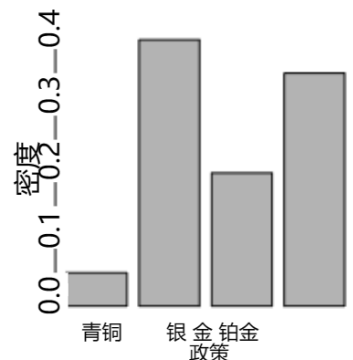
| ID    | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 |
|-------|----|----|----|----|----|----|----|----|----|----|
| SCORE | 42 | 47 | 59 | 27 | 84 | 49 | 72 | 43 | 73 | 59 |

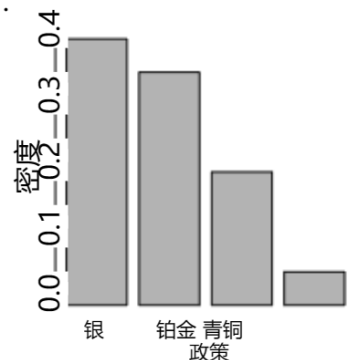
| ID    | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|-------|----|----|----|----|----|----|----|----|----|----|
| SCORE | 58 | 82 | 50 | 79 | 89 | 75 | 70 | 59 | 67 | 35 |

Using this data, perform the following tasks on the SCORE feature:

第 3 章 数据探索 (练习解决方案)



为了突出显示 mode 和 2mode , 我们可以按高度对图中的条形进行排序:



然而, 在这种情况下, 这并不是一个特别好的主意, 因为数据虽然是分类的, 但具有自然的顺序, 在可视化中更改它可能会导致混淆。

5. 下表显示了一组学生在考试中取得的分数。

|    |    |    |    |    |    |    |    |    |    |    |     |    |    |    |    |  |
|----|----|----|----|----|----|----|----|----|----|----|-----|----|----|----|----|--|
| 编号 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | S42 | 47 | 59 | 27 | 84 |  |
|    | 49 | 72 | 43 | 73 | 59 |    |    |    |    |    |     |    |    |    |    |  |
| 编号 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | S58 |    |    |    |    |  |
|    | 87 | 50 | 70 | 80 | 75 | 70 | 50 | 67 | 35 |    |     |    |    |    |    |  |

使用此数据, 对 Sfeature 执行以下任务:

规范化到0, 1区间

- (a) A **range normalization** that generates data in the range (0, 1)

To perform a range normalization, we need the minimum and maximum of the dataset and the high and low for the target range. From the data we can see that the minimum is 27 and the maximum is 89. In the question we are told that the low value of the target range is 0 and that the high value is 1. Using these values, we normalize an individual value using the following equation:

$$a'_i = \frac{a_i - \min(a)}{\max(a) - \min(a)} \times (\text{high} - \text{low}) + \text{low}$$

So, the first score in the dataset, 42, would be normalized as follows:

$$\begin{aligned} a'_i &= \frac{42 - 27}{89 - 27} \times (1 - 0) + 0 \\ &= \frac{15}{62} \\ &= 0.2419 \end{aligned}$$

This is repeated for each instance in the dataset to give the full normalized data set as

| ID    | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   |
|-------|------|------|------|------|------|------|------|------|------|------|
| SCORE | 0.24 | 0.32 | 0.52 | 0.00 | 0.92 | 0.35 | 0.73 | 0.26 | 0.74 | 0.52 |

| ID    | 11   | 12   | 13   | 14   | 15   | 16   | 17   | 18   | 19   | 20   |
|-------|------|------|------|------|------|------|------|------|------|------|
| SCORE | 0.50 | 0.89 | 0.37 | 0.84 | 1.00 | 0.77 | 0.69 | 0.52 | 0.65 | 0.13 |

- (b) A **range normalization** that generates data in the range (−1, 1)

This normalization differs from the previous range normalization only in that the high and low values are different—in this case, −1 and 1. So the first score in the dataset, 42, would be normalized as follows:

$$\begin{aligned} a'_i &= \frac{42 - 27}{89 - 27} \times (1 - (-1)) + (-1) \\ &= \frac{15}{62} \times 2 - 1 \\ &= -0.5161 \end{aligned}$$

### 第 3 章 数据探索 (练习解决方案)

对 Sfeature 执行以下任务: (a) 在 p0、lq 范围内生成数据的范围归一化

要执行范围归一化, 我们需要数据集的最小值和最大值以及目标范围的最高值和最低值。从数据中我们可以看出, 最小值为 27, 最大值为 89。在问题中, 我们被告知目标范围的低值为 0, 高值为 1。使用这些值, 我们使用以下公式对单个值进行归一化:

$$a = \frac{A' \text{ 明帕克} - \text{大腿} \text{ 'lowq' 低}}{\text{Maxpāq} \text{ '明帕克} - \text{大腿} \text{ 'lowq' 低}}$$

因此, 数据集中的第一个分数 42 将按如下方式进行归一化:

$$\begin{aligned} a &= \frac{42 - 27}{89 - 27} \times (1 - 0) + 0 \\ &= \frac{15}{62} \\ &= 0.2419 \end{aligned}$$

对数据集中的每个实例重复此操作, 以提供完整的规范化数据集

编号 1 2 3 4 5 6 7 8 9 10  
编号 11 12 13 14 15 16 17 18 19 20 S0.50 0.89  
0.37 0.84 1.00 0.77 0.69 0.52 0.65 0.13

(b) 在 p'1、lq 范围内生成数据的范围归一化

此归一化与之前的范围归一化的不同之处仅在于高值和低值不同, 在本例中为 '1 和 1。因此, 数据集中的第一个分数 42 将按如下方式进行归一化:

$$\begin{aligned} a &= \frac{42 - 27}{89 - 27} \times (1 - ('1)) + ('1) \\ &= \frac{15}{62} \times 2 - 1 \\ &= -0.5161 \end{aligned}$$

Applying this to each instance in the dataset gives the full normalized dataset as

| ID    | 1     | 2     | 3    | 4     | 5    | 6     | 7    | 8     | 9    | 10   |
|-------|-------|-------|------|-------|------|-------|------|-------|------|------|
| SCORE | -0.52 | -0.35 | 0.03 | -1.00 | 0.84 | -0.29 | 0.45 | -0.48 | 0.48 | 0.03 |

| ID    | 11   | 12   | 13    | 14   | 15   | 16   | 17   | 18   | 19   | 20    |
|-------|------|------|-------|------|------|------|------|------|------|-------|
| SCORE | 0.00 | 0.77 | -0.26 | 0.68 | 1.00 | 0.55 | 0.39 | 0.03 | 0.29 | -0.74 |

(c) A **standardization** of the data

To perform a standardization, we use the following formula for each instance in the dataset:

$$a'_i = \frac{a_i - \bar{a}}{sd(a)} \quad \begin{array}{l} \text{均值} \\ \text{标准差} \end{array}$$

So we need the mean,  $\bar{a}$ , and standard deviation,  $sd(a)$ , for the feature to be standardized. In this case, the mean is calculated from the original dataset as 60.95, and the standard deviation is 17.2519. So the standardized value for the first instance in the dataset can be calculated as

$$\begin{aligned} a'_i &= \frac{42 - 60.95}{17.2519} \\ &= -1.0984 \end{aligned}$$

Standardizing in the same way for the rest of the dataset gives us the following:

| ID    | 1     | 2     | 3     | 4     | 5    | 6     | 7    | 8     | 9    | 10    |
|-------|-------|-------|-------|-------|------|-------|------|-------|------|-------|
| SCORE | -1.10 | -0.81 | -0.11 | -1.97 | 1.34 | -0.69 | 0.64 | -1.04 | 0.70 | -0.11 |

| ID    | 11    | 12   | 13    | 14   | 15   | 16   | 17   | 18    | 19   | 20    |
|-------|-------|------|-------|------|------|------|------|-------|------|-------|
| SCORE | -0.17 | 1.22 | -0.63 | 1.05 | 1.63 | 0.81 | 0.52 | -0.11 | 0.35 | -1.50 |

6. The following table shows the IQs for a group of people who applied to take part in a television general-knowledge quiz.

| ID | 1  | 2   | 3  | 4   | 5   | 6  | 7  | 8   | 9  | 10  |
|----|----|-----|----|-----|-----|----|----|-----|----|-----|
| IQ | 92 | 107 | 83 | 101 | 107 | 92 | 99 | 119 | 93 | 106 |

| ID | 11  | 12 | 13  | 14 | 15 | 16  | 17  | 18 | 19  | 20  |
|----|-----|----|-----|----|----|-----|-----|----|-----|-----|
| IQ | 105 | 88 | 106 | 90 | 97 | 118 | 120 | 72 | 100 | 104 |

Using this dataset, generate the following **binned** versions of the IQ feature:

分箱版本

### 第 3 章 数据探索 (练习解决方案)

将此应用于数据集中的每个实例，可将完整的规范化数据集定为

| ID | 1     | 2     | 3    | 4     | 5    | 6     | 7    | 8     | 9    | 10   | S |
|----|-------|-------|------|-------|------|-------|------|-------|------|------|---|
|    | -0.52 | -0.35 | 0.03 | -1.00 | 0.84 | -0.29 | 0.45 | -0.48 | 0.48 | 0.03 |   |

| 编号 | 11   | 12   | 13    | 14   | 15   | 16   | 17   | 18   | 19   | 20    | S |
|----|------|------|-------|------|------|------|------|------|------|-------|---|
|    | 0.00 | 0.77 | -0.26 | 0.68 | 1.00 | 0.55 | 0.39 | 0.03 | 0.29 | -0.74 |   |

#### (c) 数据标准化

为了执行标准化，我们对数据集中的每个实例使用以下公式：

$$a' = \frac{a - \bar{a}}{sd}$$

因此，我们需要平均值  $\bar{a}$  和标准差  $sd$  才能对特征进行标准化。在本例中，根据原始数据集计算的平均值为 60.95，标准差为 17.2519。因此，数据集中第一个实例的标准化值可以计算为

$$a' = \frac{42 - 60.95}{17.2519} = -1.0984$$

以相同的方式对数据集的其余部分进行标准化，可以得到以下结果：

| ID | 1     | 2     | 3     | 4     | 5    | 6     | 7    | 8     | 9    | 10    | S |
|----|-------|-------|-------|-------|------|-------|------|-------|------|-------|---|
|    | -1.10 | -0.81 | -0.11 | -1.97 | 1.34 | -0.69 | 0.64 | -1.04 | 0.70 | -0.11 |   |

| 编号 | 11    | 12   | 13    | 14   | 15   | 16   | 17   | 18    | 19   | 20    | S |
|----|-------|------|-------|------|------|------|------|-------|------|-------|---|
|    | -0.17 | 1.22 | -0.63 | 1.05 | 1.63 | 0.81 | 0.52 | -0.11 | 0.35 | -1.50 |   |

6. 下表显示了申请参加电视常识测验的一组人的智商。

| ID | 1  | 2  | 3   | 4  | 5   | 6  | 7   | 8  | 9   | 10  | 智商 |
|----|----|----|-----|----|-----|----|-----|----|-----|-----|----|
|    | 92 | 99 | 119 | 93 | 106 | 92 | 107 | 83 | 101 | 107 |    |

| 身份证 | 11  | 12 | 13  | 14 | 15 | 16  | 17  | 18 | 19  | 20  | 智商 |
|-----|-----|----|-----|----|----|-----|-----|----|-----|-----|----|
|     | 105 | 88 | 106 | 90 | 97 | 118 | 120 | 72 | 100 | 104 |    |

使用此数据集，生成 IQ 功能的以下分箱版本：

- (a) An **equal-width binning** using 5 bins.

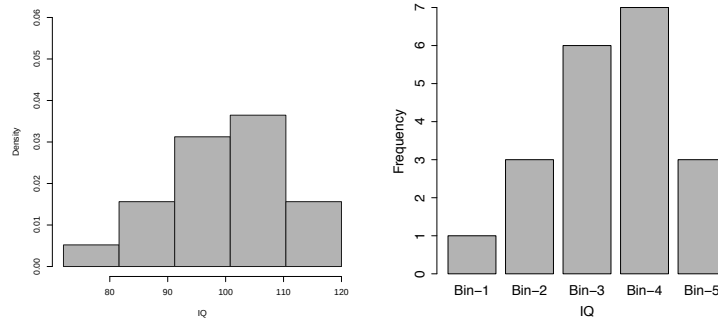
To perform an equal-width binning, we first calculate the bin size as  $\frac{\text{range}}{b}$  where  $b$  is the number of bins. In this case, this is calculated as  $\frac{120-72}{5} = 9.6$ , where 72 and 120 are the minimum and maximum values. Using the bin size we can calculate the following bin ranges.

| Bin | Low   | High  |
|-----|-------|-------|
| 1   | 72.0  | 81.6  |
| 2   | 81.6  | 91.2  |
| 3   | 91.2  | 100.8 |
| 4   | 100.8 | 110.4 |
| 5   | 110.4 | 120.0 |

Once we have calculated the boundaries, we can use these to determine the bin to which each result belongs.

| ID | IQ  | IQ (BIN) | ID | IQ  | IQ (BIN) |
|----|-----|----------|----|-----|----------|
| 1  | 92  | Bin-3    | 11 | 105 | Bin-4    |
| 2  | 107 | Bin-4    | 12 | 88  | Bin-2    |
| 3  | 83  | Bin-2    | 13 | 106 | Bin-4    |
| 4  | 101 | Bin-4    | 14 | 90  | Bin-2    |
| 5  | 107 | Bin-4    | 15 | 97  | Bin-3    |
| 6  | 92  | Bin-3    | 16 | 118 | Bin-5    |
| 7  | 99  | Bin-3    | 17 | 120 | Bin-5    |
| 8  | 119 | Bin-5    | 18 | 72  | Bin-1    |
| 9  | 93  | Bin-3    | 19 | 100 | Bin-3    |
| 10 | 106 | Bin-4    | 20 | 104 | Bin-4    |

It is interesting to graph a histogram of the values in the dataset according to the bin boundaries as well as a bar plot showing each of the bins created.



We can see from these graphs that a lot of instances belong to the middle bins and very few in the high and low bins.

### 第 3 章 数据探索 (练习解决方案)

(a) 使用 5 个箱的等宽分箱。

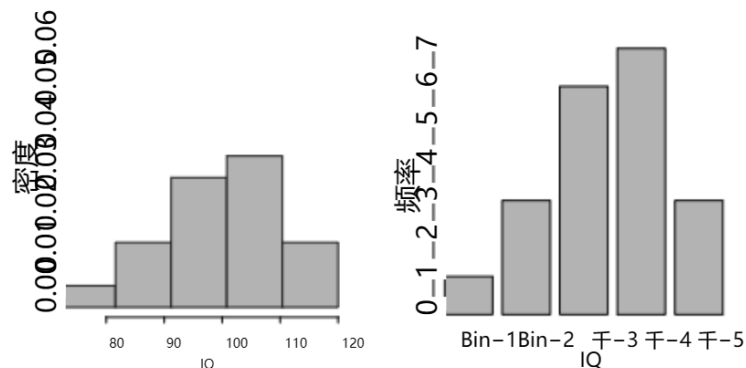
要执行等宽分箱，我们首先计算分箱大小，其中  $b$  是分箱数。在本例中，计算公式为  $\frac{120 - 72}{5} = 9.6$ ，其中 72 和 120 是最小值和最大值。使用箱，大小我们可以计算以下箱范围。

| Bin | Low   | 高     |
|-----|-------|-------|
| 1   | 72.0  | 81.6  |
| 2   | 81.6  | 91.2  |
| 3   | 91.2  | 100.8 |
| 4   | 100.8 | 110.4 |
| 5   | 110.4 | 120.0 |

一旦我们计算了边界，我们就可以使用这些边界来确定每个结果所属的条柱。

| ID | 智商  | 智商 (B) | ID | 智商  | 智商 (B) |
|----|-----|--------|----|-----|--------|
| 1  | 92  | Bin-3  | 11 | 105 | Bin-4  |
| 2  | 107 | Bin-4  | 12 | 88  | Bin-2  |
| 3  | 83  | Bin-2  | 13 | 106 | Bin-4  |
| 4  | 101 | Bin-4  | 14 | 90  | Bin-2  |
| 5  | 107 | Bin-4  | 15 | 97  | Bin-3  |
| 6  | 92  | Bin-3  | 16 | 118 | Bin-5  |
| 7  | 99  | Bin-3  | 17 | 120 | Bin-5  |
| 8  | 119 | Bin-5  | 18 | 72  | Bin-1  |
| 9  | 93  | Bin-3  | 19 | 100 | Bin-3  |
| 10 | 106 | Bin-4  | 20 | 104 | Bin-4  |

根据条柱边界绘制数据集中值的直方图以及显示每个创建的条柱的条形图是很有趣的。



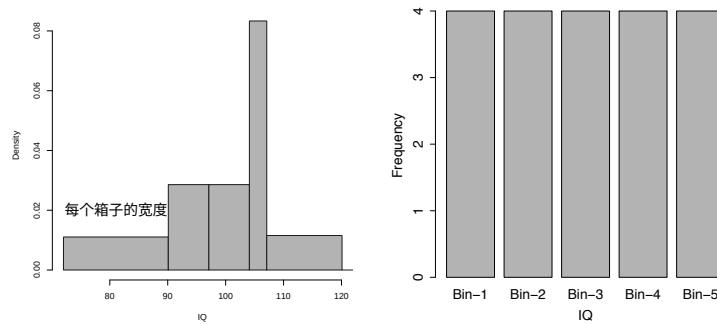
从这些图中可以看出，很多实例属于中间箱，而很少属于高箱和低箱。

(b) An **equal-frequency binning** using 5 bins

To perform an equal-frequency binning, we first determine the number of instances that will belong to each bin. This is simply the number of instances in the dataset divided by the number of bins, in this case  $\frac{20}{5} = 4$ . Next we sort the data by the binning feature and assign instances in order to the first bin until it is full, before moving to the second and so on. The table below shows how the instances in this dataset, sorted by `RESULT`, would be added to the five bins.

| ID | IQ  | IQ (BIN) | ID | IQ  | IQ (BIN) |
|----|-----|----------|----|-----|----------|
| 18 | 72  | Bin-1    | 4  | 101 | Bin-3    |
| 3  | 83  | Bin-1    | 20 | 104 | Bin-3    |
| 12 | 88  | Bin-1    | 11 | 105 | Bin-4    |
| 14 | 90  | Bin-1    | 10 | 106 | Bin-4    |
| 1  | 92  | Bin-2    | 13 | 106 | Bin-4    |
| 6  | 92  | Bin-2    | 2  | 107 | Bin-4    |
| 9  | 93  | Bin-2    | 5  | 107 | Bin-5    |
| 15 | 97  | Bin-2    | 16 | 118 | Bin-5    |
| 7  | 99  | Bin-3    | 8  | 119 | Bin-5    |
| 19 | 100 | Bin-3    | 17 | 120 | Bin-5    |

It is interesting to graph a histogram of the values in the dataset according to the bin boundaries as well as a bar plot showing each of the bins created.



Remember that in the histogram, because the bins are not of equal width, the bars are different heights. The area of a bar represents the density of the instances in the range represented by the bar. The key things to note here are that each bin is equally populated, but that the bins in the middle are much narrower than those at either end of the range.

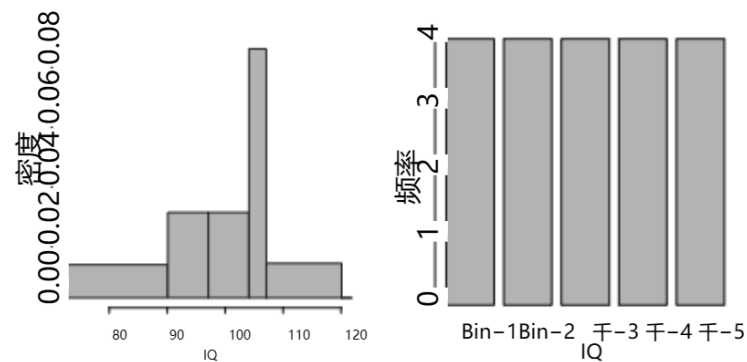
### 第 3 章 数据探索 (练习解决方案)

#### (b) 使用 5 个分档的等频分档

要执行等频分箱，我们首先确定属于每个分箱的实例数。这只是数据集中的实例数除以箱数，在本例中为 "4"。接下来，我们按装箱功能对数据进行排序，并按顺序将实例分配给第一个装箱，直到它已满，然后再移动到第二个装箱，依此类推。下表显示了如何将此数据集中的实例（按 R 排序）添加到五个条柱中。

| ID | 智商  | 智商 (B) | ID | 智商  | 智商 (B) |
|----|-----|--------|----|-----|--------|
| 18 | 72  | Bin-1  | 4  | 101 | Bin-3  |
| 3  | 83  | Bin-1  | 20 | 104 | Bin-3  |
| 12 | 88  | Bin-1  | 11 | 105 | Bin-4  |
| 14 | 90  | Bin-1  | 10 | 106 | Bin-4  |
| 1  | 92  | Bin-2  | 13 | 106 | Bin-4  |
| 6  | 92  | Bin-2  | 2  | 107 | Bin-4  |
| 9  | 93  | Bin-2  | 5  | 107 | Bin-5  |
| 15 | 97  | Bin-2  | 16 | 118 | Bin-5  |
| 7  | 99  | Bin-3  | 8  | 119 | Bin-5  |
| 19 | 100 | Bin-3  | 17 | 120 | Bin-5  |

根据条柱边界绘制数据集中值的直方图以及显示每个创建的条柱的条形图是很有趣的。



请记住，在直方图中，由于条柱的宽度不相等，因此条形的高度不同。条形图的面积表示条形图所表示的范围内实例的密度。这里需要注意的关键事项是，每个箱子的填充量都相同，但中间的箱子比范围两端的箱子窄得多。

## 4 Information-Based Learning (Exercise Solutions)

1. The image below shows a set of eight Scrabble pieces.



- (a) What is the **entropy** in bits of the letters in this set?

We can calculate the probability of randomly selecting a letter of each type from this set:  $P(O) = \frac{3}{8}$ ,  $P(X) = \frac{1}{8}$ ,  $P(Y) = \frac{1}{8}$ ,  $P(M) = \frac{1}{8}$ ,  $P(R) = \frac{1}{8}$ ,  $P(N) = \frac{1}{8}$ .

Using these probabilities, we can calculate the entropy of the set:

$$-\left(\frac{3}{8} \times \log_2\left(\frac{3}{8}\right) + \left(\frac{1}{8} \times \log_2\left(\frac{1}{8}\right)\right) \times 5\right) = 2.4056 \text{ bits}$$

Note that the contribution to the entropy for any letter that appears only once is the same and so has been included 5 times—once for each of  $X$ ,  $Y$ ,  $M$ ,  $R$ , and  $N$ .

- (b) What would be the reduction in entropy (i.e., the **information gain**) in bits if we split these letters into two sets, one containing the vowels and the other containing the consonants?

Information gain is the reduction in entropy that occurs after we split the original set. We know that the entropy of the initial set is 2.4056 *bits*. We calculate the remaining entropy after we split the original set using a weighted

#### 4信息化学习（运动方案）

1. 下图显示了一组八个拼字游戏。（甲）这组字母的熵是多少？



我们可以计算从这个集合中随机选择每种类型的字母的概率：  
PpOq “、PpXq”、PpYq “、PpMq”、PpRq “、PpNq”。

利用这些概率，我们可以计算集合的熵：

$$-\frac{3}{8} \log_2 \frac{3}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} = 2.4056 \text{ 位}$$

请注意，任何只出现一次的字母对熵的贡献是相同的，因此已经包含了 5 次——X、Y、M、R 和 N 各一次。

- (b) 如果我们把这些字母分成两组，一组包含元音，另一组包含辅音，那么熵（即信息增益）会减少多少？

信息增益是拆分原始集合后发生的熵减少。我们知道初始集的熵是 2.4056 位。我们使用加权

summation of the entropies for the new sets. The two new sets are vowels {O,O,O} and consonants {X,Y,M,R,N}.

The entropy of the vowel set is

$$-\left(\frac{3}{3} \times \log_2 \left(\frac{3}{3}\right)\right) = 0 \text{ bits}$$

The entropy of the consonant set is

$$-\left(\left(\frac{1}{5} \times \log_2 \left(\frac{1}{5}\right)\right) \times 5\right) = 2.3219 \text{ bits}$$

The weightings used in the summation of the set entropies are just the relative size of each set. So, the weighting of the vowel set entropy is  $\frac{3}{8}$ , and the weighting of the consonant set entropy is  $\frac{5}{8}$ .

This gives the entropy remaining after we split the set of letters into vowels and consonants as

$$rem = \frac{3}{8} \times 0 + \frac{5}{8} \times 2.3219 = 1.4512 \text{ bits}$$

The information gain is the difference between the initial entropy and the remainder:

$$IG = 2.4056 - 1.4512 = 0.9544 \text{ bits}$$

- (c) What is the maximum possible entropy in bits for a set of eight Scrabble pieces?

The maximum entropy occurs when there are eight different letters in the set. The entropy for a set with this distribution of letters is

$$\left(\frac{1}{8} \times \log_2 \left(\frac{1}{8}\right)\right) \times 8 = 3 \text{ bits}$$

- (d) In general, which is preferable when you are playing Scrabble: a set of letters with high entropy or a set of letters with low entropy?

In general, sets of letters with high entropy are preferable to lower entropy sets because the more diverse the letters in the set, the more words you are likely to be able to make from the set.

#### 第四章 资讯为本的学习 (练习方案)

新集合的熵求和。两组新元音  
和辅音{X, 噢, M, R, N}。  
元音集的熵是

$$-\frac{1}{3} \log_2 \frac{1}{3} - \frac{1}{3} \log_2 \frac{1}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0 \text{ 位}$$

辅音集的熵是

$$-\frac{1}{5} \log_2 \frac{1}{5} - \frac{1}{5} \log_2 \frac{1}{5} - \frac{1}{5} \log_2 \frac{1}{5} - \frac{1}{5} \log_2 \frac{1}{5} - \frac{1}{5} \log_2 \frac{1}{5} = 2.3219 \text{ 位}$$

集合熵总和中使用的权重只是每个集合的相对大小。因此，元音集熵的权重为  $\frac{3}{8}$ ，辅音集熵的权重为  $\frac{5}{8}$ 。

这给出了我们将字母集分成元音和辅音后剩余的熵为

$$\text{雷姆} = \frac{3}{8} \times 0 + \frac{5}{8} \times 2.3219 = 1.4512 \text{ 位}$$

信息增益是初始熵与余熵之间的差值：

$$IG = 2.4056 - 1.4512 = 0.9544 \text{ 位}$$

(c) 一组八个拼字游戏棋子的最大可能熵是多少？

当集合中有八个不同的字母时，就会出现最大熵。具有这种字母分布的集合的熵为

$$-\frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} = 3 \text{ 位}$$

(d) 一般来说，当你玩拼字游戏时，哪个更可取：一组高熵的字母还是一组低熵的字母？

一般来说，具有高熵的字母集比低熵集更可取，因为集合中的字母越多样化，您能够从集合中产生的单词就越多。

2. A convicted criminal who reoffends after release is known as a *recidivist*. The following table lists a dataset that describes prisoners released on parole and whether they reoffended within two years of release.<sup>1</sup>

| ID | GOOD<br>BEHAVIOR | AGE < 30 | DRUG<br>DEPENDENT | RECIDIVIST |
|----|------------------|----------|-------------------|------------|
| 1  | false            | true     | false             | true       |
| 2  | false            | false    | false             | false      |
| 3  | false            | true     | false             | true       |
| 4  | true             | false    | false             | false      |
| 5  | true             | false    | true              | true       |
| 6  | true             | false    | false             | false      |

This dataset lists six instances in which prisoners were granted parole. Each of these instances is described in terms of three binary descriptive features (GOOD BEHAVIOR, AGE < 30, DRUG DEPENDENT) and a binary target feature (RECIDIVIST). The GOOD BEHAVIOR feature has a value of *true* if the prisoner had not committed any infringements during incarceration, the AGE < 30 has a value of *true* if the prisoner was under 30 years of age when granted parole, and the DRUG DEPENDENT feature is *true* if the prisoner had a drug addiction at the time of parole. The target feature, RECIDIVIST, has a *true* value if the prisoner was arrested within two years of being released; otherwise it has a value of *false*.

- (a) Using this dataset, construct the decision tree that would be generated by the **ID3** algorithm, using entropy-based information gain.

The first step in building the decision tree is to figure out which of the three descriptive features is the best one on which to split the dataset at the root node (i.e., which descriptive feature has the highest information gain). The total entropy for this dataset is computed as follows:

$$\begin{aligned}
 H(\text{RECIDIVIST}, \mathcal{D}) &= - \sum_{l \in \{\text{true}, \text{false}\}} P(\text{RECIDIVIST} = l) \times \log_2(P(\text{RECIDIVIST} = l)) \\
 &= - \left( \left( \frac{3}{6} \times \log_2\left(\frac{3}{6}\right) \right) + \left( \frac{3}{6} \times \log_2\left(\frac{3}{6}\right) \right) \right) = 1.00 \text{ bit}
 \end{aligned}$$

1. This example of predicting recidivism is based on a real application of machine learning: parole boards do rely on machine learning prediction models to help them when they are making their decisions. See ? for a recent comparison of different machine learning models used for this task. Datasets dealing with prisoner recidivism are available online, for example, [catalog.data.gov/dataset/prisoner-recidivism/](https://catalog.data.gov/dataset/prisoner-recidivism/). The dataset presented here is not based on real data.

2. 被定罪的罪犯在获释后再犯被称为累犯。下表列出了一个数据集，该数据集描述了假释的囚犯以及他们是否在释放后两年内重新犯罪。

| ID | G  |     |    | D |   |      |
|----|----|-----|----|---|---|------|
|    | BA | A30 | DR | 假 | 真 | R    |
| 1  | 假  | 真   | -  | 假 |   | true |
| 2  | 假  | 假   |    |   |   | 假    |
| 3  | 假  | 真   |    | 假 |   | true |
| 4  | 真  | 假   | 假  |   |   | 假    |
| 5  | 真  | 假   | 真  |   |   |      |
| 6  | 真  | 假   | 假  |   |   | 假    |

该数据集列出了囚犯获得假释的六个实例。这些实例中的每一个都根据三个二进制描述性特征（GBIOR、A30、DD）和一个二进制目标特征（R）进行描述。如果囚犯在监禁期间没有犯下任何违法行为，则 GBfeature 的值为 true，如果囚犯在获得假释时未满 30 岁，则 A30 的值为 true，如果囚犯在假释时吸毒成瘾，则 DDfeature 为 true。如果囚犯在获释后两年内被捕，则目标特征 R 具有真实值；否则，它的值为 false。

- (a) 使用此数据集，使用基于熵的信息增益构建由 ID3 算法生成的决策树。

构建决策树的第一步是找出三个描述性特征中哪一个是拆分根节点数据集的最佳特征（即，哪个描述性特征具有最高的信息增益）。此数据集的总熵计算如下：

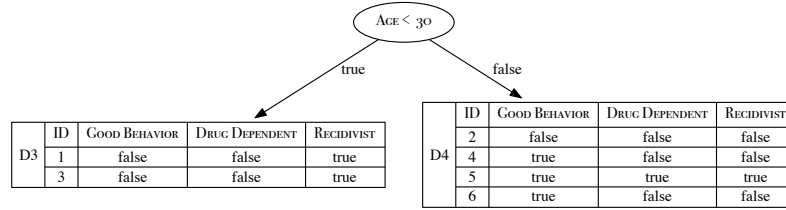
$$H(p(R), p(D)) = -\left(\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6}\right) - \left(\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6}\right) = 1.00 \text{ 位}$$

1. 这个预测累犯的例子是基于机器学习的真实应用：假释委员会确实依靠机器学习预测模型来帮助他们做出决定。看？用于此任务的不同机器学习模型的最新比较。例如，处理囚犯累犯的数据集可在线获得，例如 [catalog.data.gov/dataset/prisoner-recidivism/](https://catalog.data.gov/dataset/prisoner-recidivism/)。此外提供的数据集并非其子

The table below illustrates the computation of the information gain for each of the descriptive features:

| Split by Feature | Level        | Part.           | Instances                                                              | Partition Entropy | Rem.   | Info. Gain |
|------------------|--------------|-----------------|------------------------------------------------------------------------|-------------------|--------|------------|
| GOOD BEHAVIOR    | <i>true</i>  | $\mathcal{D}_1$ | $\mathbf{d}_4, \mathbf{d}_5, \mathbf{d}_6$                             | 0.9183            | 0.9183 | 0.0817     |
|                  | <i>false</i> | $\mathcal{D}_2$ | $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$                             | 0.9183            |        |            |
| AGE < 30         | <i>true</i>  | $\mathcal{D}_3$ | $\mathbf{d}_1, \mathbf{d}_3$                                           | 0                 | 0.5409 | 0.4591     |
|                  | <i>false</i> | $\mathcal{D}_4$ | $\mathbf{d}_2, \mathbf{d}_4, \mathbf{d}_5, \mathbf{d}_6$               | 0.8113            |        |            |
| DRUG DEPENDENT   | <i>true</i>  | $\mathcal{D}_5$ | $\mathbf{d}_5$                                                         | 0                 | 0.8091 | 0.1909     |
|                  | <i>false</i> | $\mathcal{D}_6$ | $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_4, \mathbf{d}_6$ | 0.9709            |        |            |

AGE < 30 has the largest information gain of the three features. Consequently, this feature will be used at the root node of the tree. The figure below illustrates the state of the tree after we have created the root node and split the data based on AGE < 30.



In this image we have shown how the data moves down the tree based on the split on the AGE < 30 feature. Note that this feature no longer appears in these datasets because we cannot split on it again.

The dataset on the left branch contains only instances where RECIDIVIST is *true* and so does not need to be split any further.

The dataset on the right branch of the tree ( $\mathcal{D}_4$ ) is not homogenous, so we need to grow this branch of the tree. The entropy for this dataset,  $\mathcal{D}_4$ , is calculated as follows:

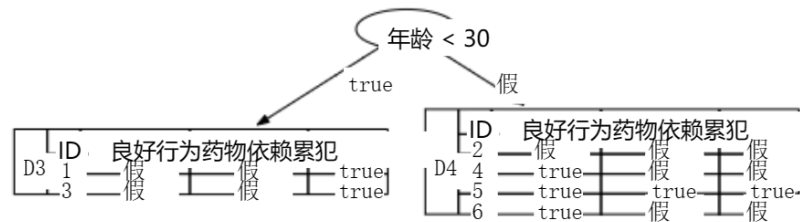
$$\begin{aligned}
 & H(\text{RECIDIVIST}, \mathcal{D}_4) \\
 &= - \sum_{l \in \{\text{true}, \text{false}\}} P(\text{RECIDIVIST} = l) \times \log_2(P(\text{RECIDIVIST} = l)) \\
 &= - \left( \left( \frac{1}{4} \times \log_2\left(\frac{1}{4}\right) \right) + \left( \frac{3}{4} \times \log_2\left(\frac{3}{4}\right) \right) \right) = 0.8113 \text{ bits}
 \end{aligned}$$

The table below shows the computation of the information gain for the GOOD BEHAVIOR and DRUG DEPENDENT features in the context of the  $\mathcal{D}_4$  dataset:

下表说明了每个描述性特征的信息增益的计算：

| 按分区信息拆分。功能级别 部分 实例 熵 雷姆 增益 Gtrue |        |   |               |
|----------------------------------|--------|---|---------------|
| Dd, d, d                         | 0.9183 |   |               |
| Bfalse Dd, d, d                  | 0.9183 |   | 0.9183 0.0817 |
| AA 30 真 Dd, d                    | 0      |   | 0.5409 0.4591 |
| 假 Dd, d, d, d                    | 0.8113 |   |               |
| Dtrue Dd                         |        | 0 | 0.8091 0.1909 |
| Dfalse Dd, d, d, d               | 0.9709 |   |               |

Aa 30 在这三个特征中具有最大的信息增益。因此，此功能将在树的根节点上使用。下图说明了我们创建根节点并根据 Aa 30 拆分数据后树的状态。



在此图像中，我们展示了数据如何根据 Aa 30 要素的分割在树上向下移动。请注意，此功能不再出现在这些数据集中，因为我们无法再次对其进行拆分。

左侧分支上的数据集仅包含 Ris 为 true 的实例，因此不需要进一步拆分。树右分支（D）上的数据集不是同质的，因此我们需要增加树的这个分支。此数据集的熵 D 的计算方法如下：

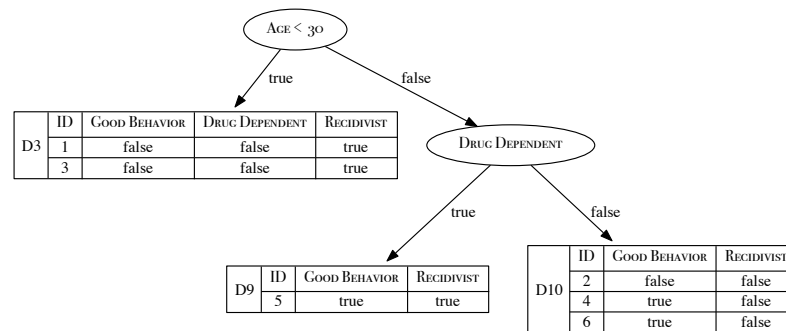
$$H(p_R, D_q) = - \sum_{i=1}^q p_i \log p_i - \sum_{j=1}^q p_j \log p_j = 0.8113 \text{ 位}$$

下表显示了 D 数据集上下文中 G 波段 DDfeatures 信息增益的计算：

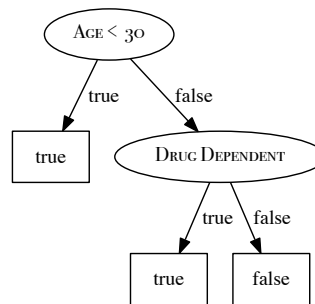
| Split by Feature | Level        | Part.              | Instances                                  | Partition Entropy | Rem.   | Info. Gain |
|------------------|--------------|--------------------|--------------------------------------------|-------------------|--------|------------|
| GOOD BEHAVIOR    | <i>true</i>  | $\mathcal{D}_7$    | $\mathbf{d}_4, \mathbf{d}_5, \mathbf{d}_6$ | 0.918295834       | 0.4591 | 0.3522     |
| DRUG DEPENDENT   | <i>true</i>  | $\mathcal{D}_9$    | $\mathbf{d}_5$                             | 0                 | 0      | 0.8113     |
| DRUG DEPENDENT   | <i>false</i> | $\mathcal{D}_{10}$ | $\mathbf{d}_2, \mathbf{d}_4, \mathbf{d}_6$ | 0                 | 0      | 0.8113     |

These calculations show that the DRUG DEPENDENT feature has a higher information gain than GOOD BEHAVIOR: 0.8113 versus 0.3522 and so should be chosen for the next split.

The image below shows the state of the decision tree after the  $\mathcal{D}_4$  partition has been split based on the feature DRUG DEPENDENT.



All the datasets at the leaf nodes are now pure, so the algorithm will stop growing the tree. The image below shows the tree that will be returned by the ID3 algorithm:



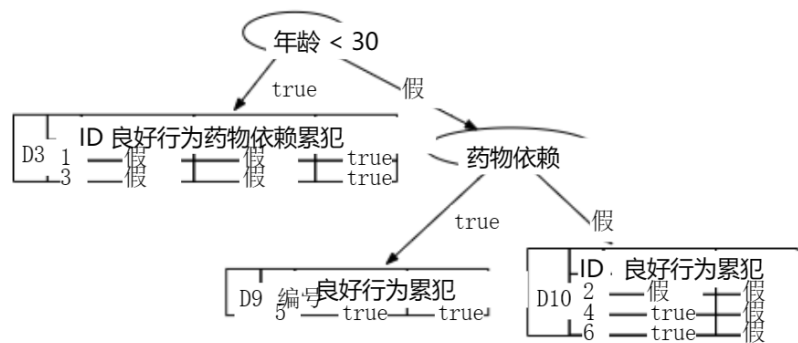
- (b) What prediction will the decision tree generated in Part (a) of this question return for the following query?

GOOD BEHAVIOR = *false*, AGE < 30 = *false*,  
 DRUG DEPENDENT = *true*

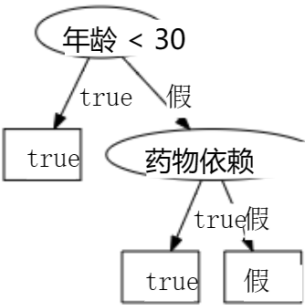
| 按分区信息拆分。特征级别 部分 实例 熵 雷姆 增益 Gtrue |             |   |   |               |
|----------------------------------|-------------|---|---|---------------|
| Dd, d, d0                        | 0.918295834 | — | — | —             |
| Bfalse Dd0                       | —           | — | — | 0.4591 0.3522 |
| Dtrue Dd0                        | —           | — | — | 0 0.8113      |
| Dfalse Dd、d、d0                   | —           | — | — | —             |

这些计算表明，DD特征具有比GB更高的信息增益：0.8113对0.3522，因此应选择用于下一次拆分。

下图显示了基于特征 DD 拆分 Dpartition 后决策树的状态。



叶节点上的所有数据集现在都是纯数据集，因此算法将停止增长树。下图显示了 ID3 算法将返回的树：



(乙) 本题 (a) 部分生成的决策树会为以下查询返回什么预测？

GB= 假, A > 30 = 假,  
DD= 真

|                          |
|--------------------------|
| RECIDIVIST = <i>true</i> |
|--------------------------|

- (c) What prediction will the decision tree generated in Part (a) of this question return for the following query?

GOOD BEHAVIOR = *true*, AGE < 30 = *true*,  
 DRUG DEPENDENT = *false*

|                          |
|--------------------------|
| RECIDIVIST = <i>true</i> |
|--------------------------|

- \* 7. The following table lists a dataset collected in an electronics shop showing details of customers and whether they responded to a special offer to buy a new laptop.

| ID | AGE     | INCOME | STUDENT | CREDIT | BUYS |
|----|---------|--------|---------|--------|------|
| 1  | < 31    | high   | no      | bad    | no   |
| 2  | < 31    | high   | no      | good   | no   |
| 3  | 31 – 40 | high   | no      | bad    | yes  |
| 4  | > 40    | med    | no      | bad    | yes  |
| 5  | > 40    | low    | yes     | bad    | yes  |
| 6  | > 40    | low    | yes     | good   | no   |
| 7  | 31 – 40 | low    | yes     | good   | yes  |
| 8  | < 31    | med    | no      | bad    | no   |
| 9  | < 31    | low    | yes     | good   | yes  |
| 10 | > 40    | med    | yes     | bad    | yes  |
| 11 | < 31    | med    | yes     | good   | yes  |
| 12 | 31 – 40 | med    | no      | good   | yes  |
| 13 | 31 – 40 | high   | yes     | bad    | yes  |
| 14 | > 40    | med    | no      | good   | no   |

|       |
|-------|
| R = 真 |
|-------|

(c) 本问题 (a) 部分生成的决策树会为以下查询返回什么预测?

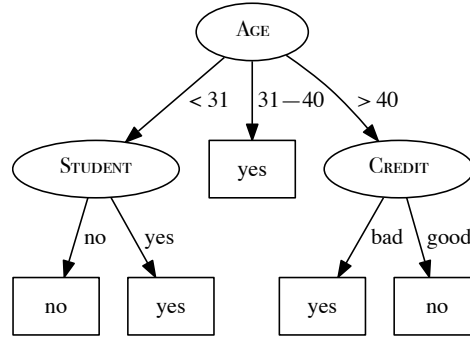
GB= 真, A 30 = 真,  
DD= 假

|       |
|-------|
| R = 真 |
|-------|

° 7.下表列出了在电子商店中收集的数据集, 该数据集显示了客户的详细信息, 以及他们是否响应了购买新笔记本电脑的特别优惠。

| 身份证 | A  | I   | S   | C    | B   |    |    |   |
|-----|----|-----|-----|------|-----|----|----|---|
| 1   | 31 | 高   | 不   | 坏    | 不   | -  | -  | - |
| 2   | 31 | 高   | 不   | 好    | 不   |    |    |   |
| 3   | 31 | '   | 40  | 高    | 不错  | 是的 |    |   |
| 4   | 40 | med | no  | bad  | yes |    |    |   |
| 5   | 40 | 低   | 是   | 坏    | 是   |    |    |   |
| 6   | 40 | 低   | 是   | 好    | 否   |    |    |   |
| 7   | 31 | '   | 40  | 低    | 是   | 好  | 是  |   |
| 8   | 31 | med | no  | bad  | no  |    |    |   |
| 9   | A  | 31  | 低   | 是    | 好   | 是  |    |   |
| 10  | 40 | med | yes | bad  | yes |    |    |   |
| 11  | 31 | med | yes | good | yes |    |    |   |
| 12  | 31 | '   | 40  | med  | 不好  | 是的 |    |   |
| 13  | 31 | '   | 40  | 高    | 是的  | 坏的 | 是的 |   |
| 14  | 40 | med | no  | good | no  |    |    |   |

This dataset has been used to build a decision tree to predict which customers will respond to future special offers. The decision tree, created using the **ID3** algorithm, is the following:



- (a) The **information gain** (calculated using entropy) of the feature AGE at the root node of the tree is 0.247. A colleague has suggested that the STUDENT feature would be better at the root node of the tree. Show that this is not the case.

To answer this question we need to calculate the information gain of the STUDENT feature at the root node and show that it is less than the information gain for AGE.

To calculate information gain for the STUDENT feature we first calculate the entropy of the entire dataset  $\mathcal{D}$  with respect to the target feature, BUYS

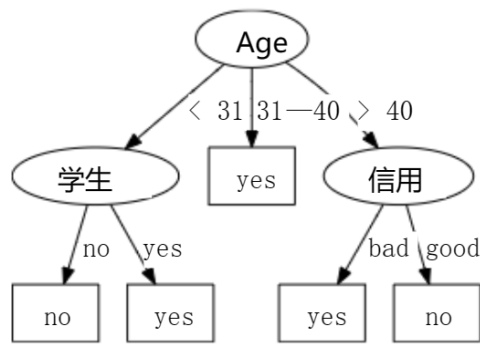
$$\begin{aligned}
 & H(\text{BUYS}, \mathcal{D}) \\
 &= - \sum_{l \in \{\text{yes}, \text{no}\}} P(\text{BUYS} = l) \times \log_2(P(\text{BUYS} = l)) \\
 &= - \left( \left( \frac{9}{14} \times \log_2 \left( \frac{9}{14} \right) \right) + \left( \frac{5}{14} \times \log_2 \left( \frac{5}{14} \right) \right) \right) \\
 &= 0.9403 \text{ bits}
 \end{aligned}$$

Next we calculate the remainder after the dataset is split based on the values of the STUDENT feature.

| Split by Feature | Feature Value | Partition | Examples         | Partition Entropy | Remainder |
|------------------|---------------|-----------|------------------|-------------------|-----------|
| STUDENT          | yes           | $DS_1$    | 5,6,7,9,10,11,13 | 0.5917            | 0.7885    |
|                  | no            | $DS_2$    | 1,2,3,4,8,12,14  | 0.9852            |           |

第四章 资讯为本的学习 (练习方案)

该数据集已用于构建决策树，以预测哪些客户将对未来的特别优惠做出反应。使用 ID3 算法创建的决策树如下：



(a) 特征A在树根节点上的信息增益（使用熵计算）为0.247。一位同事建议，Sfeature 在树的根节点上会更好。表明事实并非如此。

为了回答这个问题，我们需要计算 Sfeature 在根节点上的信息增益，并表明它小于 A 的信息增益。

为了计算 Sfeature 的信息增益，我们首先计算整个数据集 D 相对于目标特征 B 的熵

$$H_p(B, D) = - \sum_{y \in \{0,1\}} P_y \log_2 P_y = - \left( \frac{9}{14} \log_2 \frac{9}{14} + \frac{5}{14} \log_2 \frac{5}{14} \right) = 0.9403 \text{ 位}$$

接下来，我们根据 Sfeature 的值计算数据集拆分后的余数。

| 按特征分割 | 分区 | 特征值                 | 分区示例   | 熵余数    |
|-------|----|---------------------|--------|--------|
| S     | 是  | DS 5,6,7,9,10,11,13 | 0.5917 | 0.7885 |
|       | 没有 | DS 1,2,3,4,8,12,14  | 0.9852 |        |

Finally, we calculate the information gain as the difference between the original entropy and the remainder:

$$IG = 0.9403 - 0.7885 = 0.1518 \text{ bits}$$

The information gain for STUDENT is 0.1518 which is less than the 0.247 for AGE, so STUDENT is not a better feature to use at the root node.

- (b) Yet another colleague has suggested that the ID feature would be a very effective at the root node of the tree. Would you agree with this suggestion?

There is no need to calculate the information gain for the ID feature as the result will work out to be 0.9403—the total information required to make a prediction (and a value calculated in the answer to previous part of the question). We can know this without performing the calculation, however, because each instance has a unique value for the ID feature.

Because of this we know that ID would not be a good feature at the root node of the tree (or in fact anywhere in the tree) because it actually contains no information, and the resulting decision tree would be massively overfitted to the training data. Information measures such as the gain ratio are designed to address this limitation in information gain.

- \* 8. This table lists a dataset of the scores students achieved on an exam described in terms of whether the student studied for the exam (STUDIED) and the energy level of the lecturer when grading the student's exam (ENERGY).

| ID | STUDIED | ENERGY | SCORE |
|----|---------|--------|-------|
| 1  | yes     | tired  | 65    |
| 2  | no      | alert  | 20    |
| 3  | yes     | alert  | 90    |
| 4  | yes     | tired  | 70    |
| 5  | no      | tired  | 40    |
| 6  | yes     | alert  | 85    |
| 7  | no      | tired  | 35    |

Which of the two descriptive features should we use as the testing criterion at the root node of a decision tree to predict students' scores?

The target feature in this question (SCORE) is continuous. When a decision tree is predicting a continuous target, we choose as the descriptive feature to use at each node in the tree the one that results in the minimum weighted variance after the dataset has been split based on that feature. The table below shows the

#### 第四章 资讯为本的学习 (练习方案)

最后，我们将信息增益计算为原始熵与余熵之间的差值：

$$IG = 0.9403 - 0.7885 = 0.1518 \text{ 位}$$

Sis 的信息增益为 0.1518，小于 A 的 0.247，因此 Sis 不是在根节点上使用的更好功能。

(b) 另一位同事认为，ID 功能在树的根节点上非常有效。你同意这个建议吗？

无需计算 ID 特征的信息增益，因为结果将为 0.9403，即进行预测所需的总信息（以及在问题前一部分的答案中计算的值）。但是，我们可以在不执行计算的情况下知道这一点，因为每个实例都有 ID 特征的唯一值。

正因为如此，我们知道 ID 在树的根节点（或者实际上是树中的任何位置）都不是一个好的特征，因为它实际上不包含任何信息，并且生成的决策树将大量过度拟合训练数据。诸如增益比率之类的信息度量旨在解决信息增益的这一限制。

8. 此表列出了学生在考试中取得的分数数据集，这些分数以术语描述。学生是否为考试而学习 (S) 以及讲师在给学生的考试评分时的能量水平 (E)。

| ID | SES      |
|----|----------|
| 1  | 是的，累了 65 |
| 2  | 无警报 20   |
| 3  | 是警报 90   |
| 4  | 是的，累了 70 |
| 5  | 不累 40    |
| 6  | 是警报 85   |
| 7  | 不累 35    |

我们应该使用两个描述性特征中的哪一个作为决策树根节点的测试标准来预测学生的分数？

本题 (S) 中的目标特征是连续的。当决策树预测连续目标时，我们选择在树中的每个节点上使用的描述性特征，该特征在基于该特征拆分数据集后产生最小加权方差。下表显示了

calculation of the weighted variance for each of the descriptive features in this domain.

| Split by<br>Feature | Level        | Part.           | Instances                                                | $P(d = l)$    | $var(t, \mathcal{D})$ | Weighted<br>Variance |
|---------------------|--------------|-----------------|----------------------------------------------------------|---------------|-----------------------|----------------------|
| STUDIED             | <i>yes</i>   | $\mathcal{D}_1$ | $\mathbf{d}_1, \mathbf{d}_3, \mathbf{d}_4, \mathbf{d}_6$ | $\frac{4}{7}$ | $141\frac{2}{3}$      | 127.3810             |
|                     | <i>no</i>    | $\mathcal{D}_2$ | $\mathbf{d}_2, \mathbf{d}_5, \mathbf{d}_7$               | $\frac{3}{7}$ | $108\frac{1}{3}$      |                      |
| ENERGY              | <i>alert</i> | $\mathcal{D}_5$ | $\mathbf{d}_2, \mathbf{d}_3, \mathbf{d}_6$               | $\frac{3}{7}$ | 1525                  | 829.7619             |
|                     | <i>tired</i> | $\mathcal{D}_6$ | $\mathbf{d}_1, \mathbf{d}_4, \mathbf{d}_5, \mathbf{d}_7$ | $\frac{4}{7}$ | $308\frac{1}{3}$      |                      |

From these calculations we can see that splitting the dataset using the STUDIED feature results in the lowest weighted variance. Consequently, we should use the STUDIED feature at the root node of the tree.

计算此域中每个描述性特征的加权方差。

| 按加权要素级别分割 部件。实例 ppd " lq var pt, Dq 方差 |                                  |          |
|----------------------------------------|----------------------------------|----------|
| S                                      | 是 Dd、d、d、d141<br>没有 Dd、d、d108    | 127.3810 |
| E                                      | 警报 Dd、d、d1525<br>疲惫的 Dd、d、d、d308 | 829.7619 |

从这些计算中，我们可以看到，使用 S 特征拆分数据集会产生最低的加权方差。因此，我们应该在树的根节点上使用 Sfeature。

## Chapter 5

2. Email spam filtering models often use a **bag-of-words** representation for emails. In a bag-of-words representation, the descriptive features that describe a document (in our case, an email) each represent how many times a particular word occurs in the document. One descriptive feature is included for each word in a predefined dictionary. The dictionary is typically defined as the complete set of words that occur in the training dataset. The table below lists the bag-of-words representation for the following five emails and a target feature, SPAM, whether they are spam emails or genuine emails:

- “*money, money, money*”
- “*free money for free gambling fun*”
- “*gambling for fun*”
- “*machine learning for fun, fun, fun*”
- “*free machine learning*”

| ID | Bag-of-Words |      |     |          |     |         |          | SPAM  |
|----|--------------|------|-----|----------|-----|---------|----------|-------|
|    | MONEY        | FREE | FOR | GAMBLING | FUN | MACHINE | LEARNING |       |
| 1  | 3            | 0    | 0   | 0        | 0   | 0       | 0        | true  |
| 2  | 1            | 2    | 1   | 1        | 1   | 0       | 0        | true  |
| 3  | 0            | 0    | 1   | 1        | 1   | 0       | 0        | true  |
| 4  | 0            | 0    | 1   | 0        | 3   | 1       | 1        | false |
| 5  | 0            | 1    | 0   | 0        | 0   | 1       | 1        | false |

- (a) What target level would a nearest neighbor model using **Euclidean distance** return for the following email: “*machine learning for free*”?

The bag-of-words representation for this query is as follows:

Chapter

2. 垃圾邮件过滤模型通常对电子邮件使用词袋表示形式。在词袋表示中，描述文档（在我们的例子中是电子邮件）的描述性特征分别表示特定单词在文档中出现的次数。预定义词典中的每个单词都包含一个描述性特征。字典通常定义为训练数据集中出现的完整单词集。下表列出了以下五封电子邮件的词袋表示形式和目标功能  $S$ ，无论它们是垃圾邮件还是真实电子邮件：

钱，钱，钱”  
，  
免费赚钱，享受免费赌博的乐趣”  
，  
赌博的乐趣”  
，  
机器学习带来乐趣，乐趣，乐趣”  
，  
免费机器学习”  
，

|   | 同 | 上 | M | F | F | G | E | M | L | S | — | 词袋 | — |   |      |
|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|------|
| 1 |   |   | 3 |   |   |   | 0 |   | 0 |   |   | 0  | 0 | 0 | true |
| 2 |   |   | 1 |   |   |   | 2 |   | 1 |   |   | 1  | 1 | 0 | true |
| 3 |   |   | 0 |   |   |   | 0 |   | 1 |   |   | 1  | 1 | 0 | true |
| 4 |   |   | 0 |   |   |   | 0 |   | 1 |   |   | 0  | 3 | 1 | 假    |
| 5 |   |   | 0 |   |   |   | 1 |   | 0 |   |   | 0  | 0 | 1 | 假    |

(a) 使用欧几里得距离的最近邻模型会为以下电子邮件返回什么目标水平：“免费机器学习”？

此查询的词袋表示形式如下所示：

| ID    | Bag-of-Words |      |     |          |     |         |          | SPAM |
|-------|--------------|------|-----|----------|-----|---------|----------|------|
|       | MONEY        | FREE | FOR | GAMBLING | FUN | MACHINE | LEARNING |      |
| Query | 0            | 1    | 1   | 0        | 0   | 1       | 1        | ?    |

The table below shows the calculation of the Euclidean distance between the query instance and each of the instances in the training dataset:

| ID | $(\mathbf{q}[i] - \mathbf{d}_j[i])^2$ |      |     |          |     |         |          | Euclidean Distance |
|----|---------------------------------------|------|-----|----------|-----|---------|----------|--------------------|
|    | MONEY                                 | FREE | FOR | GAMBLING | FUN | MACHINE | LEARNING |                    |
| 1  | 9                                     | 1    | 1   | 0        | 0   | 1       | 1        | 3.6056             |
| 2  | 1                                     | 1    | 0   | 1        | 1   | 1       | 1        | 2.4495             |
| 3  | 0                                     | 1    | 0   | 1        | 1   | 1       | 1        | 2.2361             |
| 4  | 0                                     | 1    | 0   | 0        | 9   | 0       | 0        | 3.1623             |
| 5  | 0                                     | 0    | 1   | 0        | 0   | 0       | 0        | 1                  |

Based on these distance calculations, the nearest neighbor to the query is instance  $\mathbf{d}_5$ , for which  $\text{SPAM} = \text{false}$ . Consequently, the model will return a prediction of  $\text{SPAM} = \text{false}$  for this query.

- (b) What target level would a  $k$ -NN model with  $k = 3$  and using **Euclidean distance** return for the same query?

Based on the distance calculations in part (a) of this question, the three nearest neighbors to the query are instances  $\mathbf{d}_5$ ,  $\mathbf{d}_3$ , and  $\mathbf{d}_2$ . The majority of these three neighbors have a target value of  $\text{SPAM} = \text{true}$ . Consequently, the 3-NN model will return a prediction of  $\text{SPAM} = \text{true}$ .

- (c) What target level would a **weighted  $k$ -NN** model with  $k = 5$  and using a weighting scheme of the reciprocal of the squared Euclidean distance between the neighbor and the query, return for the query?

The weights for each of the instances in the dataset are

| ID | Weights              |            |
|----|----------------------|------------|
| 1  | $\frac{1}{3.6056^2}$ | $= 0.0769$ |
| 2  | $\frac{1}{2.4495^2}$ | $= 0.1667$ |
| 3  | $\frac{1}{2.2361^2}$ | $= 0.2$    |
| 4  | $\frac{1}{3.1623^2}$ | $= 0.1$    |
| 5  | $\frac{1}{1^2}$      | $= 1$      |

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|    | 同 | 上 | M | F | F | G | F | M | L | S | 词袋 |
|----|---|---|---|---|---|---|---|---|---|---|----|
| 查询 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | ? |   |   |    |

下表显示了查询实例与训练数据集中每个实例之间的欧几里得距离的计算:

|   | 同 | 上 | M | F | F | G | F | M | L | S | 欧氏距离   |
|---|---|---|---|---|---|---|---|---|---|---|--------|
| 1 | 9 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 3.6056 |
| 2 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2.4495 |
| 3 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2.2361 |
| 4 | 0 | 1 | 0 | 0 | 0 | 9 | 0 | 0 | 0 | 0 | 3.1623 |
| 5 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1      |

根据这些距离计算, 查询的最近邻是实例 d, 其中  $S = \text{false}$ 。因此, 对于此查询, 模型将返回  $S = \text{false}$  的预测。

- (b) k-NN模型的目标水平为k “3 和使用欧几里得距离返回相同的查询?

根据本问题 (a) 部分中的距离计算, 查询的三个最近邻是实例 d、d 和 d。这三个邻居中的大多数都具有  $S = \text{true}$  的目标值。因此, 3-NN 模型将返回  $S = \text{true}$  的预测值。

- (c) 具有 k 个最近邻和查询之间的平均目标水平是距离的倒数的加权方案, 返回查询?

数据集中每个实例的权重为

| ID | 权重                          |
|----|-----------------------------|
| 1  | $\frac{1}{3.6056} = 0.2769$ |
| 2  | $\frac{1}{2.4495} = 0.4086$ |
| 3  | $\frac{1}{2.2361} = 0.4472$ |
| 4  | $\frac{1}{3.1623} = 0.3162$ |
| 5  | $\frac{1}{1} = 1$           |

The total weight for the SPAM = *true* target level is  $0.0769 + 0.1667 + 0.2 = 0.4436$ . The total weight for the SPAM = *false* target level is  $0.1 + 1 = 1.1$ . Consequently, the SPAM = *false* has the maximum weight, and this is the prediction returned by the model.

- (d) What target level would a  $k$ -NN model with  $k = 3$  and using **Manhattan distance** return for the same query?

The table below shows the calculation of the Manhattan distance between the query bag-of-words vector and each instance in the dataset:

| ID | $abs(\mathbf{q}[i] - \mathbf{d}_j[i])$ |      |     |          |     |         |          | Manhattan Distance |
|----|----------------------------------------|------|-----|----------|-----|---------|----------|--------------------|
|    | MONEY                                  | FREE | FOR | GAMBLING | FUN | MACHINE | LEARNING |                    |
| 1  | 3                                      | 1    | 1   | 0        | 0   | 1       | 1        | 7                  |
| 2  | 1                                      | 1    | 0   | 1        | 1   | 1       | 1        | 6                  |
| 3  | 0                                      | 1    | 0   | 1        | 1   | 1       | 1        | 5                  |
| 4  | 0                                      | 1    | 0   | 0        | 3   | 0       | 0        | 4                  |
| 5  | 0                                      | 0    | 1   | 0        | 0   | 0       | 0        | 1                  |

Based on these Manhattan distance calculations, the three nearest neighbors to the query are instances  $\mathbf{d}_5$ ,  $\mathbf{d}_4$ , and  $\mathbf{d}_3$ . The majority of these three neighbors have a target value of SPAM = *false*. Consequently, the 3-NN model using Manhattan distance will return a prediction of SPAM = *false*.

- (e) There are a lot of zero entries in the spam bag-of-words dataset. This is indicative of **sparse data** and is typical for text analytics. **Cosine similarity** is often a good choice when dealing with sparse non-binary data. What target level would a 3-NN model using cosine similarity return for the query?

In order to calculate the cosine similarity between the query and each instance in the dataset, we first need to calculate the vector length of each instance and the query. The table below illustrates the calculation of these vector lengths.

| ID    | $\mathbf{d}[i]^2$ |   |   |   |   |   |   |   | Vector |        |
|-------|-------------------|---|---|---|---|---|---|---|--------|--------|
|       |                   |   |   |   |   |   |   |   | Sum    | Length |
| 1     | 9                 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 9      | 3      |
| 2     | 1                 | 4 | 1 | 1 | 1 | 0 | 0 | 0 | 8      | 2.8284 |
| 3     | 0                 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 3      | 1.7321 |
| 4     | 0                 | 0 | 1 | 0 | 9 | 1 | 1 | 1 | 12     | 3.4641 |
| 5     | 0                 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 3      | 1.7321 |
| Query | 0                 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 4      | 2      |

## 第 5 章 基于相似性的学习 (练习解决方案)

S= 真实目标水平的总权重为 0.0769  
 0.4436. S=假目标水平的总权重为0.1  
 因此, S= false 具有最大权重, 这是模型返回的预测。

- (d) k-NN模型的目标水平为k  
 返回相同的查询? „ 3.使用曼哈顿距离

下表显示了查询词袋向量与数据集中每个实例之间的曼哈顿距离的计算:

| 同上 | M | F | F | G | F | M | L | abspp | d | r | q | 曼哈顿<br>距离 |
|----|---|---|---|---|---|---|---|-------|---|---|---|-----------|
| 1  | 3 |   |   | 1 |   |   |   | 0     | 0 | 1 | 1 | 7         |
| 2  | 1 |   |   | 1 |   |   | 0 | 1     | 1 | 1 | 1 | 6         |
| 3  | 0 |   |   | 1 |   |   | 0 | 1     | 1 | 1 | 1 | 5         |
| 4  | 0 |   |   | 1 |   |   | 0 | 0     | 3 | 0 | 0 | 4         |
| 5  | 0 |   |   | 0 |   | 1 |   | 0     | 0 | 0 | 0 | 1         |

根据这些曼哈顿距离计算, 查询最近的三个邻居是实例 d、d 和 d。这三个邻居中的大多数都具有 S= false 的目标值。因此, 使用曼哈顿距离的 3-NN 模型将返回 S= false 的预测值。

- (e) 垃圾邮件词袋数据集中有很多零条目。这表示稀疏数据, 是文本分析的典型特征。在处理稀疏的非二进制数据时, 余弦相似性通常是一个不错的选择。使用余弦相似度的 3-NN 模型会为查询返回什么目标水平?

为了计算查询与数据集中每个实例之间的余弦相似度, 我们首先需要计算每个实例和查询的向量长度。下表说明了这些向量长度的计算。

| ID | D 瑞尔                | Sum | 向量<br>长度 |
|----|---------------------|-----|----------|
| 1  | 9 0 0 0 0 0 0       | 9   | 3        |
| 2  | 1 4 1 1 1 1 0       | 8   | 2.8284   |
| 3  | 0 0 1 1 1 1 0       | 3   | 1.7321   |
| 4  | 0 0 1 0 9 1         | 12  | 3.4641   |
| 5  | 0 -1 -0 -0 -0 -1 -1 | 3   | 1.7321   |
| 查询 | 0 -1 -1 -0 -0 -1 -1 | 4   | 2        |

The second component we need to calculate is the dot product between the query and each instance. The table below illustrates the calculation of these dot products.

| Pair                         | $(\mathbf{q}[i] \times \mathbf{d}_j[i])$ |   |   |   |   |   |   | Dot Product |
|------------------------------|------------------------------------------|---|---|---|---|---|---|-------------|
| $(\mathbf{q}, \mathbf{d}_1)$ | 0                                        | 0 | 0 | 0 | 0 | 0 | 0 | 0           |
| $(\mathbf{q}, \mathbf{d}_2)$ | 0                                        | 2 | 1 | 0 | 0 | 0 | 0 | 3           |
| $(\mathbf{q}, \mathbf{d}_3)$ | 0                                        | 0 | 1 | 0 | 0 | 0 | 0 | 1           |
| $(\mathbf{q}, \mathbf{d}_4)$ | 0                                        | 0 | 1 | 0 | 0 | 1 | 1 | 3           |
| $(\mathbf{q}, \mathbf{d}_5)$ | 0                                        | 1 | 0 | 0 | 0 | 1 | 1 | 3           |

We can now calculate the cosine similarity for each query-instance pair by dividing the relevant dot product by the product of the respective vector lengths. These calculations are shown below.

| Pair                         | Cosine Similarity           |            |
|------------------------------|-----------------------------|------------|
| $(\mathbf{q}, \mathbf{d}_1)$ | $\frac{0}{3 \times 2}$      | $= 0$      |
| $(\mathbf{q}, \mathbf{d}_2)$ | $\frac{3}{2.8285 \times 2}$ | $= 0.5303$ |
| $(\mathbf{q}, \mathbf{d}_3)$ | $\frac{1}{1.7321 \times 2}$ | $= 0.2887$ |
| $(\mathbf{q}, \mathbf{d}_4)$ | $\frac{3}{3.4641 \times 2}$ | $= 0.4330$ |
| $(\mathbf{q}, \mathbf{d}_5)$ | $\frac{3}{1.7321 \times 2}$ | $= 0.8660$ |

When we use a similarity index, such as cosine similarity, the higher the number, the more similar the instances. Given this, the three most similar instances in the dataset to the query are instances  $\mathbf{d}_5$ ,  $\mathbf{d}_2$ , and  $\mathbf{d}_4$ . The majority of these three neighbors have a target value of SPAM = *false*. Consequently, the 3-NN model will return a prediction of SPAM = *false*.

- \* 4. You have been given the job of building a recommender system for a large online shop that has a stock of over 100,000 items. In this domain the behavior of customers is captured in terms of what items they have bought or not bought. For example, the following table lists the behavior of two customers in this domain for a subset of the items that at least one of the customers has bought.

| ID | ITEM<br>107 | ITEM<br>498 | ITEM<br>7256 | ITEM<br>28063 | ITEM<br>75328 |
|----|-------------|-------------|--------------|---------------|---------------|
| 1  | true        | true        | true         | false         | false         |
| 2  | true        | false       | false        | true          | true          |

- (a) The company has decided to use a similarity-based model to implement the recommender system. Which of the following three similarity indexes do you think the system should be based on?

$$\text{Russell-Rao}(X,Y) = \frac{CP(X,Y)}{P}$$

$$\text{Sokal-Michener}(X,Y) = \frac{CP(X,Y) + CA(X,Y)}{P}$$

$$\text{Jaccard}(X,Y) = \frac{CP(X,Y)}{CP(X,Y) + PA(X,Y) + AP(X,Y)}$$

## 第 5 章 基于相似性的学习 (练习解决方案)

我们需要计算的第二个分量是查询和每个实例之间的点积。下表说明了这些点积的计算。

| Pair | pq | rid | ri | Dot |
|------|----|-----|----|-----|
| d d  | 0  | 0   | 0  | 0   |
| pq d | 0  | 2   | 1  | 0   |
| pq d | 0  | 0   | 1  | 0   |
| pq d | 0  | 0   | 1  | 0   |
| pq d | 0  | 0   | 1  | 0   |
| pq d | -1 | -1  | -1 | -3  |

现在，我们可以通过将相关点积除以相应向量长度的乘积来计算每个查询实例对的余弦相似度。这些计算如下所示。

| Pair | 余弦相似度  |
|------|--------|
| d d  | 0      |
| pq d | 0      |
| pq d | 0.5303 |
| pq d | 0.2887 |
| pq d | 0.4330 |
| pq d | 0.8660 |

当我们使用相似性指数时，例如余弦相似度，数字越大，实例越相似。鉴于此，数据集中与查询最相似的三个实例是实例 d、d 和 d。这三个邻居中的大多数都具有  $S = \text{false}$  的目标值。因此，3-NN 模型将返回  $S = \text{false}$  的预测值。

4. 您被赋予了为一家大型在线商店构建推荐系统的工作。库存超过 100,000 件。在这个领域中，客户的行为是根据他们购买或未购买的物品来捕捉的。例如，下表列出了此域中两个客户对至少一个客户已购买的商品子集的行为。

|     | I    | II   |      |       |       |      |
|-----|------|------|------|-------|-------|------|
| 会员号 | 107  | 498  | 7256 | 28063 | 75328 |      |
| 1   | true | true | true | 假     | 假     | 假    |
| 2   | true | 假    | 假    | true  | true  | true |

(a) 公司已决定使用基于相似性的模型来实施推荐系统。您认为该系统应该基于以下三个相似性指标中的哪一个？

罗素-饶  $(X, Y) = \frac{CPpX, Yq}{p}$

索卡尔-米切纳  $(X, Y) = \frac{CPpX, Yq + CApX, Yq}{p}$

杰卡德  $(X, Y) = \frac{CPpX, Yq + CApX, Yq}{APnX, Yq}$

In a domain where there are hundreds of thousands of items, co-absences aren't that meaningful. For example, you may be in a domain where there are so many items that most people haven't seen, listened to, bought, or visited that the majority of features will be co-absences. The technical term to describe a dataset where most of the features have zero values is **sparse data**. In these situations, you should use a metric that ignores co-absences. For a scenario such as this one, where the features are binary, the **Jaccard similarity index** is ideal as it ignores co-absences.

- (b) What items will the system recommend to the following customer? Assume that the recommender system uses the similarity index you chose in the first part of this question and is trained on the sample dataset listed above. Also assume that the system generates recommendations for query customers by finding the customer most similar to them in the dataset and then recommending the items that this similar customer has bought but that the query customer has not bought.

|       | ITEM | ITEM  | ITEM | ITEM  | ITEM  |
|-------|------|-------|------|-------|-------|
| ID    | 107  | 498   | 7256 | 28063 | 75328 |
| Query | true | false | true | false | false |

Using a similarity metric, the higher the value returned by the metric, the more similar the two items are.

Assuming you chose the **Jaccard similarity index**, then the query customer is more similar to customer  $\mathbf{d}_1$  than to customer  $\mathbf{d}_2$ :

$$\bullet \text{ Jaccard}(\mathbf{q}, \mathbf{d}_1) = \frac{2}{2+1} = 0.6667$$

$$\bullet \text{ Jaccard}(\mathbf{q}, \mathbf{d}_2) = \frac{1}{4} = 0.25$$

There is only 1 item that customer  $\mathbf{d}_1$  has bought that the query customer has not bought, item 498. As a result, the system will recommend item 498 to the query customer.

It turns out that in this instance, no matter which of the three similarity metrics we use, customer  $\mathbf{d}_1$  is more similar to the query customer than customer  $\mathbf{d}_2$ . The supporting calculations for Russell-Rao and Sokal-Michener are

$$\bullet \text{ Russell-Rao}(\mathbf{q}, \mathbf{d}_1) = \frac{2}{5} = 0.4$$

$$\bullet \text{ Russell-Rao}(\mathbf{q}, \mathbf{d}_2) = \frac{1}{5} = 0.2$$

$$\bullet \text{ Sokal-Michener}(\mathbf{q}, \mathbf{d}_1) = \frac{4}{5} = 0.8$$

$$\bullet \text{ Sokal-Michener}(\mathbf{q}, \mathbf{d}_2) = \frac{2}{5} = 0.4$$

So, the system will recommend item 498 regardless of which similarity metric is used.

## 第 5 章 基于相似性的学习 (练习解决方案)

在一个有数十万个项目的域中，共同缺席并没有那么有意义。例如，您可能所在的域中有太多项目，大多数人没有看过、听过、购买过或访问过，以至于大多数功能都是共同缺失的。描述大多数要素值为零的数据集的技术术语是稀疏数据。在这些情况下，应使用忽略共同缺席的指标。对于像这样的场景，其中特征是二进制的，Jaccard 相似性指数是理想的，因为它忽略了共缺。

(乙) 系统会向下列客户推荐什么项目？假设推荐系统使用您在此问题的第一部分中选择的相似性指数，并在上面列出的示例数据集上进行训练。此外，假设系统通过在数据集中查找与查询客户最相似的客户，然后推荐此类相似客户已购买但查询客户尚未购买的商品，为查询客户生成建议。

会员号 107 498 7256 28063 75328  
 查询 true false true false

使用相似性指标时，指标返回的值越高，两个项目就越相似。假设您选择了 Jaccard 相似性指数，那么查询 customer 与客户 d (而不是客户 d) 更相似：

$$Jaccard(p, q) = \frac{|p \cap q|}{|p \cup q|} = \frac{2}{2+1} = 0.6667$$

$$Jaccard(p, d) = \frac{|p \cap d|}{|p \cup d|} = \frac{1}{4} = 0.25$$

只有 1 件客户购买了查询客户未购买的商品，即商品 498。因此，系统将向查询客户推荐物料 498。

事实证明，在这种情况下，无论我们使用三个相似度指标中的哪一个，客户与查询客户都比客户 d 更相似。Russell-Rao 和 Sokal-Michener 的支持计算是

$$Russell-Rao(p, q) = \frac{|p \cap q|}{\sqrt{|p| \cdot |q|}} = \frac{2}{\sqrt{2 \cdot 1}} = 0.4$$

$$Russell-Rao(p, d) = \frac{|p \cap d|}{\sqrt{|p| \cdot |d|}} = \frac{1}{\sqrt{4 \cdot 1}} = 0.2$$

$$Sokal-Michener(p, q) = \frac{|p \cap q|}{|p| + |q|} = \frac{2}{5} = 0.4$$

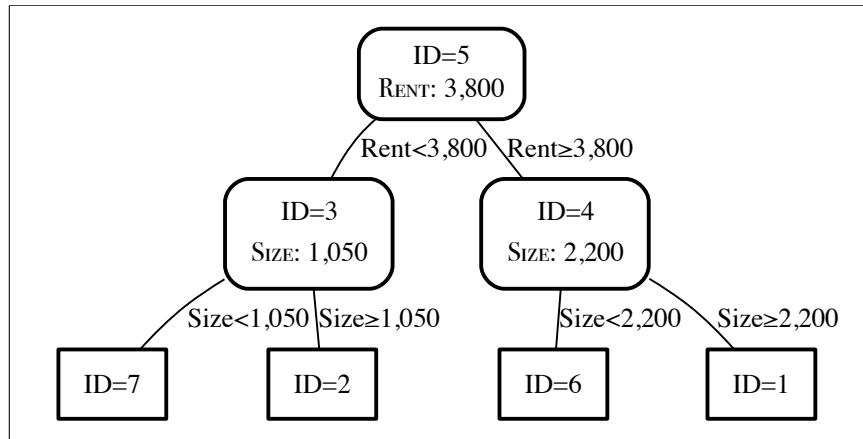
$$Sokal-Michener(p, d) = \frac{|p \cap d|}{|p| + |d|} = \frac{1}{5} = 0.2$$

因此，无论使用哪种相似性指标，系统都会推荐项目 498。

- \* 6. You have been asked by a San Francisco property investment company to create a predictive model that will generate house price estimates for properties they are considering purchasing as rental properties. The table below lists a sample of properties that have recently been sold for rental in the city. The descriptive features in this dataset are SIZE (the property size in square feet) and RENT (the estimated monthly rental value of the property in dollars). The target feature, PRICE, lists the prices that these properties were sold for in dollars.

| ID | SIZE  | RENT  | PRICE     |
|----|-------|-------|-----------|
| 1  | 2,700 | 9,235 | 2,000,000 |
| 2  | 1,315 | 1,800 | 820,000   |
| 3  | 1,050 | 1,250 | 800,000   |
| 4  | 2,200 | 7,000 | 1,750,000 |
| 5  | 1,800 | 3,800 | 1,450,500 |
| 6  | 1,900 | 4,000 | 1,500,500 |
| 7  | 960   | 800   | 720,000   |

- (a) Create a *k-d tree* for this dataset. Assume the following order over the features: RENT then SIZE.



- (b) Using the *k-d tree* that you created in the first part of this question, find the nearest neighbor to the following query: SIZE = 1,000, RENT = 2,200.

The initial step in retrieving the nearest neighbor is to descend the tree to a leaf node. For this query, this descent will terminate at the node that stores instance  $\mathbf{d}_7$ . At this point, the *current best* variable is set to the instance stored at this node  $\mathbf{d}_7$ , and the *current best-distance* variable is set to the Euclidean

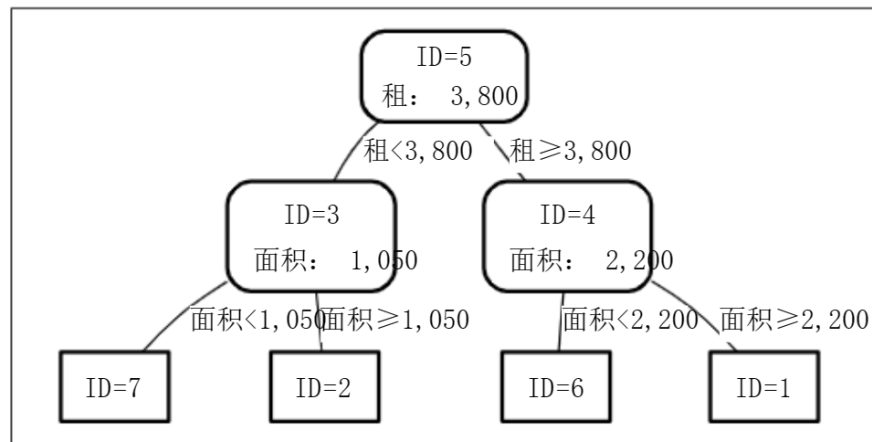
## 第 5 章 基于相似性的学习 (练习解决方案)

° 6.旧金山的一家房地产投资公司要求您创建一个

预测模型将生成他们正在考虑购买的出租物业的房价估算。下表列出了最近在该市出售出租的房产样本。该数据集中的描述性特征是 S (以平方英尺为单位的房产面积) 和 R (以美元为单位的房产估计月租金价值)。目标要素 P 列出了这些属性以美元为单位的售价。

| ID | S     | R     | P         |
|----|-------|-------|-----------|
| 1  | 2,700 | 9,235 | 2,000,000 |
| 2  | 1,315 | 1,800 | 820,000   |
| 3  | 1,050 | 1,250 | 800,000   |
| 4  | 2,200 | 7,000 | 1,750,000 |
| 5  | 1,800 | 3,800 | 1,450,500 |
| 6  | 1,900 | 4,000 | 1,500,500 |
| 7  | 960   | 800   | 720,000   |

(a) 为此数据集创建一个 k-d 树。假设要素的顺序如下: R then S。



(b) 使用在本问题的第一部分中创建的 k-d 树, 找到与以下查询最近的邻: S= 1,000, R= 2,200。

检索最近邻节点的第一步是将树下降到叶节点。对于此查询, 此下降将在存储实例 d 的节点处终止。此时, 当前最佳变量设置为存储在此节点 d 的实例, 当前最佳距离变量设置为欧几里得

distance between the query and  $\mathbf{d}_7$ :

$$\text{current best} = \mathbf{d}_7$$

$$\text{current best-distance} = \text{Euclidean}(\mathbf{q}, \mathbf{d}_7) = 1,400.5713$$

The retrieval algorithm then ascends the tree. The first node the algorithm will encounter is the node that stores instance  $\mathbf{d}_3$ . The Euclidean distance between the query and  $\mathbf{d}_3$  is less than *current best-distance*. Consequently, *current best* and *current best-distance* are updated to reflect this:

$$\text{current best} = \mathbf{d}_3$$

$$\text{current best-distance} = \text{Euclidean}(\mathbf{q}, \mathbf{d}_3) = 951.3149$$

Because the difference between the splitting feature value at this node, SIZE = 1,050, and the query, SIZE = 1,000, is less than the *current best-distance*, the algorithm descends the other branch of the tree from this node. This descent will terminate at the node  $\mathbf{d}_2$ . The Euclidean distance between the query and  $\mathbf{d}_2$  is less than the *current best-distance*. Consequently, *current best* and *current best-distance* are updated to reflect this:

$$\text{current best} = \mathbf{d}_2$$

$$\text{current best-distance} = \text{Euclidean}(\mathbf{q}, \mathbf{d}_2) = 509.1414$$

The algorithm will then ascend the tree; because it has already visited all the nodes on the path back to the root, it does not need to check for nodes closer than the *current best* until it gets back to the root. In this instance, the Euclidean distance between the query and the instance stored at the root node,  $\mathbf{d}_3$ , is greater than *current best-distance*, so neither *current best* nor *current best-distance* are updated when we reach the root node. Furthermore, because the difference between the splitting feature at the root, RENT = 3,800, and the query feature value, RENT = 2,200 is larger than the *current best-distance*, the algorithm can prune the other branch from the *k-d* tree and return  $\mathbf{d}_2$  as the nearest neighbor, which would indicate that the property is worth approximately \$820,000.

- \* 9. A lecturer is about to leave for the airport to go on vacation when they find a script for a student they forgot to mark. They don't have time to manually grade the script before the flight, so they decide to use a *k*-nearest neighbor model to grade it instead.

## 第 5 章 基于相似性的学习 (练习解决方案)

查询与 D 之间的距离:

当前最佳 “ d  
 当前最佳距离  $\text{Euclidean}(q, d) = 400.5713$

然后, 检索算法将爬上树。算法将遇到的第一个节点是存储实例 d 的节点。查询和 dis 之间的欧几里得距离小于当前最佳距离。因此, 将更新当前最佳距离和当前最佳距离以反映以下内容:

当前最佳 “ d  
 当前最佳距离  $\text{Euclidean}(q, d) = 51.3149$

由于此节点处的分割特征值  $S = 1,050$  与查询  $S = 1,000$  之间的差值小于当前最佳距离, 因此该算法从此节点降序到树的另一个分支。此下降将在节点 d 处终止。查询和 dis 之间的欧几里得距离小于当前最佳距离。因此, 将更新当前最佳距离和当前最佳距离以反映以下内容:

当前最佳 “ d  
 当前最佳距离  $\text{Euclidean}(q, d) = 509.1414$

然后, 该算法将提升树;因为它已经访问了返回根路径上的所有节点, 所以在返回根之前, 它不需要检查比当前最佳节点更近的节点。在本例中, 查询与存储在根节点 d 上的实例之间的欧几里得距离大于当前最佳距离, 因此当我们到达根节点时, 当前最佳距离和当前最佳距离都不会更新。此外, 由于根处的分割特征  $R = 3,800$  与查询特征值  $R = 2,200$  之间的差值大于当前最佳距离, 因此该算法可以从 k-d 树中修剪另一个分支并返回最近的邻居 das, 这表明该属性的价值约为 820,000 美元。

- ° 9. 一位讲师正要去机场度假时, 他们发现了一个剧本  
 对于一个学生, 他们忘记了标记。他们没有时间在飞行前手动对脚本进行评分, 因此他们决定使用 k 最近邻模型来评分。

The model is designed to award a grade to a student on the basis of how similar they are to other students in the module in terms of their grades on other modules. The following table describes a set of students in terms of their grades out of 100 on two other modules (MODULE 1 and MODULE 2) and the GRADE they got in the lecturer's module: *first-class honors*, *second-class honors*, *pass*, or *fail*.

| ID | MODULE 1 | MODULE 2 | GRADE  |
|----|----------|----------|--------|
| 1  | 55       | 85       | first  |
| 2  | 45       | 30       | fail   |
| 3  | 40       | 20       | fail   |
| 4  | 35       | 35       | fail   |
| 5  | 55       | 75       | pass   |
| 6  | 50       | 95       | second |

- (a) Looking up the results on the other modules of the student whose script hasn't been corrected, the lecturer finds that the student got the following marks: MODULE 1=60, and MODULE 2=85. Assuming that the  $k$ -nearest neighbor model uses  $k=1$  and Euclidean distance as its similarity metric, what GRADE would the model assign the student?

The Euclidean distance, rounded to 2 places of decimal, between the student and the other students in the sample are as follows:

| ID | DISTANCE | GRADE  |
|----|----------|--------|
| 1  | 5.00     | first  |
| 2  | 57.00    | fail   |
| 3  | 68.00    | fail   |
| 4  | 55.90    | fail   |
| 5  | 11.18    | pass   |
| 6  | 14.14    | second |

The student is most similar to student ID=1 and so the model will return a GRADE of *first*.

- (b) Reviewing the spread of marks for the other two modules, the lecturer notices that there is a larger variance across students in the marks for *Module 2* than there is for *Module 1*. So, the lecturer decides to update the  $k$ -nearest neighbor model to use the **Mahalanobis** distance instead of Euclidean distance as its similarity measure. Assuming that the **inverse covariance matrix** for the *Module 1* and *Module 2* results is

$$\Sigma^{-1} = \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix}$$

## 第 5 章 基于相似性的学习 (练习解决方案)

该模型旨在根据学生在其他模块上的成绩与该模块中其他学生的相似程度来授予学生成绩。下表描述了一组学生在另外两个模块 (M1 和 M2) 上的成绩 (满分 100 分) 以及讲师模块中获得的成绩: 一等荣誉、二等荣誉、通过或失败。

| 编号: | M1 | M2 | G    |
|-----|----|----|------|
| 1   | 55 | 85 | 第一   |
| 2   | 45 | 30 | fail |
| 3   | 40 | 20 | fail |
| 4   | 35 | 35 | fail |
| 5   | 55 | 75 | pass |
| 6   | 50 | 95 | 第二   |

(a) 在未改正脚本的学生的其他模块上查找结果时, 讲师发现该学生获得了以下分数:  $M_1 = 60$ ,  $M_2 = 85$ 。假设  $k$  最近邻模型使用  $k = 1$  和欧几里得距离作为其相似度指标, 该模型会给学生分配什么?

| 样本中学生与其他学生之间的欧几里得距离 (四舍五入到小数点后 2 位) 如下: |       |   |      |
|-----------------------------------------|-------|---|------|
| 编号:                                     | D     | G |      |
| 1                                       | 5.00  |   | 第一   |
| 2                                       | 57.00 |   | fail |
| 3                                       | 68.00 |   | fail |
| 4                                       | 55.90 |   | fail |
| 5                                       | 11.18 |   | pass |
| 6                                       | 14.14 |   | 第二   |

学生与学生 ID=1 最相似, 因此模型将首先返回 Gof。

(b) 在回顾其他两个单元的分数分布时, 讲师注意到模块 2 的学生之间的分数差异大于模块 1 的差异。因此, 讲师决定更新  $k$  最近邻模型, 以使用马氏距离而不是欧几里得距离作为其相似性度量。

假设模块 1 和模块 2 结果的逆协方差矩阵为

$$\Sigma^{-1} = \begin{bmatrix} 0.046 & 0.009 \\ 0.009 & 0.003 \end{bmatrix}$$

what GRADE would the  $k$ -nearest neighbor model assign the student?

The calculations of the Mahalanobis distance between the student and each of the other students is as follows:

$$\text{Mahalanobis}(\mathbf{a}, \mathbf{b}) = \sqrt{[\mathbf{a}[1] - \mathbf{b}[1], \dots, \mathbf{a}[m] - \mathbf{b}[m]] \times \Sigma^{-1} \times \begin{bmatrix} \mathbf{a}[1] - \mathbf{b}[1] \\ \vdots \\ \mathbf{a}[m] - \mathbf{b}[m] \end{bmatrix}}$$

$$\begin{aligned} \text{Mahalanobis}(\mathbf{q}, \mathbf{d}_1) &= \sqrt{[5, 0] \times \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix} \times \begin{bmatrix} 5 \\ 1 \end{bmatrix}} \\ &= 1.072380529 \end{aligned}$$

$$\begin{aligned} \text{Mahalanobis}(\mathbf{q}, \mathbf{d}_2) &= \sqrt{[15, 55] \times \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix} \times \begin{bmatrix} 15 \\ 55 \end{bmatrix}} \\ &= 2.138924964 \end{aligned}$$

$$\begin{aligned} \text{Mahalanobis}(\mathbf{q}, \mathbf{d}_3) &= \sqrt{[20, 65] \times \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix} \times \begin{bmatrix} 20 \\ 65 \end{bmatrix}} \\ &= 2.770379035 \end{aligned}$$

## 第 5 章 基于相似性的学习 (练习解决方案)

k-最近邻模型会给学生分配什么?

学生与其他每个学生之间的马哈拉诺比斯距离的计算如下:

马哈拉诺比斯帕,  $bq''$

$$d_{Mahalanobis}(p, q) = \sqrt{\frac{(a_1 - b_1)^2}{\sigma_1^2} + \dots + \frac{(a_n - b_n)^2}{\sigma_n^2}}$$

A RMS ' B RMS

Mahalanobispq,  $dq''$

$$d_{Mahalanobis}(p, q) = \sqrt{\frac{(a_1 - b_1)^2}{\sigma_1^2} + \dots + \frac{(a_n - b_n)^2}{\sigma_n^2}}$$

" 1.072380529

Mahalanobispq,  $dq''$

$$d_{Mahalanobis}(p, q) = \sqrt{\frac{(a_1 - b_1)^2}{\sigma_1^2} + \dots + \frac{(a_n - b_n)^2}{\sigma_n^2}}$$

" 2.138924964

Mahalanobispq,  $dq''$

$$d_{Mahalanobis}(p, q) = \sqrt{\frac{(a_1 - b_1)^2}{\sigma_1^2} + \dots + \frac{(a_n - b_n)^2}{\sigma_n^2}}$$

" 2.770379035

$$\text{Mahalanobis}(\mathbf{q}, \mathbf{d}_4) =$$

$$\sqrt{[25, 50] \times \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix} \times \begin{bmatrix} 25 \\ 50 \end{bmatrix}} \\ = 3.708099244$$

$$\text{Mahalanobis}(\mathbf{q}, \mathbf{d}_5) =$$

$$\sqrt{[5, 10] \times \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix} \times \begin{bmatrix} 5 \\ 10 \end{bmatrix}} \\ = 0.374165739$$

$$\text{Mahalanobis}(\mathbf{q}, \mathbf{d}_6) =$$

$$\sqrt{[10, -10] \times \begin{bmatrix} 0.046 & -0.009 \\ -0.009 & 0.003 \end{bmatrix} \times \begin{bmatrix} 10 \\ -10 \end{bmatrix}} \\ = 2.588435821$$

When the model is updated to use Mahalanobis distance the student is most similar to  $\mathbf{d}_5$  and so the model returns a GRADE of *pass*.

## 第 5 章 基于相似性的学习 (练习解决方案)

Mahalanobispq, dq "

```

0 -----«-----ff-----ff
f
e R25,50年代 0.046 '0.009 ^ 25
0.009 0.003 50

```

" 3.708099244

Mahalanobispq, dq "

```

0 -----«-----ff-----ff
f
e R5,10秒 ^ 0.046 '0.009 ^ 5
0.009 0.003 10

```

" 0.374165739

Mahalanobispq, dq "

```

0 -----«-----ff-----ff
f
e R10, '10s ^ 0.046 '0.009 ^ 10
0.009 0.003 '10

```

" 2.588435821

当模型更新为使用 Mahalanobis 距离时，学生与 dand 最相似，因此模型返回 Gof 传递。

## Chapter 6

2. The table below gives details of symptoms that patients presented and whether they were suffering from meningitis.

| ID | HEADACHE | FEVER | VOMITING | MENINGITIS |
|----|----------|-------|----------|------------|
| 1  | true     | true  | false    | false      |
| 2  | false    | true  | false    | false      |
| 3  | true     | false | true     | false      |
| 4  | true     | false | true     | false      |
| 5  | false    | true  | false    | true       |
| 6  | true     | false | true     | false      |
| 7  | true     | false | true     | false      |
| 8  | true     | false | true     | true       |
| 9  | false    | true  | false    | false      |
| 10 | true     | false | true     | true       |

Using this dataset, calculate the following probabilities:

&KDSWHU

2. 下表详细介绍了患者出现的症状以及他们是否患有脑膜炎。

| ID | H    | F    | V    | M    |
|----|------|------|------|------|
| 1  | true | true | 假    | 假    |
| 2  | 假    | true | 假    | 假    |
| 3  | true | 假    | true | 假    |
| 4  | true | 假    | true | 假    |
| 5  | 假    | true | 假    | true |
| 6  | true | 假    | true | 假    |
| 7  | true | 假    | true | 假    |
| 8  | true | 假    | true | true |
| 9  | 假    | true | 假    | 假    |
| 10 | true | 假    | true | true |

使用此数据集，计算以下概率：

- (a)  $P(\text{VOMITING} = \text{true})$

This can be calculated easily by counting:  
 $P(\text{VOMITING} = \text{true}) = \frac{6}{10} = 0.6$

- (b)  $P(\text{HEADACHE} = \text{false})$

This can be calculated easily by counting:  
 $P(\text{HEADACHE} = \text{false}) = \frac{3}{10} = 0.3$

- (c)  $P(\text{HEADACHE} = \text{true}, \text{VOMITING} = \text{false})$

This can be calculated easily by counting:  
 $P(\text{HEADACHE} = \text{true}, \text{VOMITING} = \text{false}) = \frac{1}{10} = 0.1$   
 Or using the product rule:  
 $P(\text{HEADACHE} = \text{true}, \text{VOMITING} = \text{false}) = P(\text{HEADACHE} = \text{true} \mid \text{VOMITING} = \text{false}) \times P(\text{VOMITING} = \text{false}) = \frac{1}{4} \times \frac{4}{10} = 0.1$

- (d)  $P(\text{VOMITING} = \text{false} \mid \text{HEADACHE} = \text{true})$

This can be calculated easily by counting:  
 $P(\text{VOMITING} = \text{false} \mid \text{HEADACHE} = \text{true}) = \frac{1}{7} = 0.1429$

- (e)  $P(\text{MENINGITIS} \mid \text{FEVER} = \text{true}, \text{VOMITING} = \text{false})$

This can be calculated easily by counting. First,  
 $P(\text{MENINGITIS} = \text{true} \mid \text{FEVER} = \text{true}, \text{VOMITING} = \text{false}) = \frac{1}{4} = 0.25$ .  
 Then,  
 $P(\text{MENINGITIS} = \text{false} \mid \text{FEVER} = \text{true}, \text{VOMITING} = \text{false}) = \frac{3}{4} = 0.75$   
 So,  
 $\mathbf{P}(\text{MENINGITIS} \mid \text{FEVER} = \text{true}, \text{VOMITING} = \text{false}) = \langle 0.25, 0.75 \rangle$

- \* 5. The table below lists a dataset containing details of policyholders at an insurance company. The descriptive features included in the table describe each policy holders' ID, occupation, gender, age, type of insurance policy, and preferred contact channel. The preferred contact channel is the target feature in this domain.

## 第 6 章 基于概率的学习 (练习解决方案)

(a) PpV (英语: trueq)

这可以通过计算轻松计算:

$$PpV = \frac{\text{trueq}}{q} = \frac{6}{10} = 0.6$$

(b) 聚氢酶 (PpH)

这可以通过计算轻松计算:

$$PpH = \frac{\text{假}}{q} = \frac{3}{10} = 0.3$$

(c) PPH值 „真, V” „falseq

这可以通过计算轻松计算:

$$PpH = \frac{\text{真, V}}{q} = \frac{1}{10} = 0.1$$

或者使用产品规则:

$$PpH = \frac{\text{真, V}}{q} = \frac{1}{10} = 0.1$$

(d) PPV值 „假 | H” „trueq的

这可以通过计算轻松计算:

$$PpV = \frac{\text{假 | H}}{q} = \frac{1}{7} = 0.1429$$

(e) PpM „真, V” „falseq

这可以通过计数轻松计算。第一

$$PpM = \frac{\text{trueq}}{q} = \frac{1}{4} = 0.25$$

然后

$$PpM = \frac{\text{假 | F}}{q} = \frac{3}{4} = 0.75$$

So,

$$PpM = \frac{\text{真, V}}{q} = \frac{25}{100} = 0.25$$

° 5. 下表列出了一个数据集, 其中包含保险公司投保人的详细信息。

帕尼。表中包含的描述性特征描述了每个保单持有人的 ID、职业、性别、年龄、保单类型和首选联系渠道。首选联系渠道是此域中的目标功能。

| ID | OCCUPATION   | GENDER | AGE | POLICY<br>TYPE | PREF<br>CHANNEL |
|----|--------------|--------|-----|----------------|-----------------|
| 1  | lab tech     | female | 43  | planC          | email           |
| 2  | farmhand     | female | 57  | planA          | phone           |
| 3  | biophysicist | male   | 21  | planA          | email           |
| 4  | sheriff      | female | 47  | planB          | phone           |
| 5  | painter      | male   | 55  | planC          | phone           |
| 6  | manager      | male   | 19  | planA          | email           |
| 7  | geologist    | male   | 49  | planC          | phone           |
| 8  | messenger    | male   | 51  | planB          | email           |
| 9  | nurse        | female | 18  | planC          | phone           |

- (a) Using **equal-frequency binning**, transform the AGE feature into a categorical feature with three levels: *young*, *middle-aged*, *mature*.

There are 3 bins and 9 instances. So using equal-frequency binning we know that there will be 3 instances in each bin. In the table below we have ordered the instances in ascending order by AGE

| ID | OCCUPATION   | GENDER | AGE | POLICY<br>TYPE | PREF<br>CHANNEL |
|----|--------------|--------|-----|----------------|-----------------|
| 9  | nurse        | female | 18  | planC          | phone           |
| 6  | manager      | male   | 19  | planA          | email           |
| 3  | biophysicist | male   | 21  | planA          | email           |
| 1  | lab tech     | female | 43  | planC          | email           |
| 4  | sheriff      | female | 47  | planB          | phone           |
| 7  | geologist    | male   | 49  | planC          | phone           |
| 8  | messenger    | male   | 51  | planB          | email           |
| 5  | painter      | male   | 55  | planC          | phone           |
| 2  | farmhand     | female | 57  | planA          | phone           |

If we put the first 3 instances into the *young* bin, the next 3 instances into the *middle-aged* bin, etc. we end up with the dataset in the table below.

## 第 6 章 基于概率的学习 (练习解决方案)

| 同上 O G A T C |        |      |    | P   | P    |
|--------------|--------|------|----|-----|------|
| 1            | 实验室技术  | 女性   | 43 | 计划C | 电子邮件 |
| 2            | 雇农     | 女性   | 57 | 平   | 电话   |
| 3            | 生物物理学家 | male | 21 | 平   | 电子邮件 |
| 4            | 郡治安官   | 女性   | 47 | 计划B | 电话   |
| 5            | 画家     | male | 55 | 计划C | 电话   |
| 6            | 经理     | male | 19 | 平   | 电子邮件 |
| 7            | 地质学家   | male | 49 | 计划C | 电话   |
| 8            | 信使     | male | 51 | 计划B | 电子邮件 |
| 9            | 护士     | 女性   | 18 | 计划C | 电话   |

(a) 使用等频分箱，将 Afeature 转换为具有三个级别的分类特征：年轻、中年、成熟。

有 3 个箱和 9 个实例。因此，使用等频分档，我们知道每个分档中将有 3 个实例。在下表中，我们按 A 的升序对实例进行了排序

| 同上 O G A T C |        |      |    | P   | P    |
|--------------|--------|------|----|-----|------|
| 9            | 护士     | 女性   | 18 | 计划C | 电话   |
| 6            | 经理     | male | 19 | 平   | 电子邮件 |
| 3            | 生物物理学家 | male | 21 | 平   | 电子邮件 |
| 1            | 实验室技术  | 女性   | 43 | 计划C | 电子邮件 |
| 4            | 郡治安官   | 女性   | 47 | 计划B | 电话   |
| 7            | 地质学家   | male | 49 | 计划C | 电话   |
| 8            | 信使     | male | 51 | 计划B | 电子邮件 |
| 5            | 画家     | male | 55 | 计划C | 电话   |
| 2            | 雇农     | 女性   | 57 | 平   | 电话   |

如果我们将前 3 个实例放入年轻垃圾箱，将接下来的 3 个实例放入中年垃圾箱，依此类推，我们最终会得到下表中的数据集。

| ID | OCCUPATION   | GENDER | AGE         | POLICY<br>TYPE | PREF<br>CHANNEL |
|----|--------------|--------|-------------|----------------|-----------------|
| 9  | nurse        | female | young       | planC          | phone           |
| 6  | manager      | male   | young       | planA          | email           |
| 3  | biophysicist | male   | young       | planA          | email           |
| 1  | lab tech     | female | middle-aged | planC          | email           |
| 4  | sheriff      | female | middle-aged | planB          | phone           |
| 7  | geologist    | male   | middle-aged | planC          | phone           |
| 8  | messenger    | male   | mature      | planB          | email           |
| 5  | painter      | male   | mature      | planC          | phone           |
| 2  | farmhand     | female | mature      | planA          | phone           |

The thresholds for the different bins are calculated as the mid-point between the AGE values of the two instances on either side of the boundary. So, the threshold between the *young* and *middle-aged* bins would be the mid point between  $\mathbf{d}_3 \text{with AGE} = 21$  and  $\mathbf{d}_1 \text{with AGE} = 43$ :

$$\frac{21 + 43}{2} = 32$$

Likewise, the threshold between the *middleaged* and *mature* bins would be the mid point between  $\mathbf{d}_7 \text{with AGE} = 49$  and  $\mathbf{d}_8 \text{with AGE} = 51$ :

$$\frac{49 + 51}{2} = 50$$

- (b) Examine the descriptive features in the dataset and list the features that you would exclude before you would use the dataset to build a predictive model. For each feature you decide to exclude, explain why you have made this decision.

As is always the case the ID feature should not be used as a descriptive feature during training. However, in this example there is another feature that should be removed from the dataset prior to training. The OCCUPATION feature has different and unique levels for each instance in the dataset. In other words, the OCCUPATION feature is equivalent to an id for each instance. Consequently, it should also be removed from the dataset prior to training a model.

- (c) Calculate the probabilities required by a **naive Bayes model** to represent this domain.

To train a naive Bayes model using this data, we need to compute the prior probabilities of the target feature taking each level in its domain, and the conditional probability of each feature taking each level in its domain condi-

## 第 6 章 基于概率的学习 (练习解决方案)

| 同上 O G A T C |        |      |     | P   | P    |
|--------------|--------|------|-----|-----|------|
| 9            | 护士     | 女性   | 年轻  | 计划C | 电话   |
| 6            | 经理     | male | 年轻  | 平   | 电子邮件 |
| 3            | 生物物理学家 | male | 年轻  | 平   | 电子邮件 |
| 1            | 实验室技术  | 女性   | 中年的 | 计划C | 电子邮件 |
| 4            | 郡治安官   | 女性   | 中年的 | 计划B | 电话   |
| 7            | 地质学家   | male | 中年的 | 计划C | 电话   |
| 8            | 信使     | male | 成熟  | 计划B | 电子邮件 |
| 5            | 画家     | male | 成熟  | 计划C | 电话   |
| 2            | 雇农     | 女性   | 成熟  | 平   | 电话   |

不同条柱的阈值计算为边界两侧两个实例的 A 值之间的中点。因此，年轻人和中老年人之间的阈值将是 dwithA 之间的中点

$$\frac{21 + 43}{2} = 32$$

同样，中年和成熟 bin 之间的阈值将是 dwithA 之间的中点

$$\frac{49 + 51}{2} = 50$$

(b) 检查数据集中的描述性特征，并列出在使用数据集构建预测模型之前要排除的特征。对于您决定排除的每个功能，请解释您做出此决定的原因。

与往常一样，在训练期间不应将 ID 特征用作描述性特征。但是，在此示例中，应在训练之前从数据集中删除另一个特征。对于数据集中的每个实例，Ofeature 具有不同且唯一的级别。换言之，Ofeature 等同于每个实例的 id。因此，在训练模型之前，还应将其从数据集中删除。

(c) 计算朴素贝叶斯模型表示该域所需的概率。

为了使用这些数据训练朴素贝叶斯模型，我们需要计算目标特征在其域中获取每个级别的先验概率，以及每个特征在其域中获取每个级别的条件概率条件

tioned for each level that the target feature can take. The table below lists the probabilities required by a naive Bayes model to represent this domain.

|                                      |                                       |
|--------------------------------------|---------------------------------------|
| $P(phone) = 0.56$                    | $P(email) = 0.44$                     |
| $P(GENDER = female   phone) = 0.6$   | $P(GENDER = female   email) = 0.25$   |
| $P(GENDER = male   phone) = 0.4$     | $P(GENDER = male   email) = 0.75$     |
| $P(AGE = young   phone) = 0.2$       | $P(AGE = young   email) = 0.5$        |
| $P(AGE = middle-aged   phone) = 0.4$ | $P(AGE = middle-aged   email) = 0.25$ |
| $P(AGE = mature   phone) = 0.4$      | $P(AGE = mature   email) = 0.25$      |
| $P(POLICY = planA   phone) = 0.2$    | $P(POLICY = planA   email) = 0.5$     |
| $P(POLICY = planB   phone) = 0.2$    | $P(POLICY = planB   email) = 0.25$    |
| $P(POLICY = planC   phone) = 0.6$    | $P(POLICY = planC   email) = 0.25$    |

- (d) What target level will a **naive Bayes model** predict for the following query:

GENDER = *female*, AGE = 30, POLICY = *planA*

The first step in calculating this answer is to bin the AGE feature. We know from part (a) of this question that the threshold between the *young* and *middle-aged* bins is 32. The value for the AGE feature in the query is less than 32 so it is mapped to the *young* bin. This results in the query being defined as

GENDER = *female*, AGE = *young*, POLICY = *planA*

The calculation for  $P(\text{CHANNEL} = \textit{phone} | \mathbf{q})$  is

|                                             |
|---------------------------------------------|
| $P(phone) = 0.56$                           |
| $P(GENDER = \textit{female}   phone) = 0.6$ |
| $P(AGE = \textit{young}   phone) = 0.2$     |
| $P(POLICY = \textit{planA}   phone) = 0.2$  |

$$\left( \prod_{k=1}^m P(\mathbf{q}[k] | \textit{phone}) \right) \times P(\textit{phone}) = 0.01344$$

The calculation for  $P(\text{CHANNEL} = \textit{email} | \mathbf{q})$  is

|                                              |
|----------------------------------------------|
| $P(email) = 0.44$                            |
| $P(GENDER = \textit{female}   email) = 0.25$ |
| $P(AGE = \textit{young}   email) = 0.5$      |
| $P(POLICY = \textit{planA}   email) = 0.5$   |

$$\left( \prod_{k=1}^m P(\mathbf{q}[k] | \textit{email}) \right) \times P(\textit{email}) = 0.0275$$

The target level with the highest ranking is CHANNEL = *email*, and this is the prediction returned by the model.

## 第 6 章 基于概率的学习 (练习解决方案)

针对目标功能可以采用的每个级别进行说明。下表列出了朴素贝叶斯模型表示此域所需的概率。

| PP电话 |      |    |      | PP电子邮件 |      |      |      |
|------|------|----|------|--------|------|------|------|
| PpG  | 女性   | 电话 | 0.56 | PpG    | 女性   | 电子邮件 | 0.44 |
| PpG  | male | 电话 | 0.06 | PpG    | male | 电子邮件 | 0.25 |
| PpA  | 年轻   | 电话 | 0.04 | PpA    | 年轻   | 电子邮件 | 0.75 |
| PpA  | 中年的  | 电话 | 0.02 | PpA    | 中年的  | 电子邮件 | 0.5  |
| PpA  | 成熟的  | 电话 | 0.04 | PpA    | 成熟的  | 电子邮件 | 0.25 |
| PpP  | 平    | 电话 | 0.02 | PpP    | 平    | 电子邮件 | 0.5  |
| PpP  | 计划B  | 电话 | 0.02 | PpP    | 计划B  | 电子邮件 | 0.25 |
| PpP  | 计划C  | 电话 | 0.06 | PpP    | 计划C  | 电子邮件 | 0.25 |
|      |      | q  | 0.1  |        |      | q    | 0.1  |

(d) 朴素贝叶斯模型将预测以下查询的目标水平:

G=女性, A=30, P=平面A

计算此答案的第一步是将 Afeature 装箱。我们从这个问题的 (a) 部分知道, 年轻人和中年人之间的门槛是32。查询中 Afeature 的值小于 32, 因此它映射到年轻的 bin。这会导致查询被定义为

G=雌性, A=年轻, P=平面A

PpC的计算

| PP电话 |    |    |      |
|------|----|----|------|
| PpG  | 女性 | 电话 | 0.56 |
| PpA  | 年轻 | 电话 | 0.06 |
| PpP  | 平  | 电话 | 0.02 |
|      |    | q  | 0.2  |

$$\sim \frac{Ppqr | \text{电话} Q}{\sum_{s=1}^3 Ppqr | \text{电话} Q} = \frac{0.1344}{0.1}$$

PpC的计算

| PP电子邮件 |    |      |      |
|--------|----|------|------|
| PpG    | 女性 | 电子邮件 | 0.44 |
| PpA    | 年轻 | 电子邮件 | 0.25 |
| PpP    | 平  | 电子邮件 | 0.5  |
|        |    | q    | 0.5  |

$$\sim \frac{Ppqr | \text{电子邮件} Q}{\sum_{s=1}^3 Ppqr | \text{电子邮件} Q} = \frac{0.275}{0.1}$$

排名最高的目标级别是 C= email, 这是模型返回的预测。

- \* 6. Imagine that you have been given a dataset of 1,000 documents that have been classified as being about *entertainment* or *education*. There are 700 *entertainment* documents in the dataset and 300 *education* documents in the dataset. The tables below give the number of documents from each topic that a selection of words occurred in.

| Word-document counts for the <i>entertainment</i> dataset |     |         |           |        |          |
|-----------------------------------------------------------|-----|---------|-----------|--------|----------|
| fun                                                       | is  | machine | christmas | family | learning |
| 415                                                       | 695 | 35      | 0         | 400    | 70       |
| Word-document counts for the <i>education</i> dataset     |     |         |           |        |          |
| fun                                                       | is  | machine | christmas | family | learning |
| 200                                                       | 295 | 120     | 0         | 10     | 105      |

- (a) What target level will a **naïve Bayes model** predict for the following query document: “*machine learning is fun*”?

A naïve Bayes model will label the query with the target level that has the highest probability under the assumption of conditional independence between the evidence features. So to answer this question, we need to calculate the probability of each target level given the evidence and assuming conditional independence across the evidence.

To carry out these calculations, we need to convert the raw document counts into conditional probabilities by dividing each count by the total number of documents occurring in each topic:

| $w_k$    | Count | $P(w_k \mid \text{entertainment})$ |
|----------|-------|------------------------------------|
| fun      | 415   | $\frac{415}{700} = .593$           |
| is       | 695   | $\frac{695}{700} = .99$            |
| learning | 35    | $\frac{35}{700} = .05$             |
| machine  | 70    | $\frac{70}{700} = .10$             |

## 第 6 章 基于概率的学习 (练习解决方案)

- ° 6. 想象一下，您获得了一个包含 1,000 个文档的数据集，这些文档已被确定为关于娱乐或教育。数据集中有 700 个娱乐文档，数据集中有 300 个教育文档。下表给出了每个主题中出现所选单词的文档数量。

|                  |          |              |
|------------------|----------|--------------|
| 娱乐数据集的 Word 文档计数 | 乐趣是机器圣诞节 |              |
| 家庭学习             |          |              |
| 415              | 695      | 35 0 400 70  |
| 教育数据集的 Word 文档计数 | 乐趣是机器圣诞节 |              |
| 家庭学习             |          |              |
| 200              | 295      | 120 0 10 105 |

- (a) 对于以下查询文档：“机器学习很有趣”，朴素贝叶斯模型将预测什么目标水平？

朴素贝叶斯模型将在证据特征之间具有条件独立性的假设下，使用具有最高概率的目标级别标记查询。因此，为了回答这个问题，我们需要在给定证据的情况下计算每个目标水平的概率，并假设证据之间的条件独立性。

为了进行这些计算，我们需要将原始文档计数转换为条件概率，方法是将每个计数除以每个主题中出现的文档总数：

| wCount | Ppw 娱乐Q                  |
|--------|--------------------------|
| 好玩 415 | $\frac{415}{700} = .593$ |
| 是 695  | $\frac{695}{700} = .99$  |
| 学习 35  | $\frac{35}{700} = .05$   |
| 机器 70  | $\frac{70}{700} = .10$   |

| $w_k$    | Count | $P(w_k \mid \text{education})$ |
|----------|-------|--------------------------------|
| fun      | 200   | $\frac{200}{300} = .667$       |
| is       | 295   | $\frac{295}{300} = .983$       |
| learning | 120   | $\frac{120}{300} = .40$        |
| machine  | 105   | $\frac{105}{300} = .35$        |

We can now compute the probabilities of each target level:

$$\begin{aligned}
 P(\text{entertainment} \mid \mathbf{q}) &= P(\text{entertainment}) \times P(\text{machine} \mid \text{entertainment}) \\
 &\quad \times P(\text{learning} \mid \text{entertainment}) \\
 &\quad \times P(\text{is} \mid \text{entertainment}) \\
 &\quad \times p(\text{fun} \mid \text{entertainment}) \\
 &= 0.7 \times 0.593 \times 0.99 \times 0.5 \times 0.1 \\
 &= 0.00205
 \end{aligned}$$

$$\begin{aligned}
 P(\text{education} \mid \mathbf{q}) &= P(\text{education}) \times P(\text{machine} \mid \text{education}) \\
 &\quad \times P(\text{learning} \mid \text{education}) \\
 &\quad \times P(\text{is} \mid \text{education}) \\
 &\quad \times p(\text{fun} \mid \text{education}) \\
 &= 0.3 \times 0.667 \times 0.983 \times 0.4 \times 0.35 \\
 &= 0.00275
 \end{aligned}$$

As  $P(\text{education} \mid \mathbf{q}) > P(\text{entertainment} \mid \mathbf{q})$ , the naive Bayes model will predict the target level of *education*.

- (b) What target level will a **naive Bayes model** predict for the following query document: “*christmas family fun*”?

Because the word *christmas* does not appear in any document of either topic, both conditional probabilities for this word will be equal to 0:  
 $P(\text{christmas} \mid \text{entertainment}) = 0$  and  $P(\text{christmas} \mid \text{education}) = 0$ .

## 第 6 章 基于概率的学习 (练习解决方案)

|    | wCount | Ppw 教育Q  |
|----|--------|----------|
| 趣味 | 200    | 200 .667 |
|    |        | 300      |
| 是  | 295    | 295 .983 |
|    |        | 300      |
| 学习 | 120    | 120 .40  |
|    |        | 300      |
| 机器 | 105    | 105 .35  |
|    |        | 300      |

现在，我们可以计算每个目标水平的概率：

$$PP_{\text{娱乐} | qq} = pp_{\text{entertainment}q} \wedge pp_{\text{machine} | \text{娱乐}Q}$$

$$\wedge PP_{\text{学习} | \text{娱乐}Q}$$

$$\wedge PP_{\text{IS的} | \text{娱乐}Q}$$

$$\wedge PP_{f \text{ 联合国} | \text{娱乐}Q}$$

$$= 0.7 \wedge 0.593 \wedge 0.99 \wedge 0.5 \wedge 0.1$$

$$= 0.00205$$

$$PP_{\text{教育} | qq} = pp_{\text{education}q} \wedge pp_{\text{machine} | \text{教育}Q}$$

$$\wedge PP_{\text{学习} | \text{教育}Q}$$

$$\wedge PP_{\text{IS的} | \text{教育}Q}$$

$$\wedge PP_{f \text{ 联合国} | \text{教育}Q}$$

$$= 0.3 \wedge 0.667 \wedge 0.983 \wedge 0.4 \wedge 0.35$$

$$= 0.00275$$

作为  $PP_{\text{education} | qq}$  娱乐 | qq，朴素贝叶斯模型将预测目标教育水平。

(b) 朴素贝叶斯模型将预测以下查询文档的目标水平：“圣诞家庭娱乐”？

由于“圣诞节”一词未出现在任何一个主题文档中，因此该词的两个条件概率都将等于 0： $PP_{\text{christmas} | \text{Entertainment}Q} = 0$  和  $PP_{\text{christmas} | \text{教育}Q} = 0$ 。

Consequently, the probability for both target levels will be 0, and the model will not be able to return a prediction.

- (c) What target level will a **naïve Bayes model** predict for the query document in Part (b) of this question, if **Laplace smoothing** with  $k = 10$  and a vocabulary size of 6 is used?

The table below illustrates the smoothing of the posterior probabilities for  $P(\text{word} \mid \text{entertainment})$ .

|               |                                                                                  |   |        |
|---------------|----------------------------------------------------------------------------------|---|--------|
| Raw           | $P(\text{christmas} \mid \text{entertainment})$                                  | = | 0      |
| Probabilities | $P(\text{family} \mid \text{entertainment})$                                     | = | 0.5714 |
|               | $P(\text{fun} \mid \text{entertainment})$                                        | = | 0.5929 |
| Smoothing     | $k$                                                                              | = | 10     |
| Parameters    | $\text{count}(\text{entertainment})$                                             | = | 700    |
|               | $\text{count}(\text{christmas} \mid \text{entertainment})$                       | = | 0      |
|               | $\text{count}(\text{family} \mid \text{entertainment})$                          | = | 400    |
|               | $\text{count}(\text{fun} \mid \text{entertainment})$                             | = | 415    |
|               | $ \text{Domain}(\text{vocabulary}) $                                             | = | 6      |
| Smoothed      | $P(\text{christmas} \mid \text{entertainment}) = \frac{0+10}{700+(10 \times 6)}$ | = | 0.0132 |
| Probabilities | $P(\text{family} \mid \text{entertainment}) = \frac{400+10}{700+(10 \times 6)}$  | = | 0.5395 |
|               | $P(\text{fun} \mid \text{entertainment}) = \frac{415+10}{700+(10 \times 6)}$     | = | 0.5592 |

The smoothing of the posterior probabilities for  $P(\text{word} \mid \text{education})$  is carried out in the same way:

## 第 6 章 基于概率的学习 (练习解决方案)

因此，两个目标水平的概率都将为 0，并且模型将无法返回预测值。

(c) 如果使用  $k = 10$  和词汇量为 6 的拉普拉斯平滑，朴素贝叶斯模型将预测本问题 (b) 部分中查询文档的目标水平是多少？

下表说明了  $P(\text{word} | \text{娱乐Q})$ 。

| Raw                    | 圣诞   娱乐Q = 0                                                    |
|------------------------|-----------------------------------------------------------------|
| 概率                     | $P(\text{family}   \text{娱乐Q}) = 0.5714$                        |
|                        | $P(\text{family}   \text{娱乐Q}) = 0.5929$                        |
| 平滑                     | $k = 10$                                                        |
| 参数                     | 计数娱乐Q = 700                                                     |
|                        | CountpChristmas (圣诞节)   娱乐Q = 0                                 |
|                        | Countp Family   娱乐Q = 400                                       |
|                        | 计数 Family   娱乐Q = 415                                           |
|                        | 域名pvocabularyq = 6                                              |
| 平滑的 Ppchrismas   娱乐Q = | $\frac{0+1}{700+10} = 0.0132$                                   |
| 概率 Pp family   娱乐Q =   | $\frac{400+1}{700+10} = 0.5395$                                 |
|                        | $P(\text{family}   \text{娱乐Q}) = \frac{415+1}{700+10} = 0.5592$ |

Ppword 后验概率的平滑化 | educationq 以相同的方式进行：

Chapter 6 Probability-Based Learning (Exercise Solutions)

|               |                                                                                  |   |        |
|---------------|----------------------------------------------------------------------------------|---|--------|
| Raw           | $P(\text{christmas} \mid \text{education})$                                      | = | 0      |
| Probabilities | $P(\text{family} \mid \text{education})$                                         | = | 0.5714 |
|               | $P(\text{fun} \mid \text{education})$                                            | = | 0.5929 |
| Smoothing     | $k$                                                                              | = | 10     |
| Parameters    | $\text{count}(\text{education})$                                                 | = | 300    |
|               | $\text{count}(\text{christmas} \mid \text{education})$                           | = | 0      |
|               | $\text{count}(\text{family} \mid \text{education})$                              | = | 10     |
|               | $\text{count}(\text{fun} \mid \text{education})$                                 | = | 200    |
|               | $ \text{Domain}(\text{vocabulary}) $                                             | = | 6      |
| Smoothed      | $P(\text{christmas} \mid \text{entertainment}) = \frac{0+10}{300+(10 \times 6)}$ | = | 0.0278 |
| Probabilities | $P(\text{family} \mid \text{entertainment}) = \frac{10+10}{300+(10 \times 6)}$   | = | 0.0556 |
|               | $P(\text{fun} \mid \text{entertainment}) = \frac{200+10}{300+(10 \times 6)}$     | = | 0.5833 |

We can now compute the probabilities of each target level:

$$\begin{aligned}
 P(\text{entertainment} \mid \mathbf{q}) &= P(\text{entertainment}) \\
 &\quad \times P(\text{christmas} \mid \text{entertainment}) \\
 &\quad \times P(\text{family} \mid \text{entertainment}) \\
 &\quad \times P(\text{fun} \mid \text{entertainment}) \\
 &= 0.7 \times 0.0132 \times 0.5395 \times 0.5592 \\
 &= 0.0028
 \end{aligned}$$

$$\begin{aligned}
 P(\text{education} \mid \mathbf{q}) &= P(\text{education}) \times P(\text{christmas} \mid \text{education}) \\
 &\quad \times P(\text{family} \mid \text{education}) \times P(\text{fun} \mid \text{education}) \\
 &= 0.3 \times 0.0278 \times 0.0556 \times 0.5833 \\
 &= 0.0003
 \end{aligned}$$

As  $P(\text{entertainment} \mid \mathbf{q}) > P(\text{education} \mid \mathbf{q})$ , the model will predict a label of *entertainment* for this query.

## 第 6 章 基于概率的学习 (练习解决方案)

|     |                                                               |
|-----|---------------------------------------------------------------|
| Raw | 圣诞  教育Q " 0                                                   |
| 概率  | Pp f amily  教育Q " 0.5714                                      |
|     | Pp f —  教育Q " 0.5929                                          |
| 平滑  | k " 10                                                        |
| 参数  | 计数peducationq " 300                                           |
|     | CountpChristmas (圣诞节)  教育Q " 0                                |
|     | Countp F Amily  教育Q " 10                                      |
|     | 计数 F —  教育Q " 200                                             |
|     | 域名pvocabularyq " 6                                            |
| 平滑的 | Ppchristmas  娱乐Q " $\frac{300 \cdot P_{10} + 6Q}{300}$ 0.0278 |
| 概率  | Pp f amily  娱乐Q " $\frac{300 \cdot P_{10} + 6Q}{300}$ 0.0556  |
|     | Pp f —  娱乐Q " $\frac{300 \cdot P_{10} + 6Q}{300}$ 0.5833      |

现在，我们可以计算每个目标水平的概率：

PP娱乐 |qq " ppentertainmentq  
 $\hat{\text{圣诞}} | \text{娱乐Q}$   
 $\hat{\text{Pp f amily}} | \text{娱乐Q}$   
 $\hat{\text{Pp f un}} | \text{娱乐Q}$   
 $" 0.7 \wedge 0.0132 \wedge 0.5395 \wedge 0.5592$   
 $" 0.0028$

PP教育 |qq " ppeducationq ^ ppchristmas |教育Q  
 $\hat{\text{Pp f amily}} | \text{教育Q} \wedge \text{PP F 联合国} | \text{教育Q}$   
 $" 0.3 \wedge 0.0278 \wedge 0.0556 \wedge 0.5833$   
 $" 0.0003$

作为 Ppentertainment |qq中文版教育 |QQ，模型将预测此查询的娱乐标签。

- \* 9. The following is a description of the causal relationship between storms, the behavior of burglars and cats, and house alarms:

Jim and Martha always go shopping separately. If Jim does the shopping he buys wine, but not always. If Martha does the shopping, she buys wine, but not always. If Jim tells Martha that he has done the shopping, then Martha doesn't go shopping, but sometimes Jim forgets to tell Martha, and so sometimes both Jim and Martha go shopping.

- (a) Define the topology of a Bayesian network that encodes these causal relationships between the following Boolean variables: JIM (Jim has done the shopping, *true* or *false*), MARTHA (Martha has done the shopping, *true* or *false*), WINE (wine has been purchased, *true* or *false*).

The figure below illustrates a Bayesian network that encodes the described causal relationships. JIM directly affects the behavior of MARTHA and WINE, and this is reflected by links from the JIM node to the MARTHA and WINE nodes. The MARTHA also affects whether WINE was purchased, and hence there is a link from MARTHA to WINE.

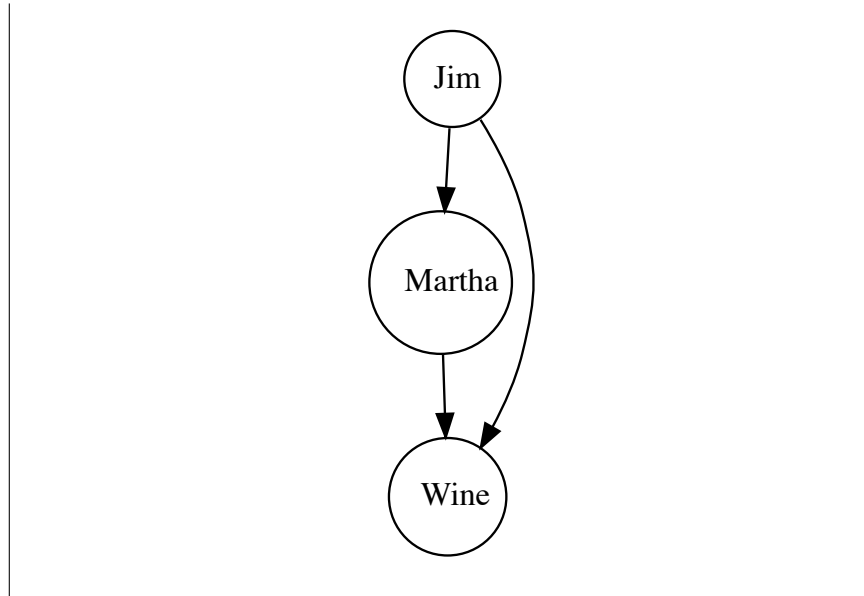
## 第 6 章 基于概率的学习 (练习解决方案)

- ° 9. 以下是对风暴之间因果关系的描述，其行为窃贼和猫，以及房屋警报器：

如果玛莎去购物，她会买酒，但并非总是如此。如果吉姆告诉玛莎他已经购物了，那么玛莎就不会去购物，但有时吉姆忘记告诉玛莎，所以有时吉姆和玛莎都去购物。

(a) 定义贝叶斯网络的拓扑结构，该拓扑对以下布尔变量之间的因果关系进行编码：J (Jim 已经完成了购物，对或假)、M (Martha 已经完成了购物，对或假)、W (购买了葡萄酒，对或假)。

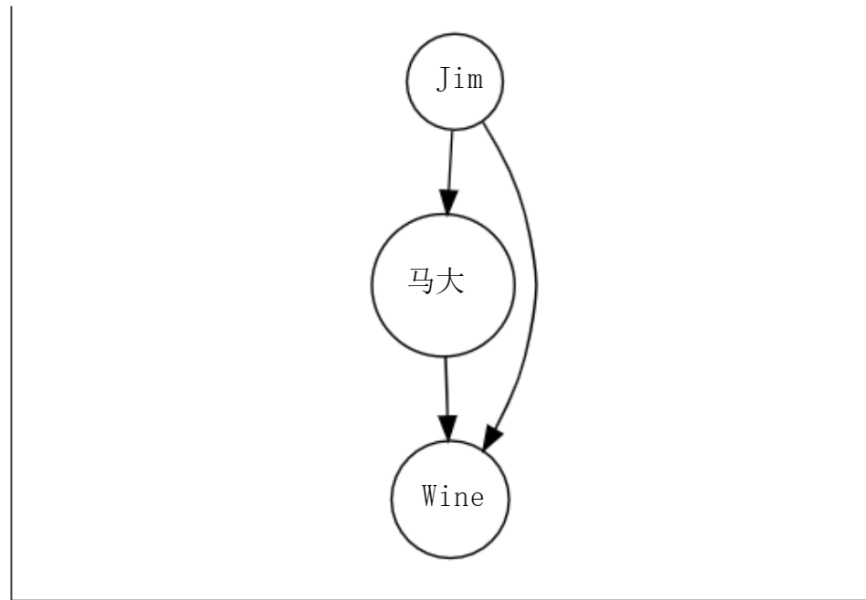
下图说明了对所描述的因果关系进行编码的贝叶斯网络。J 直接影响 M 和 W 的行为，这反映在从 J 节点到 M 和 W 节点的链接上。M 还会影响是否购买了 W，因此有一个来自 M 到 W 的链接。



- (b) The table below lists a set of instances from the house alarm domain. Using the data in this table, create the conditional probability tables (CPTs) for the network you created in the first part of this question, and round the probabilities to two places of decimal.

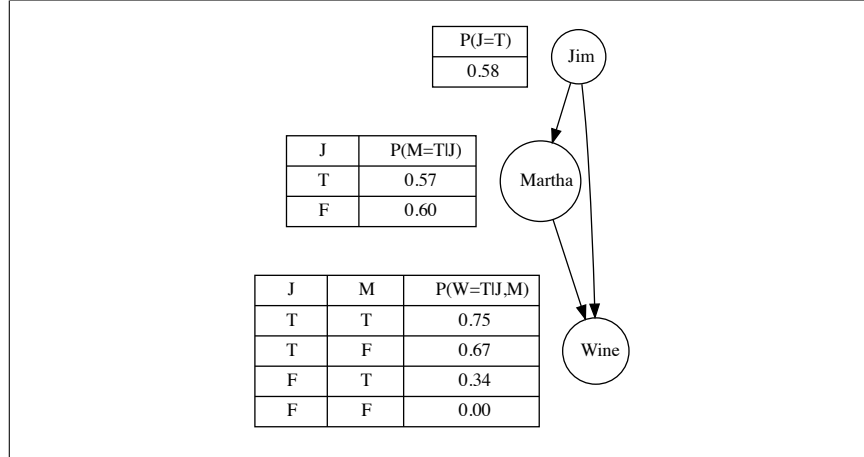
| ID | JIM   | MARTHA | WINE  |
|----|-------|--------|-------|
| 1  | false | false  | false |
| 2  | false | false  | false |
| 3  | true  | false  | true  |
| 4  | true  | false  | true  |
| 5  | true  | false  | false |
| 6  | false | true   | true  |
| 7  | false | true   | false |
| 8  | false | true   | false |
| 9  | true  | true   | true  |
| 10 | true  | true   | true  |
| 11 | true  | true   | true  |
| 12 | true  | true   | false |

## 第 6 章 基于概率的学习 (练习解决方案)



(b) 下表列出了来自房屋报警域的一组实例。使用此表中的数据，为您在本问题的第一部分中创建的网络创建条件概率表 (CPT)，并将概率四舍五入到小数点后两位。

| 编号 | J    | M    | W    |
|----|------|------|------|
| 1  | 假    | 假    | 假    |
| 2  | 假    | 假    | 假    |
| 3  | true | 假    | true |
| 4  | true | 假    | true |
| 5  | true | 假    | 假    |
| 6  | 假    | true | true |
| 7  | 假    | true | 假    |
| 8  | 假    | true | 假    |
| 9  | true | true | true |
| 10 | true | true | true |
| 11 | true | true | true |
| 12 | true | true | 假    |



- (c) What value will the Bayesian network predict for WINE if:

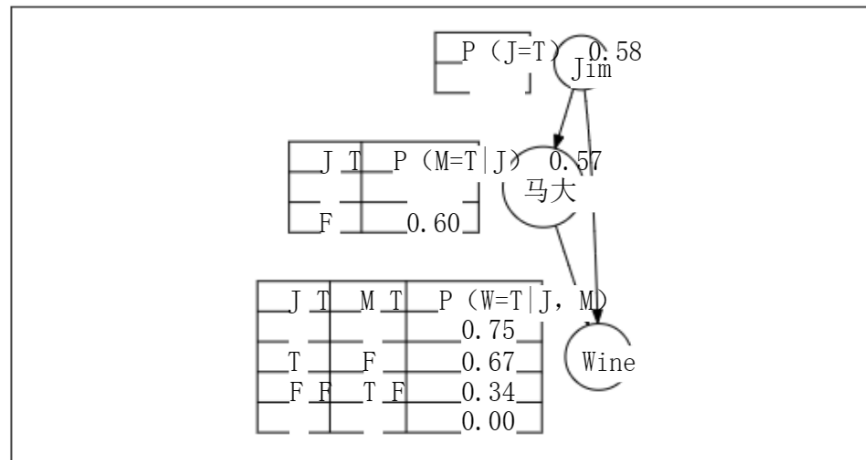
$JIM=true$  and  $MARTHA=false$

Because both the parent nodes for WINE are known, we can read the relevant probability distribution over WINE directly from the conditional probability table for the WINE node. Examining the conditional probability table, we can see that when  $JIM = true$ , and  $MARTHA = false$ , then  $WINE = true$  is the MAP prediction (0.75 versus 0.25). In other words, the network would predict that wine would be purchased in this scenario.

- (d) What is the probability that JIM went shopping given that  $WINE=true$ ?

$$\begin{aligned}
 P(JIM = true \mid WINE = true) &= \frac{P(JIM = true, WINE = true)}{P(WINE = true)} \\
 &= \frac{\sum_{M \in \{T, F\}} P(MARTHA, JIM = true, WINE = true)}{\sum_{M, J \in \{T, F\}} P(MARTHA, JIM, WINE = true)} \\
 &= \frac{0.75 + 0.67}{0.75 + 0.67 + 0.34 + 0.00} \\
 &= \frac{1.42}{1.76} \\
 &= 0.81
 \end{aligned}$$

## 第 6 章 基于概率的学习 (练习解决方案)



(c) 贝叶斯网络将预测Wif的什么值:

$J=\text{true}$  和  $M=\text{false}$

由于 Ware 的两个父节点都是已知的，因此我们可以直接从 Wnode 的条件概率表中读取 W 上的相关概率分布。检查条件概率表，我们可以看到，当  $J = \text{true}$  且  $M = \text{false}$  时， $W = \text{true}$  是 MAP 预测 (0.75 对 0.25)。换句话说，该网络将预测在这种情况下将购买葡萄酒。

(d) 假设  $W=\text{true}$ , Jwent 购物的概率是多少?

$$\begin{aligned}
 P(J=\text{true} | W=\text{true}) &= \frac{P(J=\text{true}, W=\text{true})}{P(W=\text{true})} \\
 &= \frac{P(J=\text{true}) \cdot P(M=\text{true} | J=\text{true}) \cdot P(W=\text{true} | J=\text{true}, M=\text{true})}{P(J=\text{true}) \cdot P(M=\text{true} | J=\text{true}) \cdot P(W=\text{true} | J=\text{true}, M=\text{true}) + P(J=\text{true}) \cdot P(M=\text{false} | J=\text{true}) \cdot P(W=\text{true} | J=\text{true}, M=\text{false})} \\
 &= \frac{0.58 \cdot 0.57 \cdot 0.75}{0.58 \cdot 0.57 \cdot 0.75 + 0.58 \cdot 0.60 \cdot 0.67} \\
 &= \frac{0.25065}{0.25065 + 0.23256} \\
 &= \frac{0.25065}{0.48321} \\
 &\approx 0.52
 \end{aligned}$$

## Chapter 7

2. You have been hired by the European Space Agency to build a model that predicts the amount of oxygen that an astronaut consumes when performing five minutes of intense physical work. The descriptive features for the model will be the age of the astronaut and their average heart rate throughout the work. The regression model is

$$\text{OXYCON} = \mathbf{w}[0] + \mathbf{w}[1] \times \text{AGE} + \mathbf{w}[2] \times \text{HEARTRATE}$$

The table that follows shows a historical dataset that has been collected for this task.

| ID | OXYCON | AGE | HEART | ID | OXYCON | AGE | HEART |
|----|--------|-----|-------|----|--------|-----|-------|
|    |        |     | RATE  |    |        |     | RATE  |
| 1  | 37.99  | 41  | 138   | 7  | 44.72  | 43  | 158   |
| 2  | 47.34  | 42  | 153   | 8  | 36.42  | 46  | 143   |
| 3  | 44.38  | 37  | 151   | 9  | 31.21  | 37  | 138   |
| 4  | 28.17  | 46  | 133   | 10 | 54.85  | 38  | 158   |
| 5  | 27.07  | 48  | 126   | 11 | 39.84  | 43  | 143   |
| 6  | 37.85  | 44  | 145   | 12 | 30.83  | 43  | 138   |

- (a) Assuming that the current weights in a multivariate linear regression model are  $\mathbf{w}[0] = -59.50$ ,  $\mathbf{w}[1] = -0.15$ , and  $\mathbf{w}[2] = 0.60$ , make a prediction for each training instance using this model.

The following table shows the predictions made using the given model weights.

&KDSWHU

2. 您受雇于欧洲航天局，负责构建一个模型，该模型可以预测宇航员在进行五分钟的高强度体力劳动时消耗的氧气量。该模型的描述性特征将是宇航员的年龄和他们在整个工作过程中的平均心率。回归模型是

$$\text{OC}' \text{ in } r0s' \text{ in } r1s \wedge A' \text{ in } r2s \wedge \text{HR}$$

下表显示了为此任务收集的历史数据集。

| H  |        |    |     | H  |        |    |     |
|----|--------|----|-----|----|--------|----|-----|
| ID | OCAR奥卡 |    |     | ID | OCAR奥卡 |    |     |
| 1  | 37.99  | 41 | 138 | 7  | 44.72  | 43 | 158 |
| 2  | 47.34  | 42 | 153 | 8  | 36.42  | 46 | 143 |
| 3  | 44.38  | 37 | 151 | 9  | 31.21  | 37 | 138 |
| 4  | 28.17  | 46 | 133 | 10 | 54.85  | 38 | 158 |
| 5  | 27.07  | 48 | 126 | 11 | 39.84  | 43 | 143 |
| 6  | 37.85  | 44 | 145 | 12 | 30.83  | 43 | 138 |

- (a) 假设多元线性回归模型中的当前权重为  $w_{r0s} = 59.50$ 、 $w_{r1s} = 0.15$  和  $w_{r2s} = 0.60$ ，请使用该模型对每个训练实例进行预测。

下表显示了使用给定模型权重进行的预测。

| ID | OXYCON | AGE | HEART RATE | Prediction |
|----|--------|-----|------------|------------|
| 1  | 37.99  | 41  | 138        | 17.15      |
| 2  | 47.34  | 42  | 153        | 26.00      |
| 3  | 44.38  | 37  | 151        | 25.55      |
| 4  | 28.17  | 46  | 133        | 13.40      |
| 5  | 27.07  | 48  | 126        | 8.90       |
| 6  | 37.85  | 44  | 145        | 20.90      |
| 7  | 44.72  | 43  | 158        | 28.85      |
| 8  | 36.42  | 46  | 143        | 19.40      |
| 9  | 31.21  | 37  | 138        | 17.75      |
| 10 | 54.85  | 38  | 158        | 29.60      |
| 11 | 39.84  | 43  | 143        | 19.85      |
| 12 | 30.83  | 43  | 138        | 16.85      |

- (b) Calculate the sum of squared errors for the set of predictions generated in Part (a).

The following table shows the predictions made by the model and sum of squared error calculation based on these predictions.

| Initial Weights                   |        |            |                  |               |                                                  |                                                  |                                                  |      |
|-----------------------------------|--------|------------|------------------|---------------|--------------------------------------------------|--------------------------------------------------|--------------------------------------------------|------|
| $\mathbf{w}[0]:$                  |        | -59.50     | $\mathbf{w}[1]:$ |               | -0.15                                            | $\mathbf{w}[2]:$                                 |                                                  | 0.60 |
| Iteration 1                       |        |            |                  |               |                                                  |                                                  |                                                  |      |
| ID                                | OXYCON | Prediction | Error            | Squared Error | $errorDelta$<br>( $\mathcal{D}, \mathbf{w}[0]$ ) | $errorDelta$<br>( $\mathcal{D}, \mathbf{w}[1]$ ) | $errorDelta$<br>( $\mathcal{D}, \mathbf{w}[2]$ ) |      |
| 1                                 | 37.99  | 17.15      | 20.84            | 434.41        | 20.84                                            | 854.54                                           | 2,876.26                                         |      |
| 2                                 | 47.34  | 26.00      | 21.34            | 455.41        | 21.34                                            | 896.29                                           | 3,265.05                                         |      |
| 3                                 | 44.38  | 25.55      | 18.83            | 354.60        | 18.83                                            | 696.74                                           | 2,843.45                                         |      |
| 4                                 | 28.17  | 13.40      | 14.77            | 218.27        | 14.77                                            | 679.60                                           | 1,964.93                                         |      |
| 5                                 | 27.07  | 8.90       | 18.17            | 330.09        | 18.17                                            | 872.08                                           | 2,289.20                                         |      |
| 6                                 | 37.85  | 20.90      | 16.95            | 287.35        | 16.95                                            | 745.86                                           | 2,457.94                                         |      |
| 7                                 | 44.72  | 28.85      | 15.87            | 251.91        | 15.87                                            | 682.48                                           | 2,507.71                                         |      |
| 8                                 | 36.42  | 19.40      | 17.02            | 289.72        | 17.02                                            | 782.98                                           | 2,434.04                                         |      |
| 9                                 | 31.21  | 17.75      | 13.46            | 181.26        | 13.46                                            | 498.14                                           | 1,857.92                                         |      |
| 10                                | 54.85  | 29.60      | 25.25            | 637.57        | 25.25                                            | 959.50                                           | 3,989.52                                         |      |
| 11                                | 39.84  | 19.85      | 19.99            | 399.47        | 19.99                                            | 859.44                                           | 2,858.12                                         |      |
| 12                                | 30.83  | 16.85      | 13.98            | 195.52        | 13.98                                            | 601.25                                           | 1,929.61                                         |      |
| Sum                               |        |            |                  | 4,035.56      | 216.48                                           | 9,128.90                                         | 3,1273.77                                        |      |
| Sum of squared errors ( $Sum/2$ ) |        |            |                  | 2,017.78      |                                                  |                                                  |                                                  |      |

- (c) Assuming a learning rate of 0.000002, calculate the weights at the next iteration of the gradient descent algorithm.

To calculate the updated weight values we apply the weight update rule for multivariate linear regression with gradient descent for each weight as fol-

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| ID | OCAHR | Prediction   |
|----|-------|--------------|
| 1  | 37.99 | 41 138 17.15 |
| 2  | 47.34 | 42 153 26.00 |
| 3  | 44.38 | 37 151 25.55 |
| 4  | 28.17 | 46 133 13.40 |
| 5  | 27.07 | 48 126 8.90  |
| 6  | 37.85 | 44 145 20.90 |
| 7  | 44.72 | 43 158 28.85 |
| 8  | 36.42 | 46 143 19.40 |
| 9  | 31.21 | 37 138 17.75 |
| 10 | 54.85 | 38 158 29.60 |
| 11 | 39.84 | 43 143 19.85 |
| 12 | 30.83 | 43 138 16.85 |

(b) 计算 (a) 部分中生成的一组预测的平方误差之和。

下表显示了模型所做的预测以及基于这些预测的平方误差计算。

| 初始权重                                         |       |            |       |          |            |            |            |         |         |
|----------------------------------------------|-------|------------|-------|----------|------------|------------|------------|---------|---------|
| 在 R0S 中: -59.50 在 R1S 中: -0.15 在 R2s 中: 0.60 |       |            |       |          |            |            |            |         |         |
| 迭代 1                                         |       |            |       |          |            |            |            |         |         |
| ID                                           | OC    | Prediction | 错误    | 错误 pD,   | 平方误差       | Delta      | 错误Delta    | 错误Delta | 错误Delta |
|                                              |       |            |       |          | pD, w r0sq | pD, w r1sq | pD, w r2sq |         |         |
| 1                                            | 37.99 | 17.15      | 20.84 | 434.41   | 20.84      | 854.54     | 2,876.26   |         |         |
| 2                                            | 47.34 | 26.00      | 21.34 | 455.41   | 21.34      | 896.29     | 3,265.05   |         |         |
| 3                                            | 44.38 | 25.55      | 18.83 | 354.60   | 18.83      | 696.74     | 2,843.45   |         |         |
| 4                                            | 28.17 | 13.40      | 14.77 | 218.27   | 14.77      | 679.60     | 1,964.93   |         |         |
| 5                                            | 27.07 | 8.90       | 18.17 | 330.09   | 18.17      | 872.08     | 2,289.20   |         |         |
| 6                                            | 37.85 | 20.90      | 16.95 | 287.35   | 16.95      | 745.86     | 2,457.94   |         |         |
| 7                                            | 44.72 | 28.85      | 15.87 | 251.91   | 15.87      | 682.48     | 2,507.71   |         |         |
| 8                                            | 36.42 | 19.40      | 17.02 | 289.72   | 17.02      | 782.98     | 2,434.04   |         |         |
| 9                                            | 31.21 | 17.75      | 13.46 | 181.26   | 13.46      | 498.14     | 1,857.92   |         |         |
| 10                                           | 54.85 | 29.60      | 25.25 | 637.57   | 25.25      | 959.50     | 3,989.52   |         |         |
| 11                                           | 39.84 | 19.85      | 19.99 | 399.47   | 19.99      | 859.44     | 2,858.12   |         |         |
| 12                                           | 30.83 | 16.85      | 13.98 | 195.52   | 13.98      | 601.25     | 1,929.61   | 和       |         |
|                                              |       |            |       | 4,035.56 | 216.48     | 9,128.90   | 3,1273.77  | 误差平方和   | (S      |

(c) 假设学习率为 0.000002, 计算梯度下降算法下一次迭代时的权重。

为了计算更新的权重值, 我们应用权重更新规则进行多元线性回归, 每个权重的梯度下降如下:

lows (using *errorDelta* values given in the answer to the previous part):

$$\begin{aligned}
 \mathbf{w}[0] &\leftarrow \mathbf{w}[0] + \alpha \times \text{errorDelta}(\mathcal{D}, \mathbf{w}[0]) \\
 &\leftarrow -59.50 + 0.000002 \times 216.48 \\
 &\leftarrow -59.4996 \\
 \mathbf{w}[1] &\leftarrow \mathbf{w}[1] + \alpha \times \text{errorDelta}(\mathcal{D}, \mathbf{w}[1]) \\
 &\leftarrow -0.15 + 0.000002 \times 9128.9 \\
 &\leftarrow -0.1317 \\
 \mathbf{w}[2] &\leftarrow \mathbf{w}[2] + \alpha \times \text{errorDelta}(\mathcal{D}, \mathbf{w}[2]) \\
 &\leftarrow 0.60 + 0.000002 \times 31273.77 \\
 &\leftarrow 0.6625
 \end{aligned}$$

- (d) Calculate the sum of squared errors for a set of predictions generated using the new set of weights calculated in Part (c).

The new sum of squared errors calculated using these new weights is given by the following table.

| ID                                     | OXYCON | Prediction | Squared |        |
|----------------------------------------|--------|------------|---------|--------|
|                                        |        |            | Error   | Error  |
| 1                                      | 37.99  | 26.53      | 11.46   | 131.38 |
| 2                                      | 47.34  | 36.34      | 11.00   | 121.07 |
| 3                                      | 44.38  | 35.67      | 8.71    | 75.87  |
| 4                                      | 28.17  | 22.56      | 5.61    | 31.53  |
| 5                                      | 27.07  | 17.66      | 9.41    | 88.56  |
| 6                                      | 37.85  | 30.77      | 7.08    | 50.10  |
| 7                                      | 44.72  | 39.52      | 5.20    | 27.08  |
| 8                                      | 36.42  | 29.18      | 7.24    | 52.37  |
| 9                                      | 31.21  | 27.06      | 4.16    | 17.27  |
| 10                                     | 54.85  | 40.18      | 14.67   | 215.31 |
| 11                                     | 39.84  | 29.58      | 10.26   | 105.21 |
| 12                                     | 30.83  | 26.27      | 4.57    | 20.84  |
| Sum                                    |        |            |         | 936.57 |
| Sum of squared errors ( <i>Sum</i> /2) |        |            |         | 468.29 |

- \* 5. In building multivariate logistic regression models, it is recommended that all continuous descriptive features be normalized to the range  $[-1, 1]$ . The following table shows a data quality report for the dataset used to train the model described in Question 3.

| Feature        | Count | %     |       | Min. | 1 <sup>st</sup> |       | Median | 3 <sup>rd</sup> |       | Std. Dev. |
|----------------|-------|-------|-------|------|-----------------|-------|--------|-----------------|-------|-----------|
|                |       | Miss. | Card. |      | Qrt.            | Mean  |        | Qrt.            | Max.  |           |
| AGE            | 5,200 | 6     | 40    | 18   | 22              | 32.7  | 32     | 32              | 63    | 12.2      |
| SHOP FREQUENCY | 5,200 | 0     | 316   | 0.2  | 1.0             | 2.2   | 1.3    | 4.3             | 5.4   | 1.6       |
| SHOP VALUE     | 5,200 | 0     | 3,730 | 5    | 11.8            | 101.9 | 100.14 | 174.6           | 230.7 | 72.1      |

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lows (使用上一部分答案中给出的 errorDelta 值) :

```

in r0s Ð in r0s ' α ^ errorDeltapD, in r0sq
'59.50' 0.000002 ^ 216.48
j59.4996
in r1s Ð in r1s ' α ^ errorDeltapD, in r1sq
电话 '0.15' 0.000002 ^ 9128.9
j0.1317
in r2s Ð in r2s ' α ^ errorDeltapD, in r2sq
电话 0.60×0.000002 ^31273.77
吨 0.6625

```

(d) 计算使用 (c) 部分计算的新权重生成的一组预测的平方误差之和。

使用这些新权重计算的新平方误差总和如下表所示。

| ID       | OC    | Prediction | 错误    | 平方     |
|----------|-------|------------|-------|--------|
| 1        | 37.99 | 26.53      | 11.46 | 131.38 |
| 2        | 47.34 | 36.34      | 11.00 | 121.07 |
| 3        | 44.38 | 35.67      | 8.71  | 75.87  |
| 4        | 28.17 | 22.56      | 5.61  | 31.53  |
| 5        | 27.07 | 17.66      | 9.41  | 88.56  |
| 6        | 37.85 | 30.77      | 7.08  | 50.10  |
| 7        | 44.72 | 39.52      | 5.20  | 27.08  |
| 8        | 36.42 | 29.18      | 7.24  | 52.37  |
| 9        | 31.21 | 27.06      | 4.16  | 17.27  |
| 10       | 54.85 | 40.18      | 14.67 | 215.31 |
| 11       | 39.84 | 29.58      | 10.26 | 105.21 |
| 12       | 30.83 | 26.27      | 4.57  | 20.84  |
| 总和       |       |            |       | 936.57 |
| 误差平方和    |       |            |       | —      |
| (Sum{2}) |       |            |       | 468.29 |

° 5. 在构建多变量逻辑回归模型时, 建议所有连续

将描述性特征显示为范围  
 练问题 3 中所述模型的数据  
 集的数据质量报告。

| 特征  | 伯爵小姐。 | 卡。 | 最小值   | Qrt. | 平均中位数 | Qrt.  | 麦克斯。   | 开发    | Std. |
|-----|-------|----|-------|------|-------|-------|--------|-------|------|
| A   | 5,200 | 6  | 40    | 18   | 22    | 32.7  | 32     | 63    | 12.2 |
| S F | 5,200 | 0  | 316   | 0.2  | 1.0   | 2.2   | 1.3    | 4.3   | 1.6  |
| S V | 5,200 | 0  | 3,730 | 5    | 11.8  | 101.9 | 100.14 | 174.6 | 72.1 |

| Feature            | Count | %<br>Miss. |   | Card. | Mode  | Mode<br>Count | Mode<br>% | 2 <sup>nd</sup><br>Mode | 2 <sup>nd</sup> | 2 <sup>nd</sup> |
|--------------------|-------|------------|---|-------|-------|---------------|-----------|-------------------------|-----------------|-----------------|
|                    |       |            |   |       |       |               |           |                         | Mode<br>Count   | Mode<br>%       |
| SOCIOECONOMIC BAND | 5,200 | 8          | 3 | a     | 2,664 | 51.2          | b         | 1,315                   | 25.3            |                 |
| REPEAT PURCHASE    | 5,200 | 0          | 2 | no    | 2,791 | 53.7          | yes       | 2,409                   | 46.3            |                 |

On the basis of the information in this report, all continuous features were normalized using **range normalization**, and any missing values were replaced using **mean imputation** for continuous features and **mode imputation** for categorical features. After applying these data preparation operations, a multivariate logistic regression model was trained to give the weights shown in the following table.

| Feature                       | Weight  |
|-------------------------------|---------|
| Intercept ( $\mathbf{w}[0]$ ) | 0.6679  |
| AGE                           | -0.5795 |
| SOCIOECONOMIC BAND B          | -0.1981 |
| SOCIOECONOMIC BAND C          | -0.2318 |
| SHOP VALUE                    | 3.4091  |
| SHOP FREQUENCY                | 2.0499  |

Use this model to make predictions for each of the query instances shown in the following table (question marks refer to missing values).

| ID | AGE | SOCIOECONOMIC<br>BAND | SHOP<br>FREQUENCY | SHOP<br>VALUE |
|----|-----|-----------------------|-------------------|---------------|
| 1  | 38  | a                     | 1.90              | 165.39        |
| 2  | 56  | b                     | 1.60              | 109.32        |
| 3  | 18  | c                     | 6.00              | 10.09         |
| 4  | ?   | b                     | 1.33              | 204.62        |
| 5  | 62  | ?                     | 0.85              | 110.50        |

Before inserting the descriptive feature values into the regression model, the same data preparation operations performed on the training data (range normalization and imputation) must be performed on each query instance.

The first step is to handle the missing values. Both the AGE and SOCIOECONOMIC BAND features have small numbers of missing values. In this case we have decided to use mean imputation for continuous features and mode imputation for categorical features. The mean for the AGE feature is 32.7, and the mode for the SOCIOECONOMIC BAND feature is *a*. So the query instances are changed to reflect this as shown in the following table.

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| 特征 | 计数    | %  | Mode | 计数    | %    | Mode | 计数    | %    |
|----|-------|----|------|-------|------|------|-------|------|
| SB | 5,200 | 83 | a    | 2,664 | 51.2 | b    | 1,315 | 25.3 |
| RP | 5,200 | 0  | no   | 2,791 | 53.7 | yes  | 2,409 | 46.3 |

根据本报告中的信息，使用范围归一化对所有连续特征进行归一化，并使用连续特征的均值插补和分类特征的众数插补替换任何缺失值。应用这些数据准备操作后，训练了一个多变量逻辑回归模型，以给出下表中所示的权重。

| 特征           | 重量      |
|--------------|---------|
| 截距 (w r0s)   | 0.6679  |
| A            | -0.5795 |
| SBB -0.1981年 |         |
| SBC -0.2318  |         |
| SV           | 3.4091  |
| SF           | 2.0499  |

使用此模型对下表中显示的每个查询实例进行预测（问号表示缺失值）。

| ID | A  | S | B | F | S    | V      |
|----|----|---|---|---|------|--------|
| 1  | 38 |   | a |   | 1.90 | 165.39 |
| 2  | 56 |   | b |   | 1.60 | 109.32 |
| 3  | 18 |   | c |   | 6.00 | 10.09  |
| 4  | ?  |   | b |   | 1.33 | 204.62 |
| 5  | 62 |   | ? |   | 0.85 | 110.50 |

在将描述性特征值插入回归模型之前，必须对每个查询实例执行对训练数据执行的相同数据准备操作（范围归一化和插补）。

第一步是处理缺失值。A和SNOMIC Bfeatures都有少量的缺失值。在这种情况下，我们决定对连续特征使用均值插补，对分类特征使用众数插补。Afeature 的均值为 32.7，SBfeature 的众数为 a。因此，查询实例将进行更改以反映这一点，如下表所示。

| ID | SOCIOECONOMIC |      | SHOP      | SHOP   |
|----|---------------|------|-----------|--------|
|    | AGE           | BAND | FREQUENCY | VALUE  |
| 1  | 38            | a    | 1.90      | 165.39 |
| 2  | 56            | b    | 1.60      | 109.32 |
| 3  | 18            | c    | 6.00      | 10.09  |
| 4  | 32.7          | b    | 1.33      | 204.62 |
| 5  | 62            | a    | 0.85      | 110.50 |

Each continuous descriptive feature value must then be normalized using the same normalization parameters that were used with the training data. Range normalization was used in this case, so the required minimum and maximum parameters can be obtained from the data quality report. As a reminder, the normalization of the AGE feature for the first instance is calculated as

$$\begin{aligned}
 AGE_1' &= \frac{AGE_1 - \min(AGE)}{\max(AGE) - \min(AGE)} \times (high - low) + low \\
 &= \frac{38 - 18}{63 - 18} \times (1 - (-1)) + (-1) \\
 &= -0.111
 \end{aligned}$$

These normalized parameters are shown in the following table.

| ID | SOCIOECONOMIC |      | SHOP      | SHOP   |
|----|---------------|------|-----------|--------|
|    | AGE           | BAND | FREQUENCY | VALUE  |
| 1  | -0.111        | a    | -0.346    | 0.421  |
| 2  | 0.689         | b    | -0.462    | -0.076 |
| 3  | -1.000        | c    | 1.230     | -0.955 |
| 4  | -0.347        | b    | -0.565    | 0.769  |
| 5  | 0.956         | a    | -0.750    | -0.065 |

One issue that must be considered in this case is what to do with values that are outside the range of the training data. In this case the SHOW FREQUENCY value for query instance 3 is 6, which is higher than anything that was seen in the training data. When range normalization is applied to this value, it is calculated as 1.23. This can be handled by the regression equation, but many modelers will choose to clamp this at a maximum value of 1.0 (similarly, a value below the minimum seen would be clamped at  $-1.0$ ). We choose to clamp at 1.0 in this case.

With this information, we can calculate the predictions by plugging these values into the regression equation as follows:

$$\begin{aligned}
 \mathbf{1:} \quad & Logistic(0.6679 - 0.5795 \times -0.111 - 0.1981 \times 0 - 0.2318 \times 0 + 2.0499 \times \\
 & -0.346 + 3.4091 \times 0.421)
 \end{aligned}$$

## 第7章 基于错误的学习（练习解决方案）

|    | S    |   |   |   | S    | S      |
|----|------|---|---|---|------|--------|
| 同上 | A    | B | F | V |      |        |
| 1  | 38   |   | a |   | 1.90 | 165.39 |
| 2  | 56   |   | b |   | 1.60 | 109.32 |
| 3  | 18   |   | c |   | 6.00 | 10.09  |
| 4  | 32.7 |   | b |   | 1.33 | 204.62 |
| 5  | 62   |   | a |   | 0.85 | 110.50 |

然后，必须使用与训练数据相同的归一化参数对每个连续描述性特征值进行归一化。在这种情况下使用了范围归一化，因此可以从数据质量报告中获得所需的最小和最大参数。提醒一下，第一个实例的 Afeature 规范化计算如下：

$$A = \frac{A - \min A_q}{\max A_q - \min A_q} \cdot \text{high}_q - \text{low}_q$$

$$= \frac{38 - 18}{63 - 18} \cdot 1 - 0.111$$

下表显示了这些规范化参数。

|    | S      |   |   |   | S      | S      |
|----|--------|---|---|---|--------|--------|
| 同上 | A      | B | F | V |        |        |
| 1  | -0.111 |   | a |   | -0.346 | 0.421  |
| 2  | 0.689  |   | b |   | -0.462 | -0.076 |
| 3  | -1.000 |   | c |   | 1.230  | -0.955 |
| 4  | -0.347 |   | b |   | -0.565 | 0.769  |
| 5  | 0.956  |   | a |   | -0.750 | -0.065 |

在这种情况下，必须考虑的一个问题是如何处理训练数据范围之外的值。在本例中，查询实例 3 的 SF 值为 6，高于训练数据中显示的任何值。当范围归一化应用于此值时，它的计算结果为 1.23。这可以通过回归方程来处理，但许多建模者会选择将其限制在最大值 1.0（类似地，低于最小值的值将被限制在 1.0）。在这种情况下，我们选择在 1.0 处钳制。

有了这些信息，我们可以通过将这些值代入回归方程来计算预测，如下所示：

$$1: \text{物流} p0.6679 + 0.5795 \cdot (-0.111) + 0.1981 \cdot 0 + 0.2318 \cdot 0 + 0.0499 \cdot (-0.346) + 0.4091 \cdot 0.421q$$

$$= \text{Logistic}(1.46) = \frac{1}{1+e^{-1.46}}$$

$$= 0.811 \Rightarrow \text{yes}$$

$$\mathbf{2:} \text{Logistic}(0.6679 - 0.5795 \times 0.6889 - 0.1981 \times 1 - 0.2318 \times 0 + 2.0499 \times -0.4615 + 3.4091 \times -0.0756)$$

$$= \text{Logistic}(-1.1332) = \frac{1}{1+e^{1.1332}}$$

$$= 0.2436 \Rightarrow \text{no}$$

$$\mathbf{3:} \text{Logistic}(0.6679 - 0.5795 \times -1 - 0.1981 \times 0 - 0.2318 \times 1 + 2.0499 \times 1.2308 + 3.4091 \times -0.9549)$$

$$= \text{Logistic}(0.2832) = \frac{1}{1+e^{-0.2832}}$$

$$= 0.5703 \Rightarrow \text{yes}$$

$$\mathbf{4:} \text{Logistic}(0.6679 - 0.5795 \times -0.3467 - 0.1981 \times 1 - 0.2318 \times 0 + 2.0499 \times -0.5654 + 3.4091 \times 0.7689)$$

$$= \text{Logistic}(2.1330) = \frac{1}{1+e^{-2.1330}}$$

$$= 0.8940 \Rightarrow \text{yes}$$

$$\mathbf{5:} \text{Logistic}(0.6679 - 0.5795 \times 0.9556 - 0.1981 \times 0 - 0.2318 \times 0 + 2.0499 \times -0.7500 + 3.4091 \times -0.0651)$$

$$= \text{Logistic}(-1.6453) = \frac{1}{1+e^{1.6453}}$$

$$= 0.1617 \Rightarrow \text{no}$$

One interesting thing to note about these calculations is that the second query instance in this question is the same as the first query instance in the previous question. Note that the resulting answers are almost identical. This is reassuring, as simple data preparation operations like those applied in this example should not have a significant impact on the resulting model.

- \* 8. A support vector machine has been built to predict whether a patient is at risk of cardiovascular disease. In the dataset used to train the model, there are two target levels—*high risk* (the positive level, +1) or *low risk* (the negative level, −1)—and three descriptive features—AGE, BMI, and BLOOD PRESSURE. The support vectors in the trained model are shown in the table below (all descriptive feature values have been standardized).

## 第7章 基于错误的学习 (练习解决方案)

```

„物流 p1. 46
“ 0.811 ñ yes q "1"e
2: 物流 p0.6679 , 0.5795 , 0.6889 , 0.1981 ^ 1 , 0.2318 ^ 0 , 0.0499 ^
0.4615 , 0.4091 ^ 0.0756
“ 物流 p' 1.1332 , 0.1332 q "1"e
“ 2436 ñ no q "1"e
3: 物流 p0.6679' 0.5795 ^ 1' 0.1981^0'0.2318^1'2.0499^1.2308'
3.4091 , 0.9549
“ 物流 p0.2832 q "1"e
“ 5703 ñ yes q "1"e
4: 物流 p0.6679 , 0.5795 , 0.3467 , 0.1981 ^ 1 , 0.2318 ^ 0 , 0.0499 ^
0.5654 , 0.4091 ^ 0.7689
“ 物流 P2.1330 , 0.1330 q "1"e
“ 8940 ñ yes q "1"e
5: 物流 p0.6679 , 0.5795 , 0.9556 , 0.1981 ^ 0 , 0.2318 ^ 0 , 0.0499 ^
0.7500 , 0.4091 ^ 0.0651
“ 物流 p' 1.6453 , 0.6453 q "1"e
“ 1617 ñ no q "1"e

```

关于这些计算，需要注意的一件有趣的事情是，此问题中的第二个查询实例与上一个问题中的第一个查询实例相同。请注意，生成的答案几乎相同。这是令人放心的，因为本示例中应用的简单数据准备操作不会对生成的模型产生重大影响。

### ° 8. 已经建立了一个支持向量机来预测患者是否有风险

心血管疾病。在用于训练模型的数据集中，有两个目标级别（高风险（正水平，'1）或低风险（负水平，'1））和三个描述性特征（A、BMI 和 BP）。训练模型中的支持向量如下表所示（所有描述性特征值均已标准化）。

| AGE     | BMI     | BLOOD    |           |
|---------|---------|----------|-----------|
|         |         | PRESSURE | RISK      |
| -0.4549 | 0.0095  | 0.2203   | low risk  |
| -0.2843 | -0.5253 | 0.3668   | low risk  |
| 0.3729  | 0.0904  | -1.0836  | high risk |
| 0.558   | 0.2217  | 0.2115   | high risk |

In the model the value of  $w_0$  is  $-0.0216$ , and the values of the  $\alpha$  parameters are  $\langle 1.6811, 0.2384, 0.2055, 1.7139 \rangle$ . What predictions would this model make for the following query instances?

| ID | AGE     | BMI     | BLOOD    |      |
|----|---------|---------|----------|------|
|    |         |         | PRESSURE | RISK |
| 1  | -0.8945 | -0.3459 | 0.5520   |      |
| 2  | 0.4571  | 0.4932  | -0.4768  |      |
| 3  | -0.3825 | -0.6653 | 0.2855   |      |
| 4  | 0.7458  | 0.1253  | -0.7986  |      |

For the first query instance,  $\langle -0.8945, -0.3459, 0.552 \rangle$ , the output of the support vector machine model is

$$\begin{aligned}
& \mathbb{M}_{\alpha, -0.0216}(\langle -0.8945, -0.3459, 0.552 \rangle) \\
&= (-1 \times 1.6811 \times (\langle -0.4549, 0.0095, 0.2203 \rangle \cdot \langle -0.8945, -0.3459, 0.552 \rangle) - 0.0216) \\
&\quad + (-1 \times 0.2384 \times (\langle -0.2843, -0.5253, 0.3668 \rangle \cdot \langle -0.8945, -0.3459, 0.552 \rangle) - 0.0216) \\
&\quad + (-1 \times 0.2055 \times (\langle 0.3729, 0.0904, -1.0836 \rangle \cdot \langle -0.8945, -0.3459, 0.552 \rangle) - 0.0216) \\
&\quad + (-1 \times 1.7139 \times (\langle 0.558, 0.2217, 0.2115 \rangle \cdot \langle -0.8945, -0.3459, 0.552 \rangle) - 0.0216) \\
&= -0.9045 - 0.1737 - 0.2195 - 0.8083 \\
&= -2.1061
\end{aligned}$$

Because this output is less than  $-1$ , the model predicts the negative level—*low risk*.

For the second query instance,  $\langle 0.4571, 0.4932, -0.4768 \rangle$ , the output of the sup-

## 第 7 章 基于错误的学习 (练习解决方案)

|         | B       |         |     |
|---------|---------|---------|-----|
| ABMI 公关 |         |         |     |
| -0.4549 | 0.0095  | 0.2203  | 低风险 |
| -0.2843 | -0.5253 | 0.3668  | 低风险 |
| 0.3729  | 0.0904  | -1.0836 | 高危  |
| 0.558   | 0.2217  | 0.2115  | 高危  |

在模型中,  $w$  0.0216 的值和  $\alpha$  参数的值分别为 (1.6811、0.2384、0.2055、1.7139)。此模型将对以下查询实例进行哪些预测?

|            | B       |         |         |
|------------|---------|---------|---------|
| 编号: ABMI P |         |         |         |
| 1          | -0.8945 | -0.3459 | 0.5520  |
| 2          | 0.4571  | 0.4932  | -0.4768 |
| 3          | -0.3825 | -0.6653 | 0.2855  |
| 4          | 0.7458  | 0.1253  | -0.7986 |

对于第一个查询实例 (0.8945, 0.3459, 0.552), 支持向量机模型的输出为

$M_p(0.8945, 0.3459, 0.552) q$

$P^1 \wedge 1.6811 \wedge P(0.4549, 0.0095, 0.2203) \wedge (0.8945, 0.3459, 0.552) q' 0.0216 q$   
 $P^1 \wedge 0.2384 \wedge p(0.2843, 0.5253, 0.3668) \wedge (0.8945, 0.3459, 0.552) q' 0.0216 q$   
 $P^1 \wedge 0.2055 \wedge p(0.3729, 0.0904, 1.0836) \wedge (0.8945, 0.3459, 0.552) q' 0.0216 q$   
 $P^1 \wedge 1.7139 \wedge p(0.558, 0.2217, 0.2115) \wedge (0.8945, 0.3459, 0.552) q' 0.0216 q$   
 $0.9045 \wedge 0.1737 \wedge 0.2195 \wedge 0.8083$   
 $2.1061$

由于此输出小于 1, 因此模型预测负水平 - 低风险。

对于第二个查询实例, (0.4571, 0.4932, 0.4768), sup-

port vector machine model is

$$\begin{aligned}
 & \mathbb{M}_{\mathbf{a}, -0.0216}(\langle 0.4571, 0.4932, -0.4768 \rangle) \\
 &= (-1 \times 1.6811 \times (\langle -0.4549, 0.0095, 0.2203 \rangle \cdot \langle 0.4571, 0.4932, -0.4768 \rangle) - 0.0216) \\
 &\quad + (-1 \times 0.2384 \times (\langle -0.2843, -0.5253, 0.3668 \rangle \cdot \langle 0.4571, 0.4932, -0.4768 \rangle) - 0.0216) \\
 &\quad + (-1 \times 0.2055 \times (\langle 0.3729, 0.0904, -1.0836 \rangle \cdot \langle 0.4571, 0.4932, -0.4768 \rangle) - 0.0216) \\
 &\quad + (-1 \times 1.7139 \times (\langle 0.558, 0.2217, 0.2115 \rangle \cdot \langle 0.4571, 0.4932, -0.4768 \rangle) - 0.0216) \\
 &= 0.4967 + 0.1129 + 0.1288 + 0.4302 \\
 &= 1.1686
 \end{aligned}$$

Because this output is greater than 1, the model predicts the positive level—*high risk*.

For the third query instance,  $\langle -0.3825, -0.6653, 0.2855 \rangle$ , the output of the support vector machine model is

$$\begin{aligned}
 & \mathbb{M}_{\mathbf{a}, -0.0216}(\langle -0.3825, -0.6653, 0.2855 \rangle) \\
 &= (-1 \times 1.6811 \times (\langle -0.4549, 0.0095, 0.2203 \rangle \cdot \langle -0.3825, -0.6653, 0.2855 \rangle) - 0.0216) \\
 &\quad + (-1 \times 0.2384 \times (\langle -0.2843, -0.5253, 0.3668 \rangle \cdot \langle -0.3825, -0.6653, 0.2855 \rangle) - 0.0216) \\
 &\quad + (-1 \times 0.2055 \times (\langle 0.3729, 0.0904, -1.0836 \rangle \cdot \langle -0.3825, -0.6653, 0.2855 \rangle) - 0.0216) \\
 &\quad + (-1 \times 1.7139 \times (\langle 0.558, 0.2217, 0.2115 \rangle \cdot \langle -0.3825, -0.6653, 0.2855 \rangle) - 0.0216) \\
 &= -0.4092 + -0.1557 + -0.1268 + -0.5367 \\
 &= -1.2284
 \end{aligned}$$

Because this output is less than  $-1$ , the model predicts the negative level—*low risk*.

For the final query instance,  $\langle 0.7458, 0.1253, -0.7986 \rangle$ , the output of the support

## 第 7 章 基于错误的学习 (练习解决方案)

支持向量机模型的输出为

MP <0.4571, 0.4932, '0.4768> q

" P'1 ^ 1.6811 ^ P <'0.4549, 0.0095, 0.2203> " <0.4571, 0.4932, '0.4768> q ' 0.0216q ' p'1 ^ 0.2384 ^ p <'0.2843, '0.5253, 0.3668> "

<0.4571, 0.4932, '0.4768> q ' 0.0216q ' p'1 ^ 0.2055 ^ p <0.3729, 0.0904, '1.0836> " <0.4571, 0.4932, '0.4768> q ' 0.0216q ' p'1 ^ 1.7139 ^ p <0.558, 0.2217, 0.2115> " <0.4571, 0.4932, '0.4768> q ' 0.0216q "0.4967 ' 0.1129 ' 0.1288 ' 0.4302

"1.1686

由于此输出大于 1，因此模型预测正水平 - 高风险。

对于第三个查询实例 <'0.3825, '0.6653, 0.2855>，支持向量机模型的输出为

Mp <'0.3825, '0.6653, 0.2855> q " p'1 ^ 1.6811 ^ p <'0.4549, 0.0095, 0.2203> " <'0.3825, '0.6653, 0.2855> q ' 0.0216q

' P'1 ^ 0.2384 ^ P <'0.2843, '0.5253, 0.3668> " <'0.3825, '0.6653, 0.2855> q ' 0.0216q ' p'1 ^ 0.2055 ^ p <0.3729, 0.0904, '1.0836> "

<'0.3825, '0.6653, 0.2855> q ' 0.0216q ' p'1 ^ 1.7139 ^ p <0.558, 0.2217, 0.2115> " <'0.3825, '0.6653, 0.2855> q ' 0.0216q " ' 0.4092 ' '0.1557 ' '0.1268 ' '0.5367 " ' 1.2284

由于此输出小于 '1，因此模型预测负水平 - 低风险。

对于最终的查询实例，<0.7458, 0.1253, '0.7986>，支持的输出

vector machine model is

$$\begin{aligned}
 & \mathbb{M}_{\alpha, -0.0216}(\langle 0.7458, 0.1253, -0.7986 \rangle) \\
 &= (-1 \times 1.6811 \times (\langle -0.4549, 0.0095, 0.2203 \rangle \cdot \langle 0.7458, 0.1253, -0.7986 \rangle) - 0.0216) \\
 &\quad + (-1 \times 0.2384 \times (\langle -0.2843, -0.5253, 0.3668 \rangle \cdot \langle 0.7458, 0.1253, -0.7986 \rangle) - 0.0216) \\
 &\quad + (-1 \times 0.2055 \times (\langle 0.3729, 0.0904, -1.0836 \rangle \cdot \langle 0.7458, 0.1253, -0.7986 \rangle) - 0.0216) \\
 &\quad + (-1 \times 1.7139 \times (\langle 0.558, 0.2217, 0.2115 \rangle \cdot \langle 0.7458, 0.1253, -0.7986 \rangle) - 0.0216) \\
 &= 0.8425 + 0.1145 + 0.2158 + 0.4498 \\
 &= 1.6227
 \end{aligned}$$

Because this output is greater than 1, the model predicts the positive level—*high risk*.

## 第 7 章 基于错误的学习 (练习解决方案)

向量机模型是

$Mp \langle 0.7458, 0.1253, '0.7986 \rangle q$

$" P^1 \wedge 1.6811 \wedge P \langle '0.4549, 0.0095, 0.2203 \rangle \neg \langle 0.7458, 0.1253,$

$'0.7986 \rangle q \wedge '0.0216q \wedge p^1 \wedge 0.2384 \wedge p \langle '0.2843, '0.5253, 0.3668 \rangle \neg$

$\langle 0.7458, 0.1253, '0.7986 \rangle q \wedge '0.0216q$

$p^1 \wedge 0.2055 \wedge p \langle 0.3729, 0.0904, '1.0836 \rangle \neg \langle 0.7458, 0.1253, '0.7986 \rangle q \wedge '0.0216q$

$\wedge p^1 \wedge 1.7139 \wedge p \langle 0.558, 0.2217, 0.2115 \rangle \neg \langle 0.7458, 0.1253,$

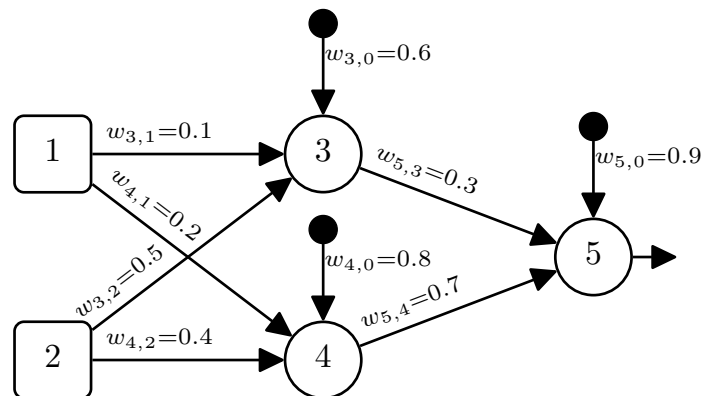
$'0.7986 \rangle q \wedge '0.0216q \quad "0.8425 \wedge 0.1145 \wedge 0.2158 \wedge 0.4498$

$"1.6227$

由于此输出大于 1, 因此模型预测正水平 - 高风险。

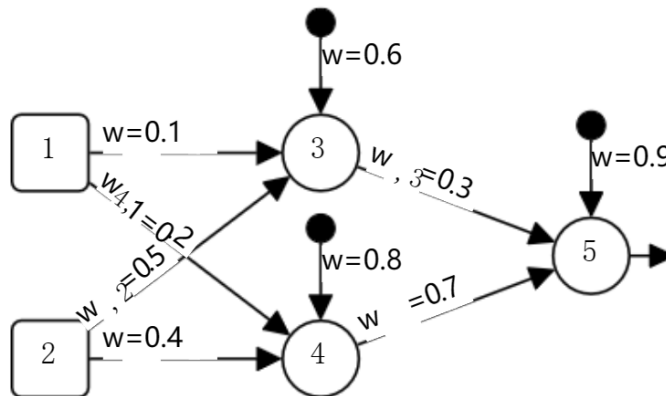
## Chapter 8

2. The following image shows an artificial neural network with two sensing neurons (Neurons 1 and 2) and 3 processing neurons (Neurons 3, 4, and 5)



&KDSWHU8

2. 下图显示了一个人工神经网络，其中包含两个传感神经元（神经元 1 和 2）和 3 个处理神经元（神经元 3、4 和 5）



- (a) Assuming that the processing neurons in this network use a **logistic** activation function, what would be the output of Neuron 5 if the network received the input vector: Neuron 1 = 0.7 and Neuron 2 = 0.3?

The weighted sum calculation for Neuron 3 is:

$$\begin{aligned} z_3 &= (0.6 \times 1) + (0.1 \times 0.7) + (0.5 \times 0.3) \\ &= (0.6) + (0.07) + (0.15) \\ &= 0.82 \end{aligned}$$

The weighted sum calculation for Neuron 4 is:

$$\begin{aligned} z_4 &= (0.8 \times 1) + (0.2 \times 0.7) + (0.4 \times 0.3) \\ &= (0.8) + (0.14) + (0.12) \\ &= 1.06 \end{aligned}$$

Using a logistic activation function the activation for Neuron 3 is:

$$\begin{aligned} a_3 &= \text{logistic}(z_3) \\ &= \frac{1}{1 + e^{-0.82}} \\ &= 0.69423634 \end{aligned}$$

Using a logistic activation function the activation for Neuron 4 is:

$$\begin{aligned} a_4 &= \text{logistic}(z_4) \\ &= \frac{1}{1 + e^{-1.06}} \\ &= 0.742690545 \end{aligned}$$

The weighted sum calculation for Neuron 5 can now be calculated as:

$$\begin{aligned} z_5 &= (0.9 \times 1) + (0.3 \times 0.69423634) + (0.7 \times 0.742690545) \\ &= (0.9) + (0.208270902) + (0.519883382) \\ &= 1.628154284 \end{aligned}$$

And the final output of the network can now be calculated by passing  $z_5$  through the activation function for Neuron 5, which for this question is the

## 第 8 章 深度学习 (练习解决方案)

(a) 假设该网络中的处理神经元使用逻辑激活函数, 如果网络接收到输入向量: 神经元 1 = 0.7, 神经元 2 = 0.3, 神经元 5 的输出是什么?

Neuron 3 的加权和计算为:

$$\begin{aligned} Z &= 0.6 \cdot 1 + 0.1 \cdot 0.7 + 0.5 \cdot 0.3 \\ &= 0.6 + 0.07 + 0.15 \\ &= 0.82 \end{aligned}$$

Neuron 4 的加权和计算为:

$$\begin{aligned} Z &= 0.8 \cdot 1 + 0.2 \cdot 0.7 + 0.4 \cdot 0.3 \\ &= 0.8 + 0.14 + 0.12 \\ &= 1.06 \end{aligned}$$

使用逻辑激活函数, Neuron 3 的激活为:

$$\begin{aligned} a &= \text{物流} \cdot z \\ &= \frac{1}{1 + e^{-z}} \\ &= 0.69423634 \end{aligned}$$

使用逻辑激活函数, Neuron 4 的激活为:

$$\begin{aligned} a &= \text{物流} \cdot z \\ &= \frac{1}{1 + e^{-z}} \\ &= 0.742690545 \end{aligned}$$

Neuron 5 的加权和计算现在可以计算为:

$$\begin{aligned} z &= 0.9 \cdot 1 + 0.3 \cdot 0.69423634 + 0.7 \cdot 0.742690545 \\ &= 0.9 + 0.208270902 + 0.519883382 \\ &= 1.628154284 \end{aligned}$$

现在可以通过将  $z$  传递给 Neuron 5 的激活函数来计算网络的最终输出, 对于这个问题, 它是

logistic function:

$$\begin{aligned} a_5 &= \text{logistic}(z_5) \\ &= \frac{1}{1 + e^{-1.628154284}} \\ &= 0.835916637 \end{aligned}$$

- (b) Assuming that the processing neurons in this network use a **ReLU** activation function, what would be the output of Neuron 5 if the network received the input vector: Neuron 1 = 0.7 and Neuron 2 = 0.3?

From the solutions to part one of this question the weighted sum calculations for Neurons 3 and 4 are as follows:

$$z_3 = 0.82$$

$$z_4 = 1.06$$

Both of these values are greater than 0 and so using the rectifier function ( $\max(0, z)$ ) as an activation function the activations for each of these neurons is:

$$a_3 = 0.82$$

$$a_4 = 1.06$$

The weighted sum calculation for Neuron 5 can now be calculated as:

$$\begin{aligned} z_5 &= (0.9 \times 1) + (0.3 \times 0.82) + (0.7 \times 1.06) \\ &= (0.9) + (0.246) + (0.742) \\ &= 1.888 \end{aligned}$$

$z_5$  is greater than 0 and so passing this value through the rectifier activation function ( $\max(0, z)$ ) does not change it, and so the activation for Neuron 5 (and the output of the network) is:

$$a_5 = 1.888$$

- (c) The following image provides a template diagram for the sequence of matrix operations that our neural network would use to process the input vector Neuron 1 =

## 第 8 章 深度学习 (练习解决方案)

物流功能:

```
a " 物流pzq
    " 1 / (1 + exp(-z))
    " 0.835916637
```

(b) 假设该网络中的处理神经元使用 ReLU 激活函数, 如果网络接收到输入向量: 神经元 1 = 0.7, 神经元 2 = 0.3, 神经元 5 的输出是什么?

根据该问题第一部分的答案, 神经元 3 和 4 的加权和计算如下:

```
z " 0.82
的 " 1.06
```

这两个值都大于 0, 因此使用整流器函数 ( $\max(0, z)$ ) 作为激活函数, 每个神经元的激活为:

```
一个 " 0.82
一个 " 1.06
```

Neuron 5 的加权和计算现在可以计算为:

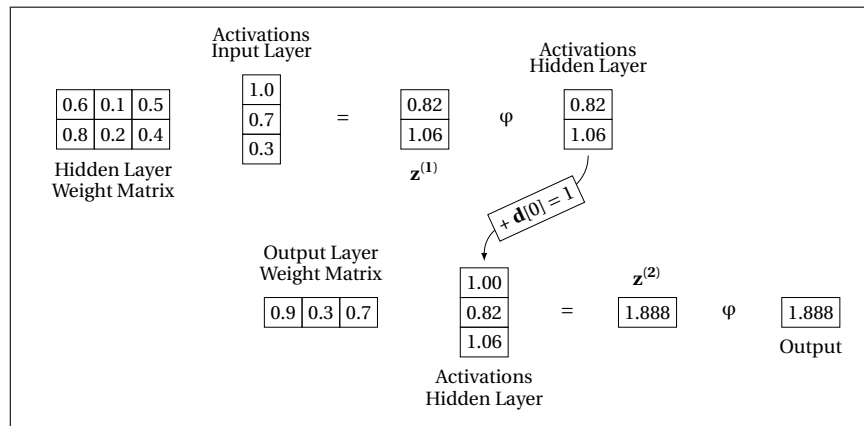
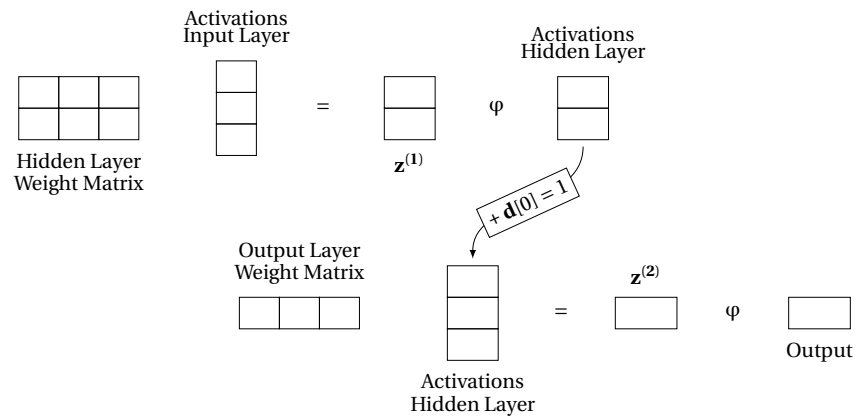
```
Z " P0.9 1Q ' P0.3 0.82Q ' P0.7 1.06Q
    " P0.9Q ' P0.246Q ' P0.742Q
    " 1.888
```

zis 大于 0, 因此通过宣叙调器激活函数 ( $\max(0, z)$ ) 传递此值不会改变它, 因此 Neuron 5 (和网络输出) 的激活为:

```
一个 " 1.888
```

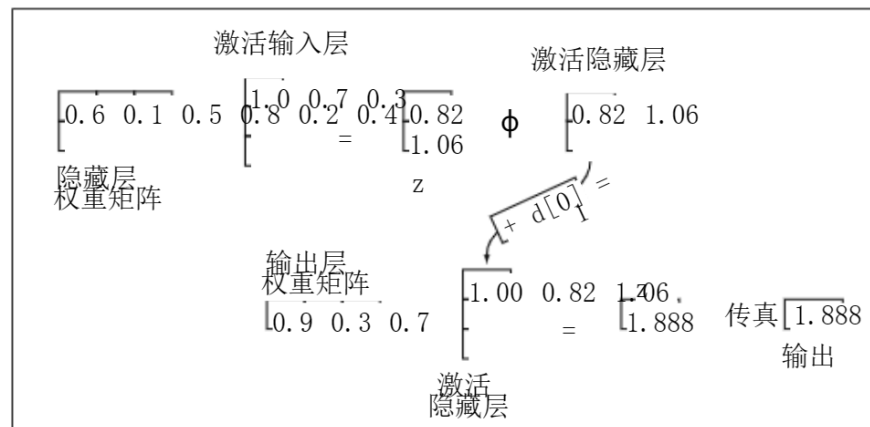
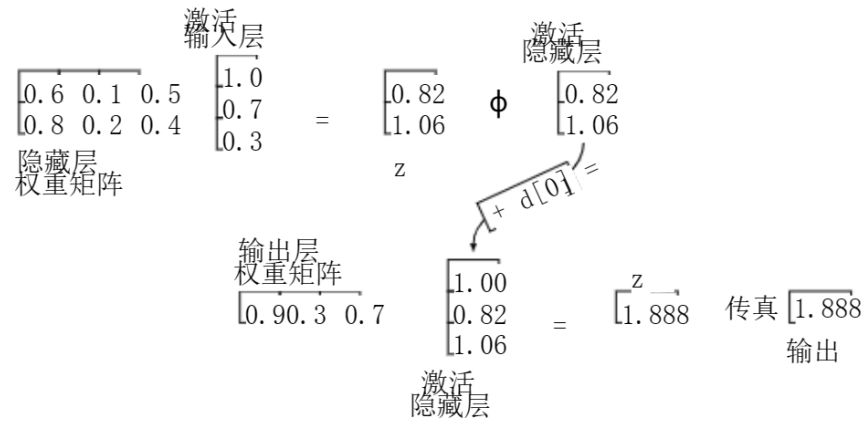
(c) 下图提供了矩阵运算序列的模板图, 我们的神经网络将使用这些矩阵运算来处理输入向量 Neuron 1 =

0.7 and Neuron 2 = 0.3. Assuming that the processing neurons in the network use a ReLU activation function, fill in the diagram with the appropriate weights, bias terms, weighted sum values, and activations.

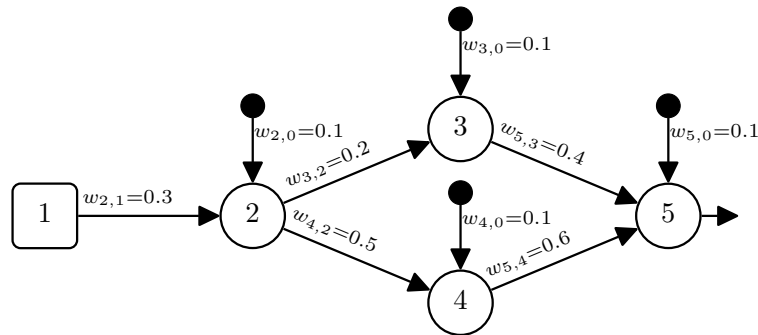


## 第 8 章 深度学习 (练习解决方案)

0.7 和神经元 2 = 0.3。假设网络中的处理神经元使用 ReLU 激活函数，则在图中填写适当的权重、偏差项、加权总和值和激活。



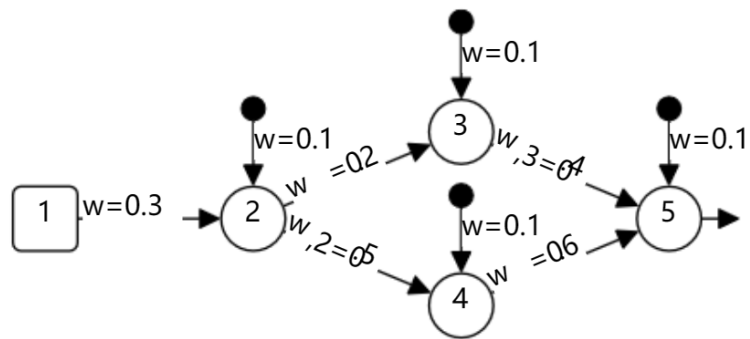
5. The following image illustrates the topology of a simple feedforward neural network that has a single sensing neuron (Neuron 1), three hidden processing neuron (Neurons 2, 3, and 4), and a single processing output neuron (Neuron 5).



- (a) Assuming that the processing neurons use **logistic** activation functions, that the input to the network is Neuron 1 = 0.5 and that the desired output for this input is 0.9:
- Calculate the output generated by the network in response to this input.

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5. 下图说明了一个简单的前馈神经网络的拓扑结构，该神经网络具有单个感知神经元（神经元 1）、三个隐藏处理神经元（神经元 2、3 和 4）和一个处理输出神经元（神经元 5）。



(a) 假设处理神经元使用逻辑激活函数，则网络的输入为 Neuron 1 = 0.5，并且该输入的期望输出为 0.9:

i. 计算网络响应此输入生成的输出。

$$\begin{aligned}
 z_2 &= ((w_{2,0} \times 1) + (w_{2,1} \times 0.5)) \\
 &= ((0.1 \times 1) + (0.3 \times 0.5)) \\
 &= 0.25
 \end{aligned}$$

$$\begin{aligned}
 a_2 &= \text{logistic}(0.25) \\
 &= 0.562176501
 \end{aligned}$$

$$\begin{aligned}
 z_3 &= ((w_{3,0} \times 1) + (w_{3,2} \times a_2)) \\
 &= ((0.1 \times 1) + (0.2 \times 0.562176501)) \\
 &= 0.2124353
 \end{aligned}$$

$$\begin{aligned}
 a_3 &= \text{logistic}(0.2124353) \\
 &= 0.552909994
 \end{aligned}$$

$$\begin{aligned}
 z_4 &= ((w_{4,0} \times 1) + (w_{4,2} \times a_2)) \\
 &= ((0.1 \times 1) + (0.5 \times 0.562176501)) \\
 &= 0.38108825
 \end{aligned}$$

$$\begin{aligned}
 a_4 &= \text{logistic}(0.38108825) \\
 &= 0.594135549
 \end{aligned}$$

$$\begin{aligned}
 z_5 &= ((w_{5,0} \times 1) + (w_{5,3} \times a_3) + (w_{5,4} \times a_4)) \\
 &= ((0.1 \times 1) + (0.4 \times 0.552909994) + (0.6 \times 0.594135549)) \\
 &= 0.677645327
 \end{aligned}$$

$$\begin{aligned}
 a_5 &= \text{logistic}(0.677645327) \\
 &= 0.663212956
 \end{aligned}$$

- ii. Calculate the  $\delta$  values for each of the processing neurons in the network (i.e.,  $\delta_5, \delta_4, \delta_3, \delta_2$ ).

## 第 8 章 深度学习 (练习解决方案)

```
Z " PPW^ 1Q ' PW^ 0.5QQ
    " pp0.1 ^ 1q ' p0.3 ^ 0.5qq
    " 0.25
```

```
A " 物流P0.25Q
    " 0.562176501
```

```
Z " PPW^ 1Q ' PW^ AQQ " PP0.1
    ^ 1Q ' P0.2 ^ 0.562176501QQ
    " 0.2124353
```

```
A " 物流P0.2124353Q
    " 0.552909994
```

```
Z " PPW^ 1Q ' PW^ AQQ
    " pp0.1 ^ 1q ' p0.5 ^ 0.552909994qq
    " 0.38108825
```

```
A " 物流P0.38108825Q
    " 0.594135549
```

```
1Q ^
    " pp0.1 ^ 1q ' p0.4 ^ 0.552909994q ' p0.6 ^ 0.594135549qq
    " 0.677645327
```

```
A " 物流P0.677645327Q
    " 0.663212956
```

ii. 计算网络中每个处理神经元 (即  $\delta$ 、 $\delta$ 、 $\delta$ 、 $\delta$ ) 的  $\delta$  值。

Calculations for  $\delta_5$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_5} &= -(t_k - a_k) \\ &= -(0.9 - 0.663212956) \\ &= -0.236787044\end{aligned}$$

$$\begin{aligned}\frac{\partial a_5}{\partial z_5} &= \text{logistic}(z_5) \times (1 - \text{logistic}(z_5)) \\ &= 0.677645327 \times (1 - 0.677645327) \\ &= 0.223361531\end{aligned}$$

$$\begin{aligned}\delta_3 &= \frac{\partial \mathcal{E}}{\partial a_5} \times \frac{\partial a_5}{\partial z_5} \\ &= -0.236787044 \times 0.223361531 \\ &= -0.052889117\end{aligned}$$

Calculations for  $\delta_4$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_4} &= w_{5,4} \times \delta_5 \\ &= 0.6 \times -0.052889117 \\ &= -0.03173347\end{aligned}$$

$$\begin{aligned}\frac{\partial a_4}{\partial z_4} &= \text{logistic}(z_4) \times (1 - \text{logistic}(z_4)) \\ &= 0.594135549 \times (1 - 0.594135549) \\ &= 0.241138498\end{aligned}$$

$$\begin{aligned}\delta_4 &= \frac{\partial \mathcal{E}}{\partial a_4} \times \frac{\partial a_4}{\partial z_4} \\ &= -0.03173347 \times 0.241138498 \\ &= -0.007652161\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

δ计算:

```
BE
Ba  " ' pt' aq
      "'P0.9'0.663212956Q
      " '0.236787044
```

```
Ba  " logisticpzq ^ p1 ' logisticpzq
Bz
      " 0.677645327 ^ p1 '
      0.677645327q " 0.223361531
```

```
δ " BE ^ Ba
    Ba  Bz
      " '0.236787044 ^ 0.223361531
      " '0.052889117
```

δ计算:

```
BE
Ba  " w^ d
      " 0.6 ^ '0.052889117
      " '0.03173347
```

```
Ba  " logisticpzq ^ p1 ' logisticpzq
Bz
      " 0.594135549 ^ p1 ' 0.594135549q
      " 0.241138498
```

```
δ " BE ^ Ba
    Ba  Bz
      " '0.03173347 ^ 0.241138498
      " '0.007652161
```

Calculations for  $\delta_3$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_3} &= w_{5,3} \times \delta_5 \\ &= 0.4 \times -0.052889117 \\ &= -0.021155647\end{aligned}$$

$$\begin{aligned}\frac{\partial a_3}{\partial z_3} &= \text{logistic}(z_3) \times (1 - \text{logistic}(z_3)) \\ &= 0.552909994 \times (1 - 0.552909994) \\ &= 0.247200532\end{aligned}$$

$$\begin{aligned}\delta_3 &= \frac{\partial \mathcal{E}}{\partial a_3} \times \frac{\partial a_3}{\partial z_3} \\ &= -0.021155647 \times 0.247200532 \\ &= -0.005229687\end{aligned}$$

Calculations for  $\delta_2$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_2} &= (w_{3,2} \times \delta_3) + (w_{4,2} \times \delta_4) \\ &= (0.2 \times -0.005229687) + (0.5 \times -0.007652161) \\ &= -0.004872018\end{aligned}$$

$$\begin{aligned}\frac{\partial a_2}{\partial z_2} &= \text{logistic}(z_2) \times (1 - \text{logistic}(z_2)) \\ &= 0.562176501 \times (1 - 0.562176501) \\ &= 0.246134083\end{aligned}$$

$$\begin{aligned}\delta_2 &= \frac{\partial \mathcal{E}}{\partial a_2} \times \frac{\partial a_2}{\partial z_2} \\ &= -0.004872018 \times 0.246134083 \\ &= -0.00119917\end{aligned}$$

- iii. Using the  $\delta$  values you have calculated, calculate the sensitivity of the error of the network to changes in each of the weights of the network (i.e.,  $\partial \mathcal{E} / \partial w_{5,4}$ ,  $\partial \mathcal{E} / \partial w_{5,3}$ ,  $\partial \mathcal{E} / \partial w_{5,0}$ ,  $\partial \mathcal{E} / \partial w_{4,2}$ ,  $\partial \mathcal{E} / \partial w_{4,0}$ ,  $\partial \mathcal{E} / \partial w_{3,2}$ ,  $\partial \mathcal{E} / \partial w_{3,0}$ ,  $\partial \mathcal{E} / \partial w_{2,1}$ ,

## 第 8 章 深度学习 (练习解决方案)

δ计算:

```

BE      " w^ d
Ba      " 0.4 ^ '0.052889117
        " '0.021155647

Ba      " logisticpzq ^ p1 ' logisticpzq
Bz      " 0.552909994 ^ p1 ' 0.552909994q
        " 0.247200532

δ      " BE ^ Ba
      " Ba      Bz
      " '0.021155647 ^ 0.247200532
      " '0.005229687
    
```

δ计算:

```

BE      " PW^ ΔQ ' PW^ ΔQ
Ba      " P0.2 ^ '0.005229687Q ' P0.5 ^ '0.007652161Q
        " '0.004872018

Ba      " logisticpzq ^ p1 ' logisticpzq
Bz      " 0.562176501 ^ p1 ' 0.562176501q
        " 0.246134083

δ      " BE ^ Ba
      " Ba      Bz
      " '0.004872018 ^ 0.246134083
      " '0.00119917
    
```

- iii. 使用您计算的δ值, 计算网络误差对网络每个权重变化的敏感性 (即  $BE\{Bw,$   
 $BE\{Bw,$   $BE\{Bw,$   $BE\{Bw,$   $BE\{Bw,$   $BE\{Bw,$   $BE\{Bw,$

$\partial\mathcal{E}/\partial w_{2,0}$ ).

To calculate the sensitivity of the error of the network to changes in a weight we multiply the  $\delta$  for the neuron that uses the weight in its weighted sum by the activation (or input) that the weight was applied to in the calculation of the weight sum. The calculations for each of the weights is shown below.

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial w_{5,4}} &= \delta_5 \times a_4 \\ &= -0.052889117 \times 0.594135549 \\ &= -0.031423304\end{aligned}$$

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial w_{5,3}} &= \delta_5 \times a_3 \\ &= -0.052889117 \times 0.552909994 \\ &= -0.029242921\end{aligned}$$

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial w_{5,0}} &= \delta_5 \times a_0 \\ &= -0.052889117 \times 1 \\ &= -0.052889117\end{aligned}$$

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial w_{4,2}} &= \delta_4 \times a_2 \\ &= -0.007652161 \times 0.562176501 \\ &= -0.004301865\end{aligned}$$

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial w_{4,0}} &= \delta_4 \times a_0 \\ &= -0.007652161 \times 1 \\ &= -0.007652161\end{aligned}$$

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial w_{3,2}} &= \delta_3 \times a_2 \\ &= -0.005229687 \times 0.562176501 \\ &= -0.002940007\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

$BE\{B_w\}$  。

为了计算网络误差对权重变化的敏感性，我们将使用其加权和中心权重的神经元的 $\delta$ 乘以在计算权重时应用权重的激活（或输入）。每个权重的计算如下所示。

$$\begin{aligned} BE \\ B_w - " d^a \\ " '0.052889117 ^ 0.594135549 \\ " '0.031423304 \end{aligned}$$

$$\begin{aligned} BE \\ B_w - " d^a \\ " '0.052889117 ^ 0.552909994 \\ " '0.029242921 \end{aligned}$$

$$\begin{aligned} BE \\ B_w - " d^a \\ " '0.052889117 ^ 1 \\ " '0.052889117 \end{aligned}$$

$$\begin{aligned} BE \\ B_w - " d^a \\ " '0.007652161 ^ 0.562176501 \\ " '0.004301865 \end{aligned}$$

$$\begin{aligned} BE \\ B_w - " d^a \\ " '0.007652161 ^ 1 \\ " '0.007652161 \end{aligned}$$

$$\begin{aligned} BE \\ B_w - " d^a \\ " '0.005229687 ^ 0.562176501 \\ " '0.002940007 \end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{E}}{\partial w_{3,0}} &= \delta_3 \times a_0 \\
&= -0.005229687 \times 1 \\
&= -0.005229687
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{E}}{\partial w_{2,1}} &= \delta_2 \times a_1 \\
&= -0.00119917 \times 0.5 \\
&= -0.000599585
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \mathcal{E}}{\partial w_{2,0}} &= \delta_2 \times a_0 \\
&= -0.00119917 \times 1 \\
&= -0.00119917
\end{aligned}$$

- iv. Assuming a learning rate of  $\alpha = 0.1$ , calculate the updated values for each of the weights in the network ( $w_{5,4}$ ,  $w_{5,3}$ ,  $w_{5,0}$ ,  $w_{4,2}$ ,  $w_{4,0}$ ,  $w_{3,2}$ ,  $w_{3,0}$ ,  $w_{2,1}$ ,  $w_{2,0}$ ) after the processing of this single training example.

We update weights as follows:

$$w_{i,k}^{t+1} = w_{i,k}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{i,k}}$$

The calculation for each weight update is below:

$$\begin{aligned}
w_{5,4}^{t+1} &= w_{5,4}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{5,4}} \\
&= 0.6 - (0.1 \times -0.031423304) \\
&= 0.60314233
\end{aligned}$$

$$\begin{aligned}
w_{5,3}^{t+1} &= w_{5,3}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{5,3}} \\
&= 0.4 - (0.1 \times -0.029242921) \\
&= 0.402924292
\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

$$w_{5,4}^{BE} = w_{5,4}^{old} + \alpha (d - a)$$

$$= 0.6 + 0.1 \times (-0.031423304)$$

$$= 0.60314233$$

$$w_{5,3}^{BE} = w_{5,3}^{old} + \alpha (d - a)$$

$$= 0.4 + 0.1 \times (-0.029242921)$$

$$= 0.402924292$$

$$w_{5,2}^{BE} = w_{5,2}^{old} + \alpha (d - a)$$

$$= 0.4 + 0.1 \times (-0.00119917)$$

$$= 0.400119917$$

假设学习率为  $\alpha = 0.1$ ，在处理此单个训练示例后，计算网络中每个权重 ( $w_1, w_2, w_3, w_4, w_5$ ) 的更新值。

我们按如下方式更新权重：

$$w_{i,k}^{BE} = w_{i,k}^{old} + \alpha (d - a)$$

每次权重更新的计算如下：

$$w_{5,4}^{BE} = w_{5,4}^{old} + \alpha (d - a)$$

$$= 0.6 + 0.1 \times (-0.031423304)$$

$$= 0.60314233$$

$$w_{5,3}^{BE} = w_{5,3}^{old} + \alpha (d - a)$$

$$= 0.4 + 0.1 \times (-0.029242921)$$

$$= 0.402924292$$

$$\begin{aligned}
w_{5,0}^{t+1} &= w_{5,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{5,0}} \\
&= 0.1 - (0.1 \times -0.052889117) \\
&= 0.105288912
\end{aligned}$$

$$\begin{aligned}
w_{4,2}^{t+1} &= w_{4,2}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{4,2}} \\
&= 0.5 - (0.1 \times -0.004301865) \\
&= 0.500430187
\end{aligned}$$

$$\begin{aligned}
w_{4,0}^{t+1} &= w_{4,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{4,0}} \\
&= 0.1 - (0.1 \times -0.007652161) \\
&= 0.100765216
\end{aligned}$$

$$\begin{aligned}
w_{3,2}^{t+1} &= w_{3,2}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{3,2}} \\
&= 0.2 - (0.1 \times -0.002940007) \\
&= 0.200294001
\end{aligned}$$

$$\begin{aligned}
w_{3,0}^{t+1} &= w_{3,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{3,0}} \\
&= 0.1 - (0.1 \times -0.005229687) \\
&= 0.100522969
\end{aligned}$$

$$\begin{aligned}
w_{2,1}^{t+1} &= w_{2,1}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{2,1}} \\
&= 0.3 - (0.1 \times -0.000599585) \\
&= 0.300059958
\end{aligned}$$

$$\begin{aligned}
w_{2,0}^{t+1} &= w_{2,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{2,0}} \\
&= 0.1 - (0.1 \times -0.00119917) \\
&= 0.100119917
\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

$w_{5,0}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.1 ' p0.1 ^ '0.052889117q  
" 0.105288912

$w_{4,2}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.5 ' p0.1 ^ '0.004301865q  
" 0.500430187

$w_{4,0}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.1 ' p0.1 ^ '0.007652161q  
" 0.100765216

$w_{3,2}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.2 ' p0.1 ^ '0.002940007q  
" 0.200294001

$w_{3,0}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.1 ' p0.1 ^ '0.005229687q  
" 0.100522969

$w_{2,1}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.3 ' p0.1 ^ '0.000599585q  
" 0.300059958

$w_{2,0}$  "  $w' a^{\text{BE}}$   $\bar{B}w^-$   
" 0.1 ' p0.1 ^ '0.00119917q  
" 0.100119917

- v. Calculate the reduction in the error of the network for this example using the new weights, compared with using the original weights.

The output for the network using the new weights is:

$$\begin{aligned} z_2 &= ((w_{2,0} \times 1) + (w_{2,1} \times 0.5)) \\ &= ((0.100119917 \times 1) + (0.300059958 \times 0.5)) \\ &= 0.250149896 \end{aligned}$$

$$\begin{aligned} a_2 &= \text{logistic}(0.250149896) \\ &= 0.562213395 \end{aligned}$$

$$\begin{aligned} z_3 &= ((w_{3,0} \times 1) + (w_{3,2} \times a_2)) \\ &= ((0.100522969 \times 1) + (0.200294001 \times 0.562213395)) \\ &= 0.213130939 \end{aligned}$$

$$\begin{aligned} a_3 &= \text{logistic}(0.213130939) \\ &= 0.55308195 \end{aligned}$$

$$\begin{aligned} z_4 &= ((w_{4,0} \times 1) + (w_{4,2} \times a_2)) \\ &= ((0.100765216 \times 1) + (0.500430187 \times 0.562213395)) \\ &= 0.38211377 \end{aligned}$$

$$\begin{aligned} a_4 &= \text{logistic}(0.38211377) \\ &= 0.594382817 \end{aligned}$$

$$\begin{aligned} z_5 &= ((w_{5,0} \times 1) + (w_{5,3} \times a_3) + (w_{5,4} \times a_4)) \\ &= ((0.105288912 \times 1) + (0.402924292 \times 0.55308195) + (0.60314233 \times 0.594382817)) \\ &= 0.686636503 \end{aligned}$$

$$\begin{aligned} a_5 &= \text{logistic}(0.686636503) \\ &= 0.665218283 \end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

v. 与使用原始权重相比, 使用新权重计算本示例中网络误差的减少。

使用新权重的网络输出为:

```
Z " PPW^ 1Q ' PW^ 0.5QQ
" pp0.100119917 ^ 1q ' p0.300059958 ^ 0.5qq
" 0.250149896
```

```
A " 物流P0.250149896Q
" 0.562213395
```

```
Z " PPW^ 1Q ' PW^ AQQ
" pp0.100522969 ^ 1q ' p0.200294001 ^ 0.562213395qq
" 0.213130939
```

```
A " 物流P0.213130939Q
" 0.55308195
```

```
Z " PPW^ 1Q ' PW^ AQQ
" pp0.100765216 ^ 1q ' p0.500430187 ^ 0.562213395qq
" 0.38211377
```

```
A " 物流P0.38211377Q
" 0.594382817
```

```
1Q ^
" pp0.105288912 ^ 1q ' p0.402924292 ^ 0.55308195q ' p0.60314233 ^
0.594382817q
0.686636503
```

```
A " 物流P0.686636503Q
" 0.665218283
```

The error with the new weights is:

$$\begin{aligned} Error' &= 0.9 - 0.665218283 \\ &0.234781717 \end{aligned}$$

Subtracting this new error from the error with the original weights (which we calculated as part of the calculation for  $\delta_3$  above) gives us the reduction in error:

$$\begin{aligned} Error\ Reduction &= 0.236787044 - 0.234781717 \\ &= 0.002005326 \end{aligned}$$

- (b) Assuming that the processing neurons are **ReLU**s, that the input to the network is Neuron 1 = 0.5 and that the desired output for this input is 0.9
- i. Calculate the output generated by the network in response to this input.

$$\begin{aligned} z_2 &= ((w_{2,0} \times 1) + (w_{2,1} \times 0.5)) \\ &= ((0.1 \times 1) + (0.3 \times 0.5)) \\ &= 0.25 \end{aligned}$$

$$\begin{aligned} a_2 &= \text{rectifier}(0.25) \\ &= \max(0, 0.25) \\ &= 0.25 \end{aligned}$$

$$\begin{aligned} z_3 &= ((w_{3,0} \times 1) + (w_{3,2} \times a_2)) \\ &= ((0.1 \times 1) + (0.2 \times 0.25)) \\ &= 0.15 \end{aligned}$$

$$\begin{aligned} a_3 &= \text{rectifier}(0.15) \\ &= \max(0, 0.15) \\ &= 0.15 \end{aligned}$$

$$\begin{aligned} z_4 &= ((w_{4,0} \times 1) + (w_{4,2} \times a_2)) \\ &= ((0.1 \times 1) + (0.5 \times 0.25)) \\ &= 0.225 \end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

新权重的错误是:

```
错误 " 0.9 ' 0.665218283
      0.234781717
```

从原始权重的误差中减去这个新误差 (我们作为上述  $\delta$  计算的一部分进行计算) 可以减少误差:

```
减少错误 " 0.236787044 ' 0.234781717
          " 0.002005326
```

(b) 假设处理神经元是 ReLU, 则网络的输入为 Neuron 1 = 0.5, 并且该输入的期望输出为 0.9

i. 计算网络响应此输入生成的输出。

```
Z " PPW^ 1Q ' PW^ 0.5QQ
  " pp0.1 ^ 1q ' p0.3 ^ 0.5qq
  " 0.25

A " 直肠 F ierp0.25Q
  " 最大P0,0.25q
  " 0.25

Z " PPW^ 1Q ' PW^ AQQ
  " pp0.1 ^ 1q ' p0.2 ^ 0.25qq
  " 0.15

a " recti f ierp0.15q
  " 最大P0,0.15q
  " 0.15

Z " PPW^ 1Q ' PW^ AQQ
  " pp0.1 ^ 1q ' p0.5 ^ 0.25qq
  " 0.225
```

$$\begin{aligned}
 a_4 &= \text{rectifier}(0.225) \\
 &= \max(0, 0.225) \\
 &= 0.225
 \end{aligned}$$

$$\begin{aligned}
 z_5 &= (w_{5,0} \times 1) + (w_{5,3} \times a_3) + (w_{5,4} \times a_4) \\
 &= ((0.1 \times 1) + (0.4 \times 0.15) + (0.6 \times 0.225)) \\
 &= 0.295
 \end{aligned}$$

$$\begin{aligned}
 a_5 &= \text{rectifier}(0.295) \\
 &= \max(0, 0.295) \\
 &= 0.295
 \end{aligned}$$

- ii. Calculate the  $\delta$  values for each of the processing neurons in the network (i.e.,  $\delta_5, \delta_4, \delta_3, \delta_2$ ).

Calculations for  $\delta_5$ :

$$\begin{aligned}
 \frac{\partial \mathcal{E}}{\partial a_5} &= -(t_k - a_k) \\
 &= -(0.9 - 0.295) \\
 &= -0.605
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial a_5}{\partial z_5} &= \begin{cases} 1 & \text{if } z_5 > 1 \\ 0 & \text{otherwise} \end{cases} \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \delta_5 &= \frac{\partial \mathcal{E}}{\partial a_5} \times \frac{\partial a_5}{\partial z_5} \\
 &= -0.605 \times 1 \\
 &= -0.605
 \end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

```
A " 直肠 f lerP0.225Q
    " 最大P0,0.225q
    " 0.225
```

```
Z " PPW^ 1Q ' PW^ AQ ' PW^ AQQ "
PP0.1 ^ 1Q ' P0.4 ^ 0.15Q ' P0.6 ^
0.225QQ
    " 0.295
```

```
A " 直肠 f ierp0.295q
    " 最大P0,0.295q
    " 0.295
```

ii. 计算网络中每个处理神经元 (即  $\delta$ 、 $\delta$ 、 $\delta$ 、 $\delta$ ) 的  $\delta$  值。

$\delta$  计算:

```
BE " ' pt' aq
Ba " ' p0.9 ' 0.295q
    " ' 0.605
```

```
#
Ba " 1 if zq 1
Bz " 0 否则
    " 1
```

```
 $\delta$  " BE ^ Ba
    Ba ' Bz
    " ' 0.605 ^ 1
    " ' 0.605
```

Calculations for  $\delta_4$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_4} &= w_{5,4} \times \delta_5 \\ &= 0.6 \times -0.605 \\ &= -0.363\end{aligned}$$

$$\begin{aligned}\frac{\partial a_4}{\partial z_4} &= \begin{cases} 1 & \text{if } z_4 > 1 \\ 0 & \text{otherwise} \end{cases} \\ &= 1\end{aligned}$$

$$\begin{aligned}\delta_4 &= \frac{\partial \mathcal{E}}{\partial a_4} \times \frac{\partial a_4}{\partial z_4} \\ &= -0.363 \times 1 \\ &= -0.363\end{aligned}$$

Calculations for  $\delta_3$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_3} &= w_{5,3} \times \delta_5 \\ &= 0.4 \times -0.605 \\ &= -0.242\end{aligned}$$

$$\begin{aligned}\frac{\partial a_3}{\partial z_3} &= \begin{cases} 1 & \text{if } z_3 > 1 \\ 0 & \text{otherwise} \end{cases} \\ &= 1\end{aligned}$$

$$\begin{aligned}\delta_3 &= \frac{\partial \mathcal{E}}{\partial a_3} \times \frac{\partial a_3}{\partial z_3} \\ &= -0.242 \times 1 \\ &= -0.242\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

δ计算:

$$\begin{aligned} \text{BE} &= w^d \\ \text{Ba} &= 0.6 \wedge 0.605 \\ &= 0.363 \end{aligned}$$

$$\begin{aligned} \text{Ba} &= 1 \text{ if } z_q = 1 \\ \text{Bz} &= 0 \text{ 否则} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \delta &= \text{BE} \wedge \text{Ba} \\ &= \text{Ba} \wedge \text{Bz} \\ &= 0.363 \wedge 1 \\ &= 0.363 \end{aligned}$$

δ计算:

$$\begin{aligned} \text{BE} &= w^d \\ \text{Ba} &= 0.4 \wedge 0.605 \\ &= 0.242 \end{aligned}$$

$$\begin{aligned} \text{Ba} &= 1 \text{ if } z_q = 1 \\ \text{Bz} &= 0 \text{ 否则} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \delta &= \text{BE} \wedge \text{Ba} \\ &= \text{Ba} \wedge \text{Bz} \\ &= 0.242 \wedge 1 \\ &= 0.242 \end{aligned}$$

Calculations for  $\delta_2$ :

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial a_2} &= (w_{3,2} \times \delta_3) + (w_{4,2} \times \delta_4) \\ &= (0.2 \times -0.242) + (0.5 \times -0.363) \\ &= -0.2299\end{aligned}$$

$$\begin{aligned}\frac{\partial a_2}{\partial z_2} &= \begin{cases} 1 & \text{if } z_2 > 1 \\ 0 & \text{otherwise} \end{cases} \\ &= 1\end{aligned}$$

$$\begin{aligned}\delta_2 &= \frac{\partial \mathcal{E}}{\partial a_2} \times \frac{\partial a_2}{\partial z_2} \\ &= -0.2299 \times 1 \\ &= -0.2299\end{aligned}$$

- iii. Using the  $\delta$  values you have calculated, calculate the sensitivity of the error of the network to changes in each of the weights of the network (i.e.,  $\partial \mathcal{E} / \partial w_{5,4}$ ,  $\partial \mathcal{E} / \partial w_{5,3}$ ,  $\partial \mathcal{E} / \partial w_{5,0}$ ,  $\partial \mathcal{E} / \partial w_{4,2}$ ,  $\partial \mathcal{E} / \partial w_{4,0}$ ,  $\partial \mathcal{E} / \partial w_{3,2}$ ,  $\partial \mathcal{E} / \partial w_{3,0}$ ,  $\partial \mathcal{E} / \partial w_{2,1}$ ,  $\partial \mathcal{E} / \partial w_{2,0}$ ).

To calculate the sensitivity of the error of the network to changes in a weight we multiply the  $\delta$  for the neuron that uses the weight in its weighted sum by the activation (or input) that the weight was applied to in the calculation of the weight sum. The calculations for each of the weights is shown below.

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{5,4}} &= \delta_5 \times a_4 \\ &= -0.605 \times 0.225 \\ &= -0.136125\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{5,3}} &= \delta_5 \times a_3 \\ &= -0.605 \times 0.15 \\ &= -0.09075\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

δ计算:

$$\begin{aligned} \delta_{Ba} &= \frac{\partial E}{\partial a} = \frac{\partial}{\partial a} (P0.2 + 0.242Q + P0.5 + 0.363Q) \\ &= 0.2299 \\ \delta_{Bz} &= \begin{cases} 1 & \text{if } z_a = 1 \\ 0 & \text{otherwise} \end{cases} \\ &= 1 \\ \delta_{Ba} &= \delta_{Bz} \cdot \frac{\partial E}{\partial a} \\ &= 0.2299 \cdot 1 \\ &= 0.2299 \end{aligned}$$

- iii. 使用您计算的δ值，计算网络误差对网络每个权重变化的敏感性（即  $\delta_{Bw}$ ， $\delta_{Bw}$ ， $\delta_{Bw}$ ， $\delta_{Bw}$ ， $\delta_{Bw}$ ， $\delta_{Bw}$ ， $\delta_{Bw}$ ， $\delta_{Bw}$ ）。

为了计算网络误差对权重变化的敏感性，我们将使用其加权和和中权重的神经元的δ乘以在计算权重和应用权重的激活（或输入）。每个权重的计算如下所示。

$$\begin{aligned} \delta_{Bw} &= \delta_{Ba} \cdot a \\ &= 0.2299 \cdot 0.605 \\ &= 0.136125 \\ \delta_{Bw} &= \delta_{Bz} \cdot a \\ &= 0.2299 \cdot 0.15 \\ &= 0.09075 \end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{5,0}} &= \delta_5 \times a_0 \\ &= -0.605 \times 1 \\ &= -0.605\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{4,2}} &= \delta_4 \times a_2 \\ &= -0.363 \times 0.25 \\ &= -0.09075\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{4,0}} &= \delta_4 \times a_0 \\ &= -0.363 \times 1 \\ &= -0.363\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{3,2}} &= \delta_3 \times a_2 \\ &= -0.242 \times 0.25 \\ &= -0.0605\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{3,0}} &= \delta_3 \times a_0 \\ &= -0.242 \times 1 \\ &= -0.242\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{2,1}} &= \delta_2 \times a_1 \\ &= -0.2299 \times 0.5 \\ &= -0.11495\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{E}}{\partial w_{2,0}} &= \delta_2 \times a_0 \\ &= -0.2299 \times 1 \\ &= -0.2299\end{aligned}$$

- iv. Assuming a learning rate of  $\alpha = 0.1$ , calculate the updated values for each of the weights in the network ( $w_{5,4}$ ,  $w_{5,3}$ ,  $w_{5,0}$ ,  $w_{4,2}$ ,  $w_{4,0}$ ,  $w_{3,2}$ ,  $w_{3,0}$ ,  $w_{2,1}$ ,  $w_{2,0}$ .)

## 第 8 章 深度学习 (练习解决方案)

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.605^1 \\ &= "0.605 \end{aligned}$$

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.363^{0.25} \\ &= "0.09075 \end{aligned}$$

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.363^1 \\ &= "0.363 \end{aligned}$$

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.242^{0.25} \\ &= "0.0605 \end{aligned}$$

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.242^1 \\ &= "0.242 \end{aligned}$$

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.2299^{0.5} \\ &= "0.11495 \end{aligned}$$

$$\begin{aligned} \bar{B}_w^{BE} &= "d^a \\ &= "0.2299^1 \\ &= "0.2299 \end{aligned}$$

假设学习率为  $\alpha = 0.1$ , 计算网络中每个权重的更新值 ( $w_1, w_2, w_3, w_4, w_5$ )

after the processing of this single training example.

We update weights as follows:

$$w_{i,k}^{t+1} = w_{i,k}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{i,k}}$$

The calculation for each weight update is below:

$$\begin{aligned} w_{5,4}^{t+1} &= w_{5,4}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{5,4}} \\ &= 0.6 - (0.1 \times -0.136125) \\ &= 0.6136125 \end{aligned}$$

$$\begin{aligned} w_{5,3}^{t+1} &= w_{5,3}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{5,3}} \\ &= 0.4 - (0.1 \times -0.09075) \\ &= 0.409075 \end{aligned}$$

$$\begin{aligned} w_{5,0}^{t+1} &= w_{5,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{5,0}} \\ &= 0.1 - (0.1 \times -0.605) \\ &= 0.1605 \end{aligned}$$

$$\begin{aligned} w_{4,2}^{t+1} &= w_{4,2}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{4,2}} \\ &= 0.5 - (0.1 \times -0.09075) \\ &= 0.509075 \end{aligned}$$

$$\begin{aligned} w_{4,0}^{t+1} &= w_{4,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{4,0}} \\ &= 0.1 - (0.1 \times -0.363) \\ &= 0.1363 \end{aligned}$$

$$\begin{aligned} w_{3,2}^{t+1} &= w_{3,2}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{3,2}} \\ &= 0.2 - (0.1 \times -0.0605) \\ &= 0.20605 \end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

在处理完这个单一的训练示例之后。

我们按如下方式更新权重：

$$w_{i,k} = w_{i,k} + \eta \frac{\partial L}{\partial w_{i,k}}$$

每次权重更新的计算如下：

$$w_{5,4} = w_{5,4} + \eta \frac{\partial L}{\partial w_{5,4}} \\ = 0.6 + 0.1 \times 0.136125 \\ = 0.6136125$$

$$w_{5,3} = w_{5,3} + \eta \frac{\partial L}{\partial w_{5,3}} \\ = 0.4 + 0.1 \times 0.09075 \\ = 0.409075$$

$$w_{5,0} = w_{5,0} + \eta \frac{\partial L}{\partial w_{5,0}} \\ = 0.1 + 0.1 \times 0.605 \\ = 0.1605$$

$$w_{4,2} = w_{4,2} + \eta \frac{\partial L}{\partial w_{4,2}} \\ = 0.5 + 0.1 \times 0.09075 \\ = 0.509075$$

$$w_{4,0} = w_{4,0} + \eta \frac{\partial L}{\partial w_{4,0}} \\ = 0.1 + 0.1 \times 0.363 \\ = 0.1363$$

$$w_{3,2} = w_{3,2} + \eta \frac{\partial L}{\partial w_{3,2}} \\ = 0.2 + 0.1 \times 0.0605 \\ = 0.20605$$

$$\begin{aligned}w_{3,0}^{t+1} &= w_{3,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{3,0}} \\&= 0.1 - (0.1 \times -0.242) \\&= 0.1242\end{aligned}$$

$$\begin{aligned}w_{2,1}^{t+1} &= w_{2,1}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{2,1}} \\&= 0.3 - (0.1 \times -0.11495) \\&= 0.311495\end{aligned}$$

$$\begin{aligned}w_{2,0}^{t+1} &= w_{2,0}^t - \alpha \times \frac{\partial \mathcal{E}}{\partial w_{2,0}} \\&= 0.1 - (0.1 \times -0.2299) \\&= 0.12299\end{aligned}$$

- v. Calculate the reduction in the error of the network for this example using the new weights, compared with using the original weights.

The output for the network using the new weights is:

$$\begin{aligned}z_2 &= ((w_{2,0} \times 1) + (w_{2,1} \times 0.5)) \\&= ((0.12299 \times 1) + (0.311495 \times 0.5)) \\&= 0.2787375\end{aligned}$$

$$\begin{aligned}a_2 &= \text{rectifier}(0.2787375) \\&= \max(0, 0.2787375) \\&= 0.2787375\end{aligned}$$

$$\begin{aligned}z_3 &= ((w_{3,0} \times 1) + (w_{3,2} \times a_2)) \\&= ((0.1242 \times 1) + (0.20605 \times 0.2787375)) \\&= 0.181633862\end{aligned}$$

## 第 8 章 深度学习 (练习解决方案)

$$w_{3,0} = w_{a^+}^{BE} - \frac{0.1}{p_{0.1}^{0.242q}} - 0.1242$$

$$w_{2,1} = w_{a^+}^{BE} - \frac{0.3}{p_{0.1}^{0.11495q}} - 0.311495$$

$$w_{2,0} = w_{a^+}^{BE} - \frac{0.1}{p_{0.1}^{0.2299q}} - 0.12299$$

v. 与使用原始权重相比，使用新权重计算本示例中网络误差的减少。

使用新权重的网络输出为：

$$Z = PPW^{1Q} + PW^{0.5QQ} - PP0.12299^{1Q} - P0.311495^{0.5QQ} - 0.2787375$$

$$a = \text{rectifier}(0.2787375q) = \max(0, 0.2787375q) = 0.2787375$$

$$Z = PPW^{1Q} + PW^{AQQ} - pp0.1242^{1q} - p0.20605^{0.2787375q} - 0.181633862$$

$$\begin{aligned}
 a_3 &= \text{rectifier}(0.181633862) \\
 &= \max(0, 0.181633862) \\
 &= 0.181633862
 \end{aligned}$$

$$\begin{aligned}
 z_4 &= ((w_{4,0} \times 1) + (w_{4,2} \times a_2)) \\
 &= ((0.1363 \times 1) + (0.509075 \times 0.2787375)) \\
 &= 0.278198293
 \end{aligned}$$

$$\begin{aligned}
 a_4 &= \text{rectifier}(0.278198293) \\
 &= \max(0, 0.278198293) \\
 &= 0.278198293
 \end{aligned}$$

$$\begin{aligned}
 z_5 &= ((w_{5,0} \times 1) + (w_{5,3} \times a_3) + (w_{5,4} \times a_4)) \\
 &= ((0.1605 \times 1) + (0.409075 \times 0.181633862) + (0.6136125 \times 0.278198293)) \\
 &= 0.405507822
 \end{aligned}$$

$$\begin{aligned}
 a_5 &= \text{rectifier}(0.405507822) \\
 &= \max(0, 0.405507822) \\
 &= 0.405507822
 \end{aligned}$$

The error with the new weights is:

$$\begin{aligned}
 \text{Error}' &= 0.9 - 0.405507822 \\
 &= 0.494492178
 \end{aligned}$$

Subtracting this new error from the error with the original weights (which we calculated as part of the calculation for  $\delta_3$  above) gives us the reduction in error:

$$\begin{aligned}
 \text{Error Reduction} &= 0.605 - 0.494492178 \\
 &= 0.110507822
 \end{aligned}$$

## Chapter 9

## 第 8 章 深度学习 (练习解决方案)

```
a " recti f ierp0.181633862q
" 最大 P0, 0.181633862Q
" 0.181633862

Z " PPW^ 1Q ' PW^ AQQ " PP0.1363 ^
1Q ' P0.509075 ^ 0.2787375QQ
" 0.278198293

a " recti f ierp0.278198293q
" 最大P0,0.278198293Q
" 0.278198293

1Q ^
" pp0.1605 ^ 1q ' p0.409075 ^ 0.181633862q ' p0.6136125 ^ 0.278198293qq
" 0.405507822

a " recti f ierp0.405507822q
" 最大P0,0.405507822Q
" 0.405507822

新权重的错误是:

错误 " 0.9 ' 0.405507822
0.494492178

从原始权重的误差中减去这个新误差 (我们作为上述  $\delta$  计算的一部分进行计算) 可以减少误差:

减少误差 " 0.605 ' 0.494492178
" 0.110507822
```

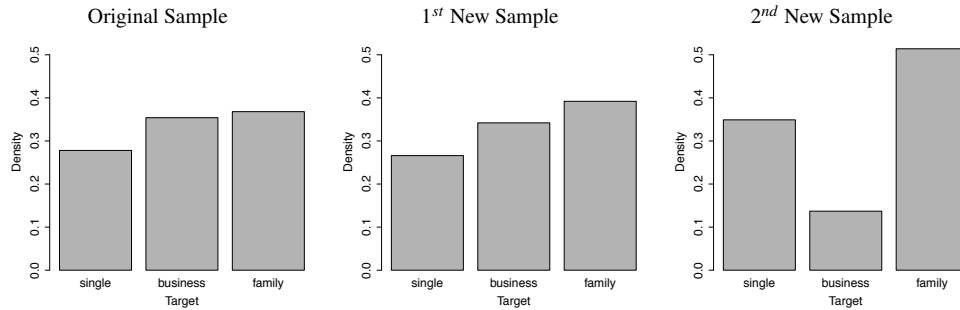
## 第9章 评估 (运动解决方案)

&KDSWHU

4. A retail supermarket chain has built a prediction model that recognizes the household that a customer comes from as being one of *single*, *business*, or *family*. After deployment, the analytics team at the supermarket chain uses the **stability index** to monitor the performance of this model. The table below shows the frequencies of predictions of the three different levels made by the model for the original validation dataset at the time the model was built, for the month after deployment, and for a monthlong period six months after deployment.

| Target          | Original Sample | 1 <sup>st</sup> New Sample | 2 <sup>nd</sup> New Sample |
|-----------------|-----------------|----------------------------|----------------------------|
| <i>single</i>   | 123             | 252                        | 561                        |
| <i>business</i> | 157             | 324                        | 221                        |
| <i>family</i>   | 163             | 372                        | 827                        |

Bar plots of these three sets of prediction frequencies are shown in the following images.



Calculate the **stability index** for the two new periods and determine whether the model should be retrained at either of these points.

The stability index is calculated as

$$stability\ index = \sum_{l \in levels} \left( \left( \frac{|\mathcal{A}_{t=l}|}{|\mathcal{A}|} - \frac{|\mathcal{B}_{t=l}|}{|\mathcal{B}|} \right) \times \log_e \left( \frac{|\mathcal{A}_{t=l}|}{|\mathcal{A}|} / \frac{|\mathcal{B}_{t=l}|}{|\mathcal{B}|} \right) \right)$$

where  $l$  is a target level,  $|\mathcal{A}|$  refers to the size of the test set on which performance measures were originally calculated,  $|\mathcal{A}_{t=l}|$  refers to the number of instances in the original test set for which the model made a prediction of level  $l$  for target  $t$ ,  $|\mathcal{B}|$  and  $|\mathcal{B}_{t=l}|$  refer to the same measurements on the newly collected dataset. The following table shows the components of calculating this for the two new periods.

| Target   | Original |        | 1 <sup>st</sup> New Sample |        |                        | 2 <sup>nd</sup> New Sample |        |                        |
|----------|----------|--------|----------------------------|--------|------------------------|----------------------------|--------|------------------------|
|          | Count    | %      | Count                      | %      | SI <sub><i>t</i></sub> | Count                      | %      | SI <sub><i>t</i></sub> |
| single   | 123      | 0.2777 | 252                        | 0.2658 | 0.00052                | 561                        | 0.3487 | 0.01617                |
| business | 157      | 0.3544 | 324                        | 0.3418 | 0.00046                | 221                        | 0.1374 | 0.20574                |
| family   | 163      | 0.3679 | 372                        | 0.3924 | 0.00157                | 827                        | 0.5140 | 0.04881                |
| Sum      | 443      |        | 948                        |        | 0.003                  | 1,609                      |        | 0.271                  |

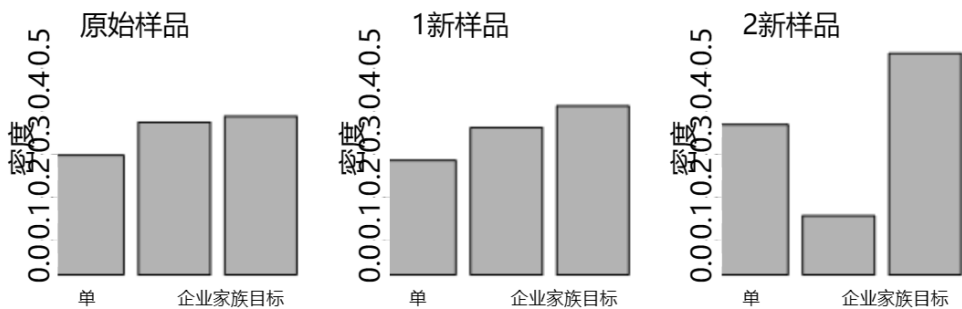
For the first new sample, the stability index is 0.003, which indicates that there is practically no difference between the distribution of target levels predicted for the original validation dataset and for the data in the newt period.

For the second sample, the stability index is 0.271, which indicates a massive difference between the distribution of target levels at this point in time compared to the distribution predicted for the original validation set. This suggests that concept drift has occurred and that the model should be retrained.

4. 一家零售连锁超市建立了一个预测模型，该模型将客户来自的家庭识别为单身、企业或家庭之一。部署后，连锁超市的分析团队使用稳定性指数来监视此模型的性能。下表显示了模型在构建模型时、部署后的一个月以及部署后六个月的一个月内对原始验证数据集所做的三个不同级别的预测频率。

|      |        | 源语言 | 1New | 2New |
|------|--------|-----|------|------|
| 目标样本 | Sample |     |      |      |
| 单    |        | 123 | 252  | 561  |
| 商    |        | 157 | 324  | 221  |
| 家庭   |        | 163 | 372  | 827  |

这三组预测频率的条形图如下图所示。



计算两个新周期的稳定性指数，并确定是否应在这两个点中的任何一个点重新训练模型。

稳定性指数计算如下：

稳定性指数 =  $\frac{\sum_{t=1}^T \frac{|A_t|}{|A|} \cdot \frac{|B_t|}{|B|}}{T}$

其中  $I$  是目标水平， $|A|$  指最初计算性能度量的测试集的大小， $|A_t|$  指原始测试集中模型对目标  $t$  的  $I$  级预测的实例数， $|B|$  以及  $|B_t|$  参考新收集的数据集上的相同测量值。下表显示了计算两个新期间的组成部分。

| 2新样本    | 目标语言    | 计数     | %       | SI  | 1新样品  | %      | SI    | 单      | 123     | 0.2777 | 252    | 0.2658  |
|---------|---------|--------|---------|-----|-------|--------|-------|--------|---------|--------|--------|---------|
| 0.00052 | 561     | 0.3487 | 0.01617 | 业务  | 157   | 0.3544 | 324   | 0.3418 | 0.00046 | 221    | 0.1374 | 0.20574 |
| 0.5140  | 0.04881 | 总和     | 443     | 948 | 0.003 | 1,609  | 0.271 |        |         |        |        |         |

对于第一个新样本，稳定性指数为 0.003，这表明原始验证数据集预测的目标水平分布与蝶螈周期数据的目标水平分布之间几乎没有差异。

对于第二个样本，稳定性指数为 0.271，这表明与原始验证集预测的分布相比，该时间点的目标水平分布之间存在巨大差异。这表明概念漂移已经发生，应该重新训练模型。

- \* 6. A marketing company working for a charity has developed two different models that predict the likelihood that donors will respond to a mailshot asking them to make a special extra donation. The prediction scores generated for a test set for these two models are shown in the table below.

| Model 1 |        |        |        | Model 2 |        |        |        |
|---------|--------|--------|--------|---------|--------|--------|--------|
| ID      | Target | Score  | Score  | ID      | Target | Score  | Score  |
| 1       | false  | 0.1026 | 0.2089 | 16      | true   | 0.7165 | 0.4569 |
| 2       | false  | 0.2937 | 0.0080 | 17      | true   | 0.7677 | 0.8086 |
| 3       | true   | 0.5120 | 0.8378 | 18      | false  | 0.4468 | 0.1458 |
| 4       | true   | 0.8645 | 0.7160 | 19      | false  | 0.2176 | 0.5809 |
| 5       | false  | 0.1987 | 0.1891 | 20      | false  | 0.9800 | 0.5783 |
| 6       | true   | 0.7600 | 0.9398 | 21      | true   | 0.6562 | 0.7843 |
| 7       | true   | 0.7519 | 0.9800 | 22      | true   | 0.9693 | 0.9521 |
| 8       | true   | 0.2994 | 0.8578 | 23      | false  | 0.0275 | 0.0377 |
| 9       | false  | 0.0552 | 0.1560 | 24      | true   | 0.7047 | 0.4708 |
| 10      | false  | 0.9231 | 0.5600 | 25      | false  | 0.3711 | 0.2846 |
| 11      | true   | 0.7563 | 0.9062 | 26      | false  | 0.4440 | 0.1100 |
| 12      | true   | 0.5664 | 0.7301 | 27      | true   | 0.5440 | 0.3562 |
| 13      | true   | 0.2872 | 0.8764 | 28      | true   | 0.5713 | 0.9200 |
| 14      | true   | 0.9326 | 0.9274 | 29      | false  | 0.3757 | 0.0895 |
| 15      | false  | 0.0651 | 0.2992 | 30      | true   | 0.8224 | 0.8614 |

- (a) Using a classification threshold of 0.5, and assuming that *true* is the positive target level, construct a **confusion matrix** for each of the models.

The first step to answering this is to apply the threshold to determine whether predictions are correct or not. For Model 1, the predictions and outcomes are as follows

|    |        | Model 1 | Model 1    | Model 1 |
|----|--------|---------|------------|---------|
| ID | Target | Score   | Prediction | Outcome |
| 1  | false  | 0.1026  | false      | TN      |
| 2  | false  | 0.2937  | false      | TN      |
| 3  | true   | 0.5120  | true       | TP      |
| 4  | true   | 0.8645  | true       | TP      |
| 5  | false  | 0.1987  | false      | TN      |
| 6  | true   | 0.7600  | true       | TP      |
| 7  | true   | 0.7519  | true       | TP      |
| 8  | true   | 0.2994  | false      | FN      |
| 9  | false  | 0.0552  | false      | TN      |
| 10 | false  | 0.9231  | true       | FP      |
| 11 | true   | 0.7563  | true       | TP      |
| 12 | true   | 0.5664  | true       | TP      |
| 13 | true   | 0.2872  | false      | FN      |
| 14 | true   | 0.9326  | true       | TP      |
| 15 | false  | 0.0651  | false      | TN      |
| 16 | true   | 0.7165  | true       | TP      |
| 17 | true   | 0.7677  | true       | TP      |
| 18 | false  | 0.4468  | false      | TN      |
| 19 | false  | 0.2176  | false      | TN      |
| 20 | false  | 0.9800  | true       | FP      |
| 21 | true   | 0.6562  | true       | TP      |
| 22 | true   | 0.9693  | true       | TP      |
| 23 | false  | 0.0275  | false      | TN      |
| 24 | true   | 0.7047  | true       | TP      |
| 25 | false  | 0.3711  | false      | TN      |
| 26 | false  | 0.4440  | false      | TN      |
| 27 | true   | 0.5440  | true       | TP      |
| 28 | true   | 0.5713  | true       | TP      |
| 29 | false  | 0.3757  | false      | TN      |
| 30 | true   | 0.8224  | true       | TP      |

Using this, we can build the following confusion matrix for Model 1

|        |              | Prediction  |              |
|--------|--------------|-------------|--------------|
|        |              | <i>true</i> | <i>false</i> |
| Target | <i>true</i>  | 15          | 2            |
|        | <i>false</i> | 2           | 11           |

- ° 6. 一家为慈善机构工作的营销公司开发了两种不同的模型，预测捐赠者回复要求他们进行特殊额外捐赠的邮件的可能性。为这两个模型的测试集生成的预测分数如下表所示。

| ID | 目标   | 模型 1<br>得分 | 模型 2<br>得分 | ID | 目标   | 模型 1<br>得分 | 模型 2<br>得分 |
|----|------|------------|------------|----|------|------------|------------|
| 1  | 假    | 0.1026     | 0.2089     | 16 | true | 0.7165     | 0.4569     |
| 2  | 假    | 0.2937     | 0.0080     | 17 | true | 0.7677     | 0.8086     |
| 3  | true | 0.5120     | 0.8378     | 18 | 假    | 0.4468     | 0.1458     |
| 4  | true | 0.8645     | 0.7160     | 19 | 假    | 0.2176     | 0.5809     |
| 5  | 假    | 0.1987     | 0.1891     | 20 | 假    | 0.9800     | 0.5783     |
| 6  | true | 0.7600     | 0.9398     | 21 | true | 0.6562     | 0.7843     |
| 7  | true | 0.7519     | 0.9800     | 22 | true | 0.9693     | 0.9521     |
| 8  | true | 0.2994     | 0.8578     | 23 | 假    | 0.0275     | 0.0377     |
| 9  | 假    | 0.0552     | 0.1560     | 24 | true | 0.7047     | 0.4708     |
| 10 | 假    | 0.9231     | 0.5600     | 25 | 假    | 0.3711     | 0.2846     |
| 11 | true | 0.7563     | 0.9062     | 26 | 假    | 0.4440     | 0.1100     |
| 12 | true | 0.5664     | 0.7301     | 27 | true | 0.5440     | 0.3562     |
| 13 | true | 0.2872     | 0.8764     | 28 | true | 0.5713     | 0.9200     |
| 14 | true | 0.9326     | 0.9274     | 29 | 假    | 0.3757     | 0.0895     |
| 15 | 假    | 0.0651     | 0.2992     | 30 | true | 0.8224     | 0.8614     |

(a) 使用0.5的分类阈值，并假设true是正目标水平，为每个模型构建一个混淆矩阵。

回答这个问题的第一步是应用阈值来确定预测是否正确。对于模型1，预测和结果如下

| ID | 目标   | 模型 1<br>分数 | 模型 1<br>预测结果 | 模型 1 |
|----|------|------------|--------------|------|
| 1  | 假    | 0.1026     | 假            | TN   |
| 2  | 假    | 0.2937     | 假            | TN   |
| 3  | true | 0.5120     | true         | TP   |
| 4  | true | 0.8645     | true         | TP   |
| 5  | 假    | 0.1987     | 假            | TN   |
| 6  | true | 0.7600     | true         | TP   |
| 7  | true | 0.7519     | true         | TP   |
| 8  | true | 0.2994     | 假            | FN   |
| 9  | 假    | 0.0552     | 假            | TN   |
| 10 | 假    | 0.9231     | true         | FP   |
| 11 | true | 0.7563     | true         | TP   |
| 12 | true | 0.5664     | true         | TP   |
| 13 | true | 0.2872     | 假            | FN   |
| 14 | true | 0.9326     | true         | TP   |
| 15 | 假    | 0.0651     | 假            | TN   |
| 16 | true | 0.7165     | true         | TP   |
| 17 | true | 0.7677     | true         | TP   |
| 18 | 假    | 0.4468     | 假            | TN   |
| 19 | 假    | 0.2176     | 假            | TN   |
| 20 | 假    | 0.9800     | true         | FP   |
| 21 | true | 0.6562     | true         | TP   |
| 22 | true | 0.9693     | true         | TP   |
| 23 | 假    | 0.0275     | 假            | TN   |
| 24 | true | 0.7047     | true         | TP   |
| 25 | 假    | 0.3711     | 假            | TN   |
| 26 | 假    | 0.4440     | 假            | TN   |
| 27 | true | 0.5440     | true         | TP   |
| 28 | true | 0.5713     | true         | TP   |
| 29 | 假    | 0.3757     | 假            | TN   |
| 30 | true | 0.8224     | true         | TP   |

使用它，我们可以为模型1构建以下混淆矩阵

|    |   | 预测 |    |
|----|---|----|----|
|    |   | 真  | 假  |
| 目标 | 对 | 15 | 2  |
|    | 错 | 2  | 11 |

We then repeat the same process for Model 2.

| ID | Target | Model 2<br>Score | Model 2<br>Prediction | Model 2<br>Outcome |
|----|--------|------------------|-----------------------|--------------------|
| 1  | false  | 0.2089           | false                 | TN                 |
| 2  | false  | 0.0080           | false                 | TN                 |
| 3  | true   | 0.8378           | true                  | TP                 |
| 4  | true   | 0.7160           | true                  | TP                 |
| 5  | false  | 0.1891           | false                 | TN                 |
| 6  | true   | 0.9398           | true                  | TP                 |
| 7  | true   | 0.9800           | true                  | TP                 |
| 8  | true   | 0.8578           | true                  | TP                 |
| 9  | false  | 0.1560           | false                 | TN                 |
| 10 | false  | 0.5600           | true                  | FP                 |
| 11 | true   | 0.9062           | true                  | TP                 |
| 12 | true   | 0.7301           | true                  | TP                 |
| 13 | true   | 0.8764           | true                  | TP                 |
| 14 | true   | 0.9274           | true                  | TP                 |
| 15 | false  | 0.2992           | false                 | TN                 |
| 16 | true   | 0.4569           | false                 | FN                 |
| 17 | true   | 0.8086           | true                  | TP                 |
| 18 | false  | 0.1458           | false                 | TN                 |
| 19 | false  | 0.5809           | true                  | FP                 |
| 20 | false  | 0.5783           | true                  | FP                 |
| 21 | true   | 0.7843           | true                  | TP                 |
| 22 | true   | 0.9521           | true                  | TP                 |
| 23 | false  | 0.0377           | false                 | TN                 |
| 24 | true   | 0.4708           | false                 | FN                 |
| 25 | false  | 0.2846           | false                 | TN                 |
| 26 | false  | 0.1100           | false                 | TN                 |
| 27 | true   | 0.3562           | false                 | FN                 |
| 28 | true   | 0.9200           | true                  | TP                 |
| 29 | false  | 0.0895           | false                 | TN                 |
| 30 | true   | 0.8614           | true                  | TP                 |

Using this we can build the following confusion matrix for Model 2.

|        |              | Prediction  |              |
|--------|--------------|-------------|--------------|
|        |              | <i>true</i> | <i>false</i> |
| Target | <i>true</i>  | 14          | 3            |
|        | <i>false</i> | 3           | 10           |

- (b) Calculate the simple accuracy and **average class accuracy** (using an **arithmetic mean**) for each model.

然后，我们对模型 2 重复相同的过程。

| ID | 目标   | 模型 2<br>分数 | 模型 2<br>预测结果 | 模型 2<br>结果 |
|----|------|------------|--------------|------------|
| 1  | 假    | 0.2089     | 假            | TN         |
| 2  | 假    | 0.0080     | 假            | TN         |
| 3  | true | 0.8378     | true         | TP         |
| 4  | true | 0.7160     | true         | TP         |
| 5  | 假    | 0.1891     | 假            | TN         |
| 6  | true | 0.9398     | true         | TP         |
| 7  | true | 0.9800     | true         | TP         |
| 8  | true | 0.8578     | true         | TP         |
| 9  | 假    | 0.1560     | 假            | TN         |
| 10 | 假    | 0.5600     | true         | FP         |
| 11 | true | 0.9062     | true         | TP         |
| 12 | true | 0.7301     | true         | TP         |
| 13 | true | 0.8764     | true         | TP         |
| 14 | true | 0.9274     | true         | TP         |
| 15 | 假    | 0.2992     | 假            | TN         |
| 16 | true | 0.4569     | 假            | FN         |
| 17 | true | 0.8086     | true         | TP         |
| 18 | 假    | 0.1458     | 假            | TN         |
| 19 | 假    | 0.5809     | true         | FP         |
| 20 | 假    | 0.5783     | true         | FP         |
| 21 | true | 0.7843     | true         | TP         |
| 22 | true | 0.9521     | true         | TP         |
| 23 | 假    | 0.0377     | 假            | TN         |
| 24 | true | 0.4708     | 假            | FN         |
| 25 | 假    | 0.2846     | 假            | TN         |
| 26 | 假    | 0.1100     | 假            | TN         |
| 27 | true | 0.3562     | 假            | FN         |
| 28 | true | 0.9200     | true         | TP         |
| 29 | 假    | 0.0895     | 假            | TN         |
| 30 | true | 0.8614     | true         | TP         |

使用它，我们可以为模型 2 构建以下混淆矩阵。

|    |   | 预测 |    |
|----|---|----|----|
|    |   | 真  | 假  |
| 目标 | 对 | 14 | 3  |
|    | 错 | 3  | 10 |

(b) 计算每个模型的简单准确率和平均类准确率 (使用算术平均值)。

We can calculate average class accuracy simply from the confusion matrices. For Model 1 we calculate the recall for the *true* target level as

$$recall_{true} = \frac{14}{(14 + 3)} = 0.8235$$

For the *false* target level, recall is

$$recall_{false} = \frac{12}{(12 + 1)} = 0.9231$$

We can then calculate average class accuracy as

$$average\ class\ accuracy_{AM} = \frac{(0.8235 + 0.9231)}{2} = 0.8733$$

For Model 2 we calculate the recalls for the two target levels as

$$recall_{true} = \frac{14}{(14 + 3)} = 0.8235$$

and

$$recall_{false} = \frac{10}{(10 + 3)} = 0.7692$$

We can then calculate average class accuracy as

$$average\ class\ accuracy_{AM} = \frac{(0.8235 + 0.7692)}{2} = 0.7964$$

- (c) Based on the average class accuracy measures, which model appears to perform best at this task?

Based on average class accuracy, Model 1 appears to be doing a better job than Model 2. The main difference is in the better ability of Model 1 to identify instances of the *false* target level.

- (d) Generate a **cumulative gain chart** for each model.

To generate a cumulative gain chart, we first have to reorder the predictions in descending order of scores and divide the test instances into deciles. The table below shows this for Model 1 (because there are 30 instances in the test set, there are 3 instances per decile).

我们可以简单地从混淆矩阵中计算出平均类准确度。对于模型 1，我们将真实目标水平的召回率计算为

$$\text{召回} = \frac{14}{14 + 3q} = 0.8235$$

对于错误目标水平，召回率为

$$\text{召回} = \frac{12}{12 + 1q} = 0.9231$$

我们可以计算平均类精度为

$$\text{平均班级准确度} = \frac{0.8235 + 0.9231}{2} = 0.8733$$

对于模型 2，我们将两个目标水平的召回率计算为

$$\text{召回} = \frac{14}{14 + 3q} = 0.8235$$

and

$$\text{召回} = \frac{10}{10 + 3q} = 0.7692$$

我们可以计算平均类精度为

$$\text{平均班级准确度} = \frac{0.8235 + 0.7692}{2} = 0.7964$$

- (c) 根据平均班级准确度测量，哪个模型似乎在这项任务中表现最好？

根据平均类准确率，模型 1 似乎比模型 2 做得更好。主要区别在于模型 1 能够更好地识别虚假目标级别的实例。

- (d) 为每个模型生成累积增益图。

要生成累积增益图，我们首先必须按分数降序对预测进行重新排序，并将测试实例划分为十分位数。下表显示了模型 1 的情况（因为测试集中有 30 个实例，所以每个十分位数有 3 个实例）。

| Decile           | ID | Target | Model 1<br>Prediction | Model 1<br>Score | Model 1<br>Outcome |
|------------------|----|--------|-----------------------|------------------|--------------------|
| 1 <sup>st</sup>  | 20 | false  | 0.9800                | true             | FP                 |
|                  | 22 | true   | 0.9693                | true             | TP                 |
|                  | 14 | true   | 0.9326                | true             | TP                 |
| 2 <sup>nd</sup>  | 10 | false  | 0.9231                | true             | FP                 |
|                  | 4  | true   | 0.8645                | true             | TP                 |
|                  | 30 | true   | 0.8224                | true             | TP                 |
| 3 <sup>rd</sup>  | 17 | true   | 0.7677                | true             | TP                 |
|                  | 6  | true   | 0.7600                | true             | TP                 |
|                  | 11 | true   | 0.7563                | true             | TP                 |
| 4 <sup>th</sup>  | 7  | true   | 0.7519                | true             | TP                 |
|                  | 16 | true   | 0.7165                | true             | TP                 |
|                  | 24 | true   | 0.7047                | true             | TP                 |
| 5 <sup>th</sup>  | 21 | true   | 0.6562                | true             | TP                 |
|                  | 28 | true   | 0.5713                | true             | TP                 |
|                  | 12 | true   | 0.5664                | true             | TP                 |
| 6 <sup>th</sup>  | 27 | true   | 0.5440                | true             | TP                 |
|                  | 3  | true   | 0.5120                | true             | TP                 |
|                  | 18 | false  | 0.4468                | false            | TN                 |
| 7 <sup>th</sup>  | 26 | false  | 0.4440                | false            | TN                 |
|                  | 29 | false  | 0.3757                | false            | TN                 |
|                  | 25 | false  | 0.3711                | false            | TN                 |
| 8 <sup>th</sup>  | 8  | true   | 0.2994                | false            | FN                 |
|                  | 2  | false  | 0.2937                | false            | TN                 |
|                  | 13 | true   | 0.2872                | false            | FN                 |
| 9 <sup>th</sup>  | 19 | false  | 0.2176                | false            | TN                 |
|                  | 5  | false  | 0.1987                | false            | TN                 |
|                  | 1  | false  | 0.1026                | false            | TN                 |
| 10 <sup>th</sup> | 15 | false  | 0.0651                | false            | TN                 |
|                  | 9  | false  | 0.0552                | false            | TN                 |
|                  | 23 | false  | 0.0275                | false            | TN                 |

Based on this table, we can calculate the gain and lift (lift is not required to answer the question but is included for completeness) for each decile as well as the cumulative gain and cumulative lift for each decile. Recall that the definitions for gain, cumulative gain, lift, and cumulative lift always refer to the actual target levels rather than the predictions and are defined as follows:

## 第9章 评估 (运动解决方案)

| 十分位数 | 目标 | 模型 1 预测           | 模型 1 得分 | 模型 1 结果 |
|------|----|-------------------|---------|---------|
| 1st  | 20 | 假 0.9800 真 FP     |         |         |
|      | 22 | true 0.9693       | true    | TP      |
|      | 14 | true 0.9326       | true    | TP      |
|      | 10 | false 0.9231 真 FP |         |         |
| 2    | 4  | true 0.8645       | true    | TP      |
|      | 30 | true 0.8224       | true    | TP      |
|      | 17 | 真 0.7677 真 TP     |         |         |
| 3    | 6  | true 0.7600       | true    | TP      |
|      | 11 | true 0.7563       | true    | TP      |
|      | 7  | true 0.7519       | true    | TP      |
| 4    | 16 | true 0.7165       | true    | TP      |
|      | 24 | true 0.7047       | true    | TP      |
|      | 21 | true 0.6562       | true    | TP      |
| 5    | 28 | true 0.5713       | true    | TP      |
|      | 12 | true 0.5664       | true    | TP      |
|      | 27 | 真 0.5440 真 TP     |         |         |
| 6    | 3  | true 0.5120       | true    | TP      |
|      | 18 | 假 0.4468          | 假       | TN      |
|      | 26 | 假 0.4440 假 TN     |         |         |
| 7    | 29 | 假 0.3757 假 TN     |         |         |
|      | 25 | 假 0.3711          | 假       | TN      |
|      | 8  | 真 0.2994 假 FN     |         |         |
| 8    | 2  | 假 0.2937 假 TN     |         |         |
|      | 13 | true 0.2872       | 假       | FN      |
|      | 19 | 假 0.2176 假 TN     |         |         |
| 9    | 5  | 假 0.1987 假 TN     |         |         |
|      | 1  | 假 0.1026          | 假       | TN      |
|      | 15 | 假 0.0651 假 TN     |         |         |
| 10   | 9  | 假 0.0552 假 TN     |         |         |
|      | 23 | 假 0.0275          | 假       | TN      |

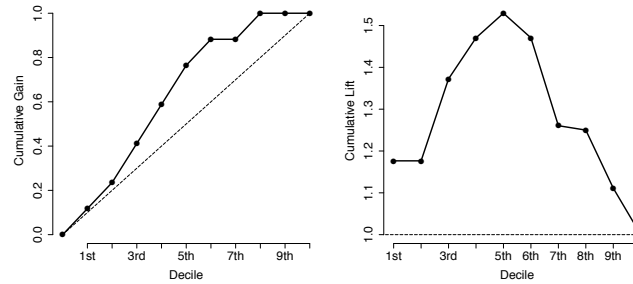
根据此表，我们可以计算每个十分位数的增益和提升（回答问题不需要提升，但为了完整性而包括在内）以及每个十分位数的累积增益和累积提升。回想一下，增益、累积增益、提升和累积提升的定义始终是指实际目标水平，而不是预测，定义如下：

$$\begin{aligned}
 \text{gain}(dec) &= \frac{\text{num positive test instances in decile } dec}{\text{num positive test instances}} \\
 \text{cumulative gain}(dec) &= \frac{\text{num positive test instances in all deciles up to } dec}{\text{num positive test instances}} \\
 \text{lift}(dec) &= \frac{\% \text{ of positive test instances in decile } dec}{\% \text{ of positive test instances}} \\
 \text{cumulative lift}(dec) &= \frac{\% \text{ of positive instances in all deciles up to } dec}{\% \text{ of positive test instances}}
 \end{aligned}$$

The measures for each decile for Model 1 are shown in the table below.

| Decile           | Positive<br>(true)<br>Count | Negative<br>(false)<br>Count | Gain   | Cum.<br>Gain | Lift   | Cum.<br>Lift |
|------------------|-----------------------------|------------------------------|--------|--------------|--------|--------------|
| 1 <sup>st</sup>  | 2                           | 1                            | 0.1176 | 0.1176       | 1.1765 | 1.1765       |
| 2 <sup>nd</sup>  | 2                           | 1                            | 0.1176 | 0.2353       | 1.1765 | 1.1765       |
| 3 <sup>rd</sup>  | 3                           | 0                            | 0.1765 | 0.4118       | 1.7647 | 1.3725       |
| 4 <sup>th</sup>  | 3                           | 0                            | 0.1765 | 0.5882       | 1.7647 | 1.4706       |
| 5 <sup>th</sup>  | 3                           | 0                            | 0.1765 | 0.7647       | 1.7647 | 1.5294       |
| 6 <sup>th</sup>  | 0                           | 3                            | 0.0000 | 0.7647       | 0.0000 | 1.2745       |
| 7 <sup>th</sup>  | 1                           | 2                            | 0.0588 | 0.8235       | 0.5882 | 1.1765       |
| 8 <sup>th</sup>  | 1                           | 2                            | 0.0588 | 0.8824       | 0.5882 | 1.1029       |
| 9 <sup>th</sup>  | 1                           | 2                            | 0.0588 | 0.9412       | 0.5882 | 1.0458       |
| 10 <sup>th</sup> | 0                           | 3                            | 0.0000 | 0.9412       | 0.0000 | 0.9412       |

Using this table, we can now draw the required cumulative gain chart as follows (the cumulative lift chart is also shown).



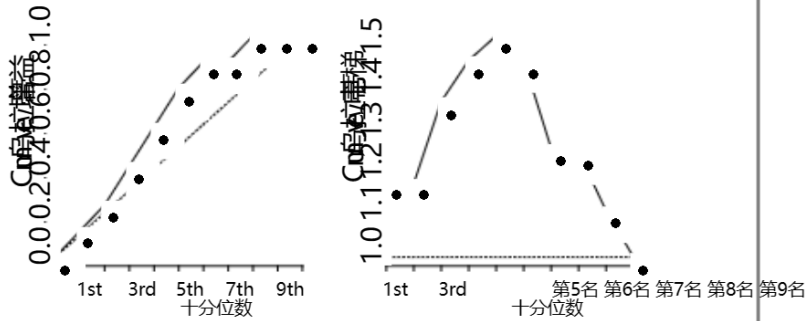
We then repeat for Model 2.

增益 $PDEC_q$  十分位数 DEC 中的阳性测试实例数  
NUM 阳性测试实例数  
累积增益 $PDEC_q$  截至 12 月, 所有十分位数的阳性测试实例数  
NUM 阳性测试实例数  
李芳  $TPDEC_q$  十分之一 Dec 的阳性检测实例百分比  
阳性检测实例的百分比  
累积 LI F  $TPDEC_q$  截至 12 月, 所有十分位数的阳性病例百分比  
阳性检测实例的百分比

Model 1 每个十分位数的测量值如下表所示。

| 正负   |     | Cum. |        | Cum.   |        |
|------|-----|------|--------|--------|--------|
| (真)  | (假) |      |        |        |        |
| 十分位数 | 计数  | 增益   | 提升     |        |        |
| 1    | 2   | 1    | 0.1176 | 0.1176 | 1.1765 |
| 2    | 2   | 1    | 0.1176 | 0.2353 | 1.1765 |
| 3    | 3   | 0    | 0.1765 | 0.4118 | 1.7647 |
| 4    | 3   | 0    | 0.1765 | 0.5882 | 1.7647 |
| 5    | 3   | 0    | 0.1765 | 0.7647 | 1.7647 |
| 6    | 0   | 3    | 0.0000 | 0.7647 | 0.0000 |
| 7    | 1   | 2    | 0.0588 | 0.8235 | 0.5882 |
| 8    | 1   | 2    | 0.0588 | 0.8824 | 0.5882 |
| 9    | 1   | 2    | 0.0588 | 0.9412 | 0.5882 |
| 10   | 0   | 3    | 0.0000 | 0.9412 | 0.0000 |

使用此表，我们现在可以绘制所需的累积增益图表，如下所示（还显示了累积提升图表）。



然后，我们对模型 2 重复上述步骤。

| Decile           | ID | Target | Model 1<br>Prediction | Model 1<br>Score | Model 1<br>Outcome |
|------------------|----|--------|-----------------------|------------------|--------------------|
| 1 <sup>st</sup>  | 7  | true   | 0.9800                | true             | TP                 |
|                  | 22 | true   | 0.9521                | true             | TP                 |
|                  | 6  | true   | 0.9398                | true             | TP                 |
| 2 <sup>nd</sup>  | 14 | true   | 0.9274                | true             | TP                 |
|                  | 28 | true   | 0.9200                | true             | TP                 |
|                  | 11 | true   | 0.9062                | true             | TP                 |
| 3 <sup>rd</sup>  | 13 | true   | 0.8764                | true             | TP                 |
|                  | 30 | true   | 0.8614                | true             | TP                 |
|                  | 8  | true   | 0.8578                | true             | TP                 |
| 4 <sup>th</sup>  | 3  | true   | 0.8378                | true             | TP                 |
|                  | 17 | true   | 0.8086                | true             | TP                 |
|                  | 21 | true   | 0.7843                | true             | TP                 |
| 5 <sup>th</sup>  | 12 | true   | 0.7301                | true             | TP                 |
|                  | 4  | true   | 0.7160                | true             | TP                 |
|                  | 19 | false  | 0.5809                | true             | FP                 |
| 6 <sup>th</sup>  | 20 | false  | 0.5783                | true             | FP                 |
|                  | 10 | false  | 0.5600                | true             | FP                 |
|                  | 24 | true   | 0.4708                | false            | FN                 |
| 7 <sup>th</sup>  | 16 | true   | 0.4569                | false            | FN                 |
|                  | 27 | true   | 0.3562                | false            | FN                 |
|                  | 15 | false  | 0.2992                | false            | TN                 |
| 8 <sup>th</sup>  | 25 | false  | 0.2846                | false            | TN                 |
|                  | 1  | false  | 0.2089                | false            | TN                 |
|                  | 5  | false  | 0.1891                | false            | TN                 |
| 9 <sup>th</sup>  | 9  | false  | 0.1560                | false            | TN                 |
|                  | 18 | false  | 0.1458                | false            | TN                 |
|                  | 26 | false  | 0.1100                | false            | TN                 |
| 10 <sup>th</sup> | 29 | false  | 0.0895                | false            | TN                 |
|                  | 23 | false  | 0.0377                | false            | TN                 |
|                  | 2  | false  | 0.0080                | false            | TN                 |

We can now calculate gain, cumulative gain, lift, and cumulative lift for each decile for Model 2.

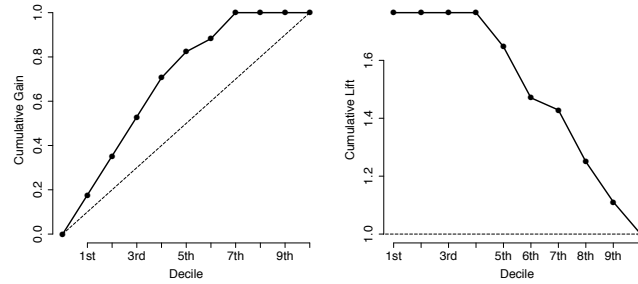
## 第9章 评估 (运动解决方案)

| 十分位数 | 目标 | 模型 1 预测           | 模型 1 得分 | 模型 1 结果 |
|------|----|-------------------|---------|---------|
| 1st  | 7  | true 0.9800       | true    | TP      |
|      | 22 | 真 0.9521 真 TP     |         |         |
|      | 6  | true 0.9398       | true    | TP      |
|      | 14 | true 0.9274       | true    | TP      |
| 2    | 28 | true 0.9200       | true    | TP      |
|      | 11 | true 0.9062       | true    | TP      |
|      | 13 | true 0.8764       | true    | TP      |
| 3    | 30 | 真 0.8614 真 TP     |         |         |
|      | 8  | true 0.8578       | true    | TP      |
|      | 3  | true 0.8378       | true    | TP      |
| 4    | 17 | true 0.8086       | true    | TP      |
|      | 21 | true 0.7843       | true    | TP      |
|      | 12 | 真 0.7301 真 TP     |         |         |
| 5    | 4  | true 0.7160       | true    | TP      |
|      | 19 | 假 0.5809          | true    | FP      |
|      | 20 | false 0.5783 真 FP |         |         |
| 6    | 10 | false 0.5600 真 FP |         |         |
|      | 24 | true 0.4708       | 假       | FN      |
|      | 16 | true 0.4569       | 假       | FN      |
| 7    | 27 | true 0.3562       | 假       | FN      |
|      | 15 | 假 0.2992          | 假       | TN      |
|      | 25 | 假 0.2846 假 TN     |         |         |
| 8    | 1  | 假 0.2089 假 TN     |         |         |
|      | 5  | 假 0.1891          | 假       | TN      |
|      | 9  | 假 0.1560 假 TN     |         |         |
| 9    | 18 | 假 0.1458 假 TN     |         |         |
|      | 26 | 假 0.1100          | 假       | TN      |
|      | 29 | 假 0.0895 假 TN     |         |         |
| 10   | 23 | 假 0.0377 假 TN     |         |         |
|      | 2  | 假 0.0080          | 假       | TN      |

现在，我们可以计算 Model 2 每个十分位数的增益、累积增益、提升和累积提升。

| Decile           | Positive<br>(true)<br>Count | Negative<br>(false)<br>Count | Gain   | Cum.<br>Gain | Lift   | Cum.<br>Lift |
|------------------|-----------------------------|------------------------------|--------|--------------|--------|--------------|
| 1 <sup>st</sup>  | 3                           | 0                            | 0.1765 | 0.1765       | 1.7647 | 1.7647       |
| 2 <sup>nd</sup>  | 2                           | 1                            | 0.1176 | 0.2941       | 1.1765 | 1.4706       |
| 3 <sup>rd</sup>  | 3                           | 0                            | 0.1765 | 0.4706       | 1.7647 | 1.5686       |
| 4 <sup>th</sup>  | 3                           | 0                            | 0.1765 | 0.6471       | 1.7647 | 1.6176       |
| 5 <sup>th</sup>  | 3                           | 0                            | 0.1765 | 0.8235       | 1.7647 | 1.6471       |
| 6 <sup>th</sup>  | 0                           | 3                            | 0.0000 | 0.8235       | 0.0000 | 1.3725       |
| 7 <sup>th</sup>  | 1                           | 2                            | 0.0588 | 0.8824       | 0.5882 | 1.2605       |
| 8 <sup>th</sup>  | 1                           | 2                            | 0.0588 | 0.9412       | 0.5882 | 1.1765       |
| 9 <sup>th</sup>  | 1                           | 2                            | 0.0588 | 1.0000       | 0.5882 | 1.1111       |
| 10 <sup>th</sup> | 0                           | 3                            | 0.0000 | 1.0000       | 0.0000 | 1.0000       |

Using this table, we can now draw the required cumulative gain chart as follows (the cumulative lift chart is also shown).

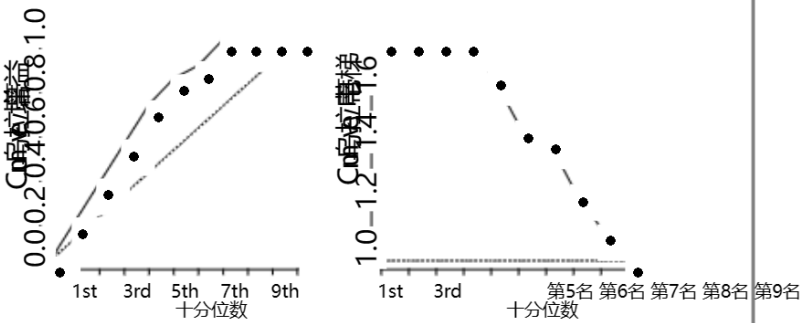


- (e) The charity for which the model is being built typically has only enough money to send a mailshot to the top 20% of its contact list. Based on the cumulative gain chart generated in the previous part, would you recommend that Model 1 or Model 2 would perform best for the charity?

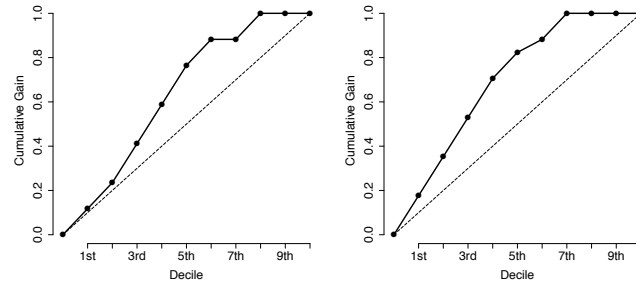
The cumulative gain charts are repeated here.

|      | 正<br>(真) | 负<br>(假) |        |        |        |        |      |
|------|----------|----------|--------|--------|--------|--------|------|
| 十分位数 | 计数       | 计数       | 增益     | 增益提升   | Cum.   | Cum.   | Cum. |
| 1    | 3        | 0        | 0.1765 | 0.1765 | 1.7647 | 1.7647 |      |
| 2    | 2        | 1        | 0.1176 | 0.2941 | 1.1765 | 1.4706 |      |
| 3    | 3        | 0        | 0.1765 | 0.4706 | 1.7647 | 1.5686 |      |
| 4    | 3        | 0        | 0.1765 | 0.6471 | 1.7647 | 1.6176 |      |
| 5    | 3        | 0        | 0.1765 | 0.8235 | 1.7647 | 1.6471 |      |
| 6    | 0        | 3        | 0.0000 | 0.8235 | 0.0000 | 1.3725 |      |
| 7    | 1        | 2        | 0.0588 | 0.8824 | 0.5882 | 1.2605 |      |
| 8    | 1        | 2        | 0.0588 | 0.9412 | 0.5882 | 1.1765 |      |
| 9    | 1        | 2        | 0.0588 | 1.0000 | 0.5882 | 1.1111 |      |
| 10   | 0        | 3        | 0.0000 | 1.0000 | 0.0000 | 1.0000 |      |

使用此表，我们现在可以绘制所需的累积增益图表，如下所示（还显示了累积提升图表）。

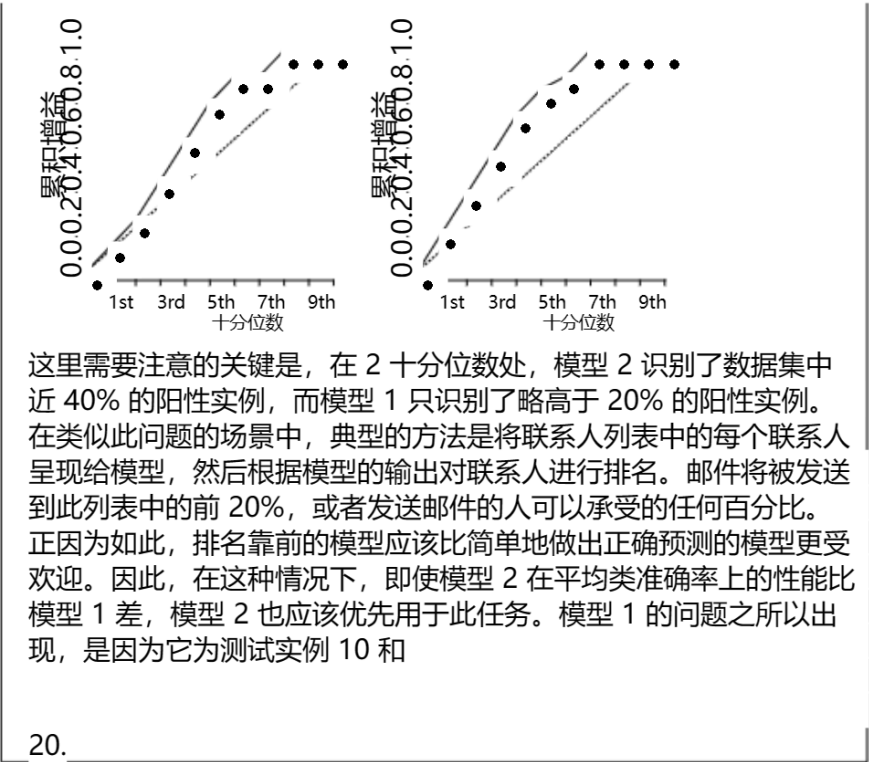


(e) 建立模型的慈善机构通常只有足够的钱向其联系人列表中的前 20% 发送邮件。根据上一部分生成的累积增益图表，您认为模型 1 或模型 2 对慈善机构来说表现最好吗？累积增益图表在此处重复。



The key thing to notice here is that at the 2<sup>nd</sup> decile, Model 2 has identified almost 40% of the positive instances in the dataset, whereas Model 1 has identified only just over 20%. In a scenario like the one in this question, the typical approach would be to present every contact in the contact list to a model and then rank the contacts by the output of the model. Mailshots would be sent to the top 20% of this list, or whatever percentage the people sending the mailshot could afford. Because of this, a model that ranks well should be preferred over one that simply makes correct predictions. So in this case, even though Model 2 performs worse than Model 1 based on average class accuracy, Model 2 should be preferred for this task. The issue for Model 1 arises because it gives very high prediction scores to test instances 10 and 20.

- \* 9. The following table lists the scores returned by a prediction model for a test set of 12 examples. The prediction task is a binary classification task, and the instances in the test set are labeled as belonging to the *positive* or *negative* class. For ease of reading, the instances have been ordered in descending order of the score the model assigned to each instance.



° 9.下表列出了预测模型返回的 12 个测试集的分  
例子。预测任务是二元分类任务，测试集中的实例被标记为属于正类或负类。为了便于阅读，实例已按模型分配给每个实例的分数的降序排序。

| ID | Target   | Score |
|----|----------|-------|
| 1  | positive | 0.75  |
| 2  | positive | 0.72  |
| 3  | positive | 0.64  |
| 4  | negative | 0.62  |
| 5  | negative | 0.55  |
| 6  | positive | 0.48  |
| 7  | negative | 0.45  |
| 8  | negative | 0.44  |
| 9  | negative | 0.38  |
| 10 | negative | 0.35  |
| 11 | negative | 0.32  |
| 12 | negative | 0.31  |

- (a) Calculate the **ROC index** for this model using the **trapezoidal method** and the following set of thresholds: 1.0, 0.5, and 0.0.

| ID | Target   | Score | Predictions by Threshold |          |          |
|----|----------|-------|--------------------------|----------|----------|
|    |          |       | T= 1.0                   | T= 0.5   | T= 0.0   |
| 1  | positive | 0.75  | negative                 | positive | positive |
| 2  | positive | 0.72  | negative                 | positive | positive |
| 3  | positive | 0.64  | negative                 | positive | positive |
| 4  | negative | 0.62  | negative                 | positive | positive |
| 5  | negative | 0.55  | negative                 | positive | positive |
| 6  | positive | 0.48  | negative                 | negative | positive |
| 7  | negative | 0.45  | negative                 | negative | positive |
| 8  | negative | 0.44  | negative                 | negative | positive |
| 9  | negative | 0.38  | negative                 | negative | positive |
| 10 | negative | 0.35  | negative                 | negative | positive |
| 11 | negative | 0.32  | negative                 | negative | positive |
| 12 | negative | 0.31  | negative                 | negative | positive |

The true positive rate and false positive rate for threshold 1 (1.0) are:

$$TPR = \frac{0}{4} = 0$$

$$FPR = \frac{0}{8} = 0$$

第 9 章 评估 (运动解决方案)

| ID | 目标分数    |
|----|---------|
| 1  | 好评 0.75 |
| 2  | 正数 0.72 |
| 3  | 好评 0.64 |
| 4  | 负数 0.62 |
| 5  | 负 0.55  |
| 6  | 好评 0.48 |
| 7  | 负 0.45  |
| 8  | 负数 0.44 |
| 9  | 负 0.38  |
| 10 | 负 0.35  |
| 11 | 负数 0.32 |
| 12 | 负 0.31  |

(a) 使用梯形方法和以下一组阈值计算该模型的 ROC 指数: 1.0、0.5 和 0.0。

| 阈值      | ID      | 预测    | 目标得分           |
|---------|---------|-------|----------------|
| T = 1.0 | T = 0.5 | T = 0 |                |
| 1       | 0       | 阳性    | 0.75 阴性 阳性 阳性  |
| 2       | 0       | 阳性    | 0.72 阴性 阳性 阳性  |
| 3       | 0       | 阳性    | 0.64 阴性 阳性 阳性  |
| 4       | 0       | 负数    | 0.62 负数 正数 正数  |
| 5       | 0       | 负数    | 0.55 负 正数 正数   |
| 6       | 0       | 阳性    | 0.48 阴性 阴性 阳性  |
| 7       | 0       | 负数    | 0.45 负负数 负数 正数 |
| 8       | 0       | 负数    | 0.44 负负负数 正数   |
| 9       | 0       | 负数    | 0.38 负数 负数 正数  |
| 10      | 0       | 负数    | 0.35 负负负数 正数   |
| 11      | 0       | 负数    | 0.32 负数 负数 正数  |
| 12      | 0       | 负数    | 0.31 负负数 负数 正数 |

阈值 1 (1.0) 的真阳性率和假阳性率为:

T 公关

FPR

0

4

0

8

"

"

0

0

The true positive rate and false positive rate for threshold 2 (0.5) are:

$$TPR = \frac{3}{4} = 0.75$$

$$FPR = \frac{2}{8} = 0.25$$

The true positive rate and false positive rate for threshold 3 (0.0) are:

$$TPR = \frac{4}{4} = 1.00$$

$$FPR = \frac{8}{8} = 1.00$$

The ROC index is then calculated as:

$$\begin{aligned} ROC\ index &= \frac{(0.25 - 0) \times (0.75 + 0)}{2} + \frac{(1.00 - 0.25) \times (1.00 + 0.75)}{2} \\ &= 0.09375 + 0.65625 \\ &= 0.75 \end{aligned}$$

- (b) The following table lists the scores returned by the same prediction model for a new test set of 12 examples. Again, the prediction task is a binary classification task, and the instances in the test set are labeled as belonging to the *positive* or *negative* class. For ease of reading, the instances have been ordered in descending order of the score the model assigned to each instance. Calculate the ROC index for this model using the **trapezoidal method** and the following set of thresholds: 1.0, 0.5, and 0.0.

| ID | Target   | Score |
|----|----------|-------|
| 1  | positive | 0.71  |
| 2  | positive | 0.70  |
| 3  | positive | 0.66  |
| 4  | positive | 0.65  |
| 5  | positive | 0.62  |
| 6  | positive | 0.60  |
| 7  | negative | 0.58  |
| 8  | positive | 0.48  |
| 9  | positive | 0.34  |
| 10 | negative | 0.30  |
| 11 | negative | 0.28  |
| 12 | negative | 0.25  |

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阈值 2 的真阳性率和假阳性率 (0.5) 为:

$$T \text{ 公关} = \frac{3}{4} = 0.75$$

$$FPR = \frac{2}{8} = 0.25$$

阈值 3 的真阳性率和假阳性率 (0.0) 为:

$$T \text{ 公关} = \frac{4}{4} = 1.00$$

$$FPR = \frac{8}{8} = 1.00$$

然后, ROC指数的计算公式为:

$$\begin{aligned} \text{ROC指数} &= \frac{p(0.25) - p(0.75)}{2} - \frac{p(1.00) - p(0.75)}{2} \\ &= 0.09375 - 0.65625 \\ &= 0.75 \end{aligned}$$

(b) 下表列出了同一预测模型为一组新的12个样本测试返回的分数。同样, 预测任务是二元分类任务, 测试集中的实例被标记为属于正类或负类。为了便于阅读, 实例已按模型分配给每个实例的分数的降序排序。计算 ROC 指数

对于此模型, 使用梯形方法和以下阈值集: 1.0、0.5 和 0.0。

| ID | 目标分数    |
|----|---------|
| 1  | 正 0.71  |
| 2  | 正数 0.70 |
| 3  | 正 0.66  |
| 4  | 正 0.65  |
| 5  | 正数 0.62 |
| 6  | 正数 0.60 |
| 7  | 负 0.58  |
| 8  | 好评 0.48 |
| 9  | 正 0.34  |
| 10 | 负 0.30  |
| 11 | 负 0.28  |
| 12 | 负数 0.25 |

| ID | Target   | Score | Predictions by Threshold |          |          |
|----|----------|-------|--------------------------|----------|----------|
|    |          |       | T= 1.0                   | T= 0.5   | T= 0.0   |
| 1  | positive | 0.71  | negative                 | positive | positive |
| 2  | positive | 0.70  | negative                 | positive | positive |
| 3  | positive | 0.66  | negative                 | positive | positive |
| 4  | positive | 0.65  | negative                 | positive | positive |
| 5  | positive | 0.62  | negative                 | positive | positive |
| 6  | positive | 0.60  | negative                 | positive | positive |
| 7  | negative | 0.58  | negative                 | positive | positive |
| 8  | positive | 0.48  | negative                 | negative | positive |
| 9  | positive | 0.34  | negative                 | negative | positive |
| 10 | negative | 0.30  | negative                 | negative | positive |
| 11 | negative | 0.28  | negative                 | negative | positive |
| 12 | negative | 0.25  | negative                 | negative | positive |

The true positive rate and false positive rate for threshold 1 (1.0) are:

$$TPR = \frac{0}{8} = 0$$

$$FPR = \frac{0}{4} = 0$$

The true positive rate and false positive rate for threshold 2 (0.5) are:

$$TPR = \frac{6}{8} = 0.75$$

$$FPR = \frac{1}{4} = 0.25$$

The true positive rate and false positive rate for threshold 3 (0.0) are:

$$TPR = \frac{8}{8} = 1.00$$

$$FPR = \frac{4}{4} = 1.00$$

## 第 9 章 评估 (运动解决方案)

| 阈值 | ID    | 预测 | 目标得分 |    |    |    |
|----|-------|----|------|----|----|----|
| 1  | "1.0" | T  | 0.5  | T  | "  |    |
| 2  | 0.0   | 阳性 | 0.71 | 阴性 | 阳性 | 阳性 |
| 3  |       | 阳性 | 0.70 | 阴性 | 阳性 | 阳性 |
| 4  |       | 阳性 | 0.66 | 阴性 | 阳性 | 阳性 |
| 5  |       | 阳性 | 0.65 | 阴性 | 阳性 | 阳性 |
| 6  |       | 阳性 | 0.62 | 阴性 | 阳性 | 阳性 |
| 7  |       | 阳性 | 0.60 | 阴性 | 阳性 | 阳性 |
| 8  |       | 负数 | 0.58 | 负数 | 正数 | 正数 |
| 9  |       | 阳性 | 0.48 | 阴性 | 阴性 | 阳性 |
| 10 |       | 阳性 | 0.34 | 阴性 | 阴性 | 阳性 |
| 11 |       | 负数 | 0.30 | 负数 | 负数 | 正数 |
| 12 |       | 负数 | 0.28 | 负数 | 负数 | 正数 |
| 12 |       | 负数 | 0.25 | 负数 | 负数 | 正数 |

阈值 1 (1.0) 的真阳性率和假阳性率为:

$$\begin{aligned} \text{T 公关} &= \frac{0}{8} = 0 \\ \text{FPR} &= \frac{0}{4} = 0 \end{aligned}$$

阈值 2 的真阳性率和假阳性率 (0.5) 为:

$$\begin{aligned} \text{T 公关} &= \frac{6}{8} = 0.75 \\ \text{FPR} &= \frac{1}{4} = 0.25 \end{aligned}$$

阈值 3 的真阳性率和假阳性率 (0.0) 为:

$$\begin{aligned} \text{T 公关} &= \frac{8}{8} = 1.00 \\ \text{FPR} &= \frac{4}{4} = 1.00 \end{aligned}$$

The ROC index is then calculated as:

$$\begin{aligned} \text{ROC index} &= \frac{(0.25 - 0) \times (0.75 + 0)}{2} + \frac{(1.00 - 0.25) \times (1.00 + 0.75)}{2} \\ &= 0.09375 + 0.65625 \\ &= 0.75 \end{aligned}$$

- (c) The ROC index is insensitive to changes in class distribution within the test set. This means that if the proportion of positive to negative instances changes in a test set, the ROC index will remain the same if the performance of the models on each class is constant. Consequently, the ROC index is robust to class imbalance or skew in the test set. Why do you think this is the case?<sup>1</sup>

In a binary confusion matrix the proportion of positive and negative instances is represented by the ratio between the first row (containing the true positive and false negatives counts) to the second row (containing the false positives and true negative counts).

The ROC index is based on the true positive rate (TPR) and the false positive rate (FPR), and each of these rates is calculated as a ratio within a single class: calculating TPR involves dividing the number of true positives by the total number of positive instances in the test set, and calculating FPR involves dividing the number of false positive by the total number of negative instances in the test set. Consequently, if the performance of the model on both the positive and negative classes remain constant (in terms of the ratio of positive examples it get correct to the total number of positive examples, and the ratio of the negative examples it get wrong to the total number of negative examples) then the ROC index will remain constant.

We can see this if we compare the ROC indices we calculated in the first two parts of this question. The ROC indices are the same for this model on both of these test sets even though in the first test set the ratio of positive to negative instances was 4:8 and in the second test set the ratio was 8:4.

1. We recommend ? as an excellent introduction and overview to ROC analysis that covers the topic of imbalance in the test set. Note also that in very highly imbalanced data where there is a very large number of negative examples, the false positive rate is not very sensitive to changes in the number of false positives (because the denominator is so large) and in these contexts, if the focus of the model is on detecting the positive class, it is probably advisable to use precision as the evaluation metric, as it focuses more on the ability to detect the positive class, rather than on the ability to distinguish between classes (which the ROC index captures).

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然后, ROC指数的计算公式为:

$$\text{ROC指数} = \frac{p0.25 \cdot 0q - p0.75 \cdot 0q}{2} - \frac{P1.00 \cdot 0.25Q - P1.00 \cdot 0.75Q}{2} \\ = \frac{0.09375 - 0.65625}{0.75}$$

(c) ROC指数对测试集中类别分布的变化不敏感。

这意味着, 如果测试集中正负实例的比例发生变化, 则如果模型在每个类上的性能保持不变, 则 ROC 指数将保持不变。因此, ROC 指数对测试集中的类别不平衡或偏斜具有鲁棒性。你认为为什么会这样?

在二元混淆矩阵中, 正负实例的比例由第一行 (包含真阳性和假阴性计数) 与第二行 (包含假阳性和真阴性计数) 之间的比率表示。

ROC 指数基于真阳性率 (TPR) 和假阳性率 (FPR), 每个率都计算为单个类别中的比率: 计算 TPR 涉及将真阳性数除以测试集中的阳性实例总数, 计算 FPR 涉及将假阳性数除以测试集中的阴性实例总数。因此, 如果模型在正类和负类上的性能保持不变 (就正样本数与正样本总数的比率而言, 以及它出错的负样本数与负样本总数的比率), 则 ROC 指数将保持不变。

如果我们比较我们在这个问题的前两部分中计算的 ROC 指数, 我们可以看到这一点。该模型在这两个测试集上的 ROC 指数是相同的, 即使在第一个测试集中, 阳性与阴性实例的比例为 4: 8, 而在第二个测试集中, 该比例为 8: 4。

1.我们推荐? 作为对 ROC 分析的出色介绍和概述, 涵盖了测试集中的不平衡主题。另请注意, 在存在大量负样本的非常不平衡的数据中, 误报率对误报数量的变化不是很敏感 (因为分母太大), 在这些上下文中, 如果模型的重点是检测正类, 建议使用精确度作为评估指标, 因为它更侧重于检测正类的能力, 而不是区分正负类的能力 (ROC 曲线指标)。

This is because the model's TPR and FPR was constant across both test sets (TPR=0.75 and FPR=0.25).

This contrasts with other metrics such as accuracy, precision and F-score which use counts from both classes in their calculation. For example, calculating precision involves the true positive count (which is a count of a portion of the set of positive instances in the test set) and the false positive count (which is a count of the portion of negative instances in the test set). In fact, if we calculate the  $F_1$  for this model on both of these test sets assuming a threshold of 0.5 we see that the  $F_1$  measure for the same model is different on both these datasets even though its TPR and FPR is consistent across the test sets. This shows that the  $F_1$  is sensitive to class distribution in the test set. The calculations are as follows:

$F_1$  for test set 1:

$$\begin{aligned} F_1 &= 2 \times \frac{\frac{3}{3+2} \times \frac{3}{3+1}}{\frac{3}{3+2} + \frac{3}{3+1}} \\ &= 2 \times \frac{0.6 \times 0.75}{0.6 + 0.75} \\ &= 0.6667 \end{aligned}$$

$F_1$  for test set 2:

$$\begin{aligned} F_1 &= 2 \times \frac{\frac{6}{6+1} \times \frac{6}{6+2}}{\frac{3}{3+2} + \frac{3}{3+1}} \\ &= 2 \times \frac{0.857142857 \times 0.75}{0.857142857 + 0.75} \\ &= 0.8 \end{aligned}$$

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这是因为模型的 TPR 和 FPR 在两个测试集中都是恒定的 (TPR=0.75 和 FPR=0.25)。这与准确度、精确度和 F 分数等其他指标形成鲜明对比,后者在计算中使用两个类别的计数。例如,计算精度涉及真阳性计数 (即测试集中部分阳性实例的计数) 和假阳性计数 (即测试集中部分阴性实例的计数)。事实上,如果我们在这两个测试集上计算这个模型的 F, 假设阈值为 0.5, 我们会看到同一模型的 Fmeasure 在这两个数据集上是不同的, 即使它的 TPR 和 FPR 在测试集中是一致的。这表明 Fis 对测试集中的类分布很敏感。计算方法如下:

F对于测试集 1:

$$F = 2 \times \frac{0.6 \times 0.75}{0.6 + 0.75} = 0.6667$$

F对于测试集 2:

$$F = 2 \times \frac{0.61 \times 0.857142857}{0.61 + 0.857142857} = 0.8$$